The Chinese University of Hong Kong

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Introduction to Axiomatic Set Theory

Final Year Project Report

Author: Chan Kwan Ho (1155194495)

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Supervisor: Dr. Ng Ming Ho

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Declaration

I hereby declare that this Final Year Project titled **"Introduction to Axiomatic Set Theory"** is the result of my own work, except where explicit reference is made to the work of others or course materials.

This work has not been submitted, either in whole or in part, for any other degree or professional qualification.

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Abstract

This report presents a formal examination of foundational concepts in propositional logic, well-ordered sets, ordinals, first-order logic, and ZF set theory. The theoretical framework directly expands upon the *Part II Logic and Set Theory* lecture series delivered by Dr. András Zsák at the University of Cambridge (Lent 2025).

Key deliverables include constructing the ZF set theory from First-Order predicate logic, which requires the formal definition and introduction on Propositional Logic, Well-Ordered Sets, and also with the help of Axiom of Choice via Zorn's Lemma.

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Contents

Declaration i

Abstract ii

Acknowledgements iii

1 Propositional Logic 1
- 1.1 Introduction . . . 1
- 1.2 Semantic Entailment . . . 3
- 1.3 Syntactic Entailment . . . 6

2 Well-Ordering and Ordinals 26
- 2.1 Introduction . . . 26
- 2.2 Ordinals . . . 44
- 2.3 Ordinal Arithmetics . . . 50
  - 2.3.1 Ordinal Addition . . . 50
  - 2.3.2 Ordinal Multiplication . . . 55
  - 2.3.3 Ordinal Exponentiation . . . 58

3 Posets and Zorn's Lemma 59
- 3.1 Introduction . . . 59
- 3.2 Zorn's Lemma . . . 64
- 3.3 The Axiom of Choice (AC) . . . 67

4 First-Order Predicate Logic 68
- 4.1 Introduction . . . 68
- 4.2 The Language . . . 69
- 4.3 Structures . . . 72
- 4.4 Theories and Models . . . 78
- 4.5 Semantic entailment . . . 81
- 4.6 Syntactic entailment . . . 84
  - 4.6.1 Axiom . . . 84
  - 4.6.2 Rules of deduction . . . 84
- 4.7 Relationship between $S \vdash p$ and $S \models p$ . . . 86
- 4.8 Applications of completeness/compactness . . . 104

4.9 Peano Arithmetic 105

5 Set Theory **107**

5.1 Construction of ZF 107

5.1.1 Axiom of Extensionality (Ext) 107

5.1.2 Axiom of Separation (Sep) 108

5.1.3 Empty-Set Axiom (Emp) 108

5.1.4 Pair-Set Axiom (Pair) 109

5.1.5 Union Axiom(Un) 109

5.1.6 Power-Set Axiom (Pow) 109

5.1.7 Axiom of Infinity (Inf) 109

5.1.8 Axiom of Replacement (Rep) 109

5.1.9 Axiom of Foundation(Fnd) 109

5.1.10 ZF 109

References **110**

List of Figures

List of Tables

1.1 The law of excluded middle 5
1.2 Modus Ponens: $\{p, p \implies q\} \models q$ 6
1.3 Sum of evens: $\{p, q\} \models r$ 6
1.4 Modus Tollens: $\{p \implies q, \neg q\} \models \neg p$ 6

Chapter 1

Propositional Logic

1.1 Introduction

Definition 1.1. The language of propositional logic consists of a set P of *primitive propositions* and the set $L = L(P)$ of propositions (or compound propositions), which is defined inductively as follows.

1. $P \subset L$
2. $\bot \in L$ (the symbol '$\bot$' is read 'false' or 'bottom')
3. if $p, q \in L$, then $(p \Rightarrow q) \in L$.

Remark. The set $P = \{p_1, p_2, ...\}$ of primitive propositions has no internal logical structure, it simply stands for elementary statement that assign truth value (0 or 1).

The false symbol $\bot$ is a constant statement where it is a standalone proposition and it's truth value is always 0.

Lastly, if we have propositions p, q, the compound proposition $(p \Rightarrow q)$ is also in L.

Example 1.1. Suppose we have $P = \{p_1, p_2, p_3, ...\}$. By using the above 3 clauses, we can generate some interesting examples which is also in L: for $p_1, p_2, p_3, \bot \in L$,

1. $(p_1 \Rightarrow p_2) \in L$ by 3.
2. since $\bot, p_1 \in L$, $(p_1 \Rightarrow \bot) \in L$, and since $p_3 \in L$, we have $((p_1 \Rightarrow \bot) \Rightarrow p_3) \in L$
3. since $(p_1 \Rightarrow p_2), (p_1 \Rightarrow p_3) \in L$, then by 3. again that $((p_1 \Rightarrow p_2) \Rightarrow (p_1 \Rightarrow p_3)) \in L$
4. if $p \in L$, then $(p \Rightarrow \bot), \bot \in L$, then $((p \Rightarrow \bot) \Rightarrow \bot) \in L$

The reason for providing the above examples will be justified later.

Remark.
- For the set of propositions L, 'L is defined inductively', or $L = \bigcup_{n \in \mathbb{N}} L_n$.

  By following the 3 clauses, we can construct such rule of induction.

Define the initial set

$$L_1 := P \cup \{\bot\}$$

and for $n \in \mathbb{N}$,

$$L_{n+1} = L_n \cup \{(p \Rightarrow q) : p, q \in L_n\}$$

- A proposition is a finite string from the alphabet $P \cup \{\bot, \Rightarrow, (,)\}$:

  We define a symbol of length to be length 1. Suppose from the base case $L_1 = P \cup \{\bot\}$, which consists of only 1 symbol, hence length 1.

  Suppose by induction that propositions in L_n have finite length. Using above inductive constructions, $L_{n+1} = L_n \cup \{(p \Rightarrow q) : p, q \in L_n\}$. If $p, q \in L_n$ has length $|p|, |q|$, then $(p \Rightarrow q)$ has length $1 + |p| + 1 + |q| + 1 = |p| + |q| + 3$, which is also finite.Now since every $p \in L$ is belongs to L_n for some finite n. Since it is generated after a finite number of inductive steps, its length must remain finite.

- Define $\sum$ to be the set of all finite strings of symbol from the alphabet $P \cup \{\bot, \Rightarrow, (,)\}$. Then, L is the smallest subset of the set $\sum$ which satisfies the above 3 clauses.

  Suppose that there is a subset $M \subset \sum$ such that it satisfies the 3 clauses. We use induction to show that $L_n \subseteq M$ for all $n \in \mathbb{N}$. $L_1 = P \cup \{\bot\} \subset M$. Suppose $L_n \subseteq M$,for some $n \geq 1$. Let $s \in L_{n+1}$. If $s \in L_n$, it is done. Suppose not, then $s \in L_{n+1} \backslash L_n$. By definition, $s = (p \Rightarrow q)$ for some $p, q \in L_n$. Since from induction hypothesis, $L_n \subseteq M$, then $p, q \in M$. And by clause 3, $(p \Rightarrow q) \in M$. So we have $s \in M$, concluding that $L_{n+1} \subseteq M$.We conclude that L is the smallest subset of $\sum$.

  Note that $L \subsetneq \sum$, for example, $\Rightarrow)(p_1 \in \sum$ but is not a proposition.

- Every proposition is built uniquely using clauses 1-3 above, i.e., for every $p \in L$, $p \in P$ or $p = \bot$ or $p = (q \Rightarrow r)$ for unique $q, r \in L$.

1.2 Semantic Entailment

Given $S \subset L$, whenever S is true, MUST t also be true? Semantic Entailment is about TRUTH.

Definition 1.2. A *valuation* on L is a function $v : L \to \{0, 1\}$ such that :

1. $v(\bot) = 0$
2. $v(p \Rightarrow q) = \begin{cases} 0 & \text{if } v(p) = 1, v(q) = 0, \\ 1 & \text{otherwise} \end{cases}$ for all $p, q \in L$

Example 1.2. Suppose that $v(p_1) = 1, v(p_2) = 0$, then to find $v((\bot \Rightarrow p_1) \Rightarrow (p_1 \Rightarrow p_2))$, first note that $v(p_1 \Rightarrow p_2) = 0, v(\bot \Rightarrow p_1) = 1$, hence $v((\bot \Rightarrow p_1) \Rightarrow (p_1 \Rightarrow p_2)) = 0$.

Remark. If $v : L \to \{0, 1\}$ is a valuation defined on the set L, and $P \subset L$ is the set of primitive proposition. Then we define $v \restriction_P : P \to \{0, 1\}$.

Proposition 1.1. *1. if v, v' are valuations with $v \restriction_P = v' \restriction_P$, then $v = v'$*

2. For any function $w : P \to \{0, 1\}$, there exist a valuation v on L such that $v \restriction_P = w$.

Remark. The proposition says that a valuation is determined by its values on P, and any values will do. The second part of the proposition tells us that for if you defines a function on $w : P \to \{0, 1\}$, we can always construct the extension of it to entire L.(Uniqueness and Existence of extension).

Proof. 1. We should show that $L_n \subset L$ for every $n \in \mathbb{N}$. Let $L' := \{p \in L : v(p) = v'(p)\}$. By assumption,for $L_1 := P \cup \{\bot\}$, since $v(\bot) = v'(\bot) = 0, v(p) = v'(p)$, then $L_1 \subset L'$. If $p, q \in L'$, then $v(p) = v'(p), v(q) = v'(q)$, hence $v(p \Rightarrow q) = v'(p \Rightarrow q)$. It follows that by induction, supposing $L_n \subset L'$, then $L_{n+1} \subset L'$ for every $n \in \mathbb{N}$. By induction, $L_n \subset L'$ for all $n \in \mathbb{N}$, and thus $v = v'$.

2. First suppose we have a function $w : P \to \{0, 1\}$. Then set $v(p) = w(p)$ for every $p \in P$, $v(\bot) = 0$. This defines v on L_1. Having defined v on L_n for some $n \in \mathbb{N}$, for $p \in L_{n+1} \backslash L_n$, write $p = (q \Rightarrow r)$ for unique $q, r \in L_n$, and define

$$v(p) = \begin{cases} 0 & \text{if } v(q) = 1, v(r) = 0 \\ 1 & \text{otherwise} \end{cases}$$

This defines v on L_{n+1}. Thus we obtain a valuation v on L with $v \restriction_P = w$.

□

Definition 1.3. $t \in L$ is said to be a tautology if $v(t) = 1$ for all valuations v.

Definition 1.4. The symbols $\top, \wedge, \vee, \neg, \iff$ can be abbreviated as follows.

$$\begin{aligned} \top &= (\bot \Rightarrow \bot) \\ \neg p &= (p \Rightarrow \bot) \\ (p \vee q) &= (\neg p \Rightarrow q) \\ (p \wedge q) &= \neg(p \Rightarrow \neg q) \\ (p \iff q) &= ((p \Rightarrow q) \wedge (q \Rightarrow p)) \end{aligned}$$

for any $p, q \in L$. We have $v(\top) = 1$ for any valuation v, and so $\top$ is a tautology. Similarly, $v(\neg p), v(p \vee q)$ and $v(p \wedge q)$ have the expected values.

Remark. Note that the following: $\bot \Rightarrow B$ for $B \in L$ is forced to be a tautology, since $v(\bot) = 0$ for all valuation.

Example 1.3. This 3 examples of tautologies will paly a role later.

1. $(p \Rightarrow (q \Rightarrow p))$ ('a true statement is implied by anything').

   we check:

$v(p)$	$v(q)$	$v(q \Rightarrow p)$	$v(p \Rightarrow (q \Rightarrow p))$
0	0	1	1
0	1	0	1
1	0	1	1
1	1	1	1

   Note that the last row is a tuatology.

2. $(\neg\neg p \Rightarrow p)$, i.e. ,$(((p \Rightarrow \bot) \Rightarrow \bot) \Rightarrow p)$ ('the law of excluded middle').

   This can also be written as $(\neg p \vee p)$.

$v(p)$	$v(p \Rightarrow \bot)$	$v((p \Rightarrow \bot) \Rightarrow \bot)$	$v(((p \Rightarrow \bot) \Rightarrow \bot) \Rightarrow p)$
0	1	0	1
1	0	1	1

Table 1.1: The law of excluded middle

3. $((p \Rightarrow (q \Rightarrow r)) \Rightarrow ((p \Rightarrow q) \Rightarrow (p \Rightarrow r)))$

   Suppose that it is not a tautology. Then there must be a 0 valuation.

$$v((p \Rightarrow (q \Rightarrow r))) = 1 \ , \ v((p \Rightarrow q) \Rightarrow (p \Rightarrow r)) = 0$$

   for some valuation v. Then $v(p \Rightarrow q) = 1, v(p \Rightarrow r) = 0$. Thus, $v(p) = 1, v(r) = 0$, hence using $v(q) = 1$, and since $v((p \Rightarrow (q \Rightarrow r))) = 1, v(p) = 1, v(q) = 1$, we have $v(q \Rightarrow r) = 1$, $v(r) = 1$, which leads to a contradiction.

Definition 1.5. For $S \subset L$ and $t \in L$, say S entails t(or semantically entails t), written $S \models t$, if for every valuation v,

$$v(s) = 1 \text{ for all } s \in S \text{ implies } v(t) = 1$$

i.e. whenever all of S is true, t is true as well.

Example 1.4. 1. $\{p, p \Rightarrow q\} \models q$

   Let v be an arbitrary valuation such that $v(p) = 1, v(p \Rightarrow q) = 1$. since by definition, we must have $v(q) = 1$. Since v was arbitrary, this holds for all valuations, so $\{p, p \Rightarrow q\} \models q$.

p	q	$p \implies q$	Model of S? $(v(p) = 1 \wedge v(p \implies q) = 1)$	$v(t) = v(q)$
1	**1**	**1**	**Yes**	**1**
1	0	0	No	0
0	1	1	No	1
0	0	1	No	0

Table 1.2: Modus Ponens: $\{p, p \implies q\} \models q$

p (x even)	q (y even)	r ($x + y$ even)	Model of S? $(v(p) = 1 \wedge v(q) = 1)$	$v(t) = v(r)$
1	**1**	**1**	**Yes**	**1**
1	0	0	No	0
0	1	0	No	0
0	0	1	No	1

Table 1.3: Sum of evens: $\{p, q\} \models r$

2. $\{p \Rightarrow q, q \Rightarrow r\} \models (p \Rightarrow r)$

   Let v be an arbitrary valuation such that $v(p \Rightarrow q) = 1, v(q \Rightarrow r) = 1$, then we only need to check the valuation on q. Suppose that $v(q) = 1$, then we must have $v(r) = 1$, which gives $v(p \Rightarrow r) = 1$. If $v(q) = 0$, then we must have $v(p) = 0$, which also yield $v(p \Rightarrow r) = 1$.

3. Modus Ponen:

Note. t is a tautology if and only if $\emptyset \models t$, which we abbreviate to $\models t$.

Note that the compound proposition '$v(s) = 1$ for all $s \in \emptyset$ implies $v(t) = 1$' is vacuously true: since empty set contains no elements, the '$\forall s \in \emptyset, v(s) = 1$' is false.

Definition 1.6. For $t \in L$, if $v(t) = 1$, then say t is true in v or v is a model of t.

For $S \subset L$, a valuation v is a model of S if $v(s) = 1$ for all $s \in S$.

Note. $S \models t$ says that t is true in every model of S.

1.3 Syntactic Entailment

A notion of proof consists of axioms and deduction rules. In Propositional Logic, we adopt the following propositions as axioms.

p	q	$p \implies q$	$\neg q$	Model of S? $(v(p \implies q) = 1 \wedge v(\neg q) = 1)$	$v(t) = v(\neg p)$
1	1	1	0	No	0
1	0	0	1	No	0
0	1	1	0	No	1
0	**0**	**1**	**1**	**Yes**	**1**

Table 1.4: Modus Tollens: $\{p \implies q, \neg q\} \models \neg p$

(A1) $p \Rightarrow (q \Rightarrow p) \quad (p, q \in L)$
(A2) $(p \Rightarrow (q \Rightarrow r)) \Rightarrow ((p \Rightarrow q) \Rightarrow (p \Rightarrow r)) \quad (p, q, r \in L)$
(A3) $\neg\neg p \Rightarrow p \quad (p \in L)$

These are more accurately called 'axiom-schemes', as each is infinite collection of axioms. Notice that the $p, q \in L$ are not just premise proposition($p, q \in P$), but arbitrary compound/ premise statements from L. Hence we can generated infinitely many distinct formulas from the above 3 'axioms' and there are infinitely many way to replace the p, q in the 'axioms'.

Note. Note that the 3 axioms $A1 - A3$ are all tautologies: meaning that given any arbitrary valuation v, $v(A_{Ai}) = 1$ for $i = 1, 2, 3$.

Example 1.5.

- p : it is raining outside.

  q : You step outside without umbrella

  r : You get wet.

  If it is raining outside, you go outside without umbrella, then you get wet $\Rightarrow$ If we also know that whenever it rains, you are forced to step outside without an umbrella $(p \Rightarrow q)$, then whenever it rains, you get wet $(p \Rightarrow r)$.

- Let (x_n) be a sequence in a metric space (X, d) :

  $p : (x_n)$ is convergent

  $q : (x_n)$ is Cauchy sequence

  $r : (x_n)$ is bounded

  Since If (x_n) is convergent, then (if (x_n) is Cauchy, then (x_n) is bounded). (All Convergent is Cauchy, then hence is bounded). Note that in metric space, Cauchy is not always convergent: $x_n = \frac{1}{n}$ in $(0, 1)$ as metric space. And that: If (x_n) is convergent, then (x_n) is Cauchy. Therefore, if (x_n) is convergent, then (x_n) is bounded.

Note. The axioms are all tautologies proved above.

we will have only one deduction rule:
Modus Ponens:'from p and $p \Rightarrow q$, can deduce q'.

For $S \subset L$ and $t \in L$, a proof of t from S is a finite sequence $t_1, t_2, ..., t_n$ of propositions such that $t_n = t$ and for each i,

1. either t_i is an axiom.
2. or t_i is a member of S(premiss or hypothesis)
3. or t_i follows from earlier lines by modus ponens (MP) : there exist $j, k < i$ with $t_k = (t_j \Rightarrow t_i)$

Remark. Meaning that the syntactic entailment proof from S to t lies within 1 from 3 of the above possibility.

Definition 1.7. If there exists a proof of t from S, say S proves t, or syntactically entails t, written $S \vdash t$.

Say t is a **theorem** if $\emptyset \vdash t$, which we simply denote $\vdash t$.

Example 1.6. 1. $\{p \Rightarrow q\,,\, q \Rightarrow r\} \vdash (p \Rightarrow r)$

We have the premise be $S : \{p \Rightarrow q\,,\, q \Rightarrow r\}$, and conclusion $t : p \Rightarrow r$.

$(P1)$	$(p \Rightarrow (q \Rightarrow r)) \Rightarrow ((p \Rightarrow q) \Rightarrow (p \Rightarrow r))$	$(A2)$
$(P2)$	$(q \Rightarrow r) \Rightarrow (p \Rightarrow (q \Rightarrow r))$	$(A1)$
(1)	$q \Rightarrow r$	premiss
(2)	$p \Rightarrow (q \Rightarrow r)$	MP : $P2, 1$
(3)	$(p \Rightarrow q) \Rightarrow (p \Rightarrow r)$	MP : $P1, 2$
(4)	$p \Rightarrow q$	premiss
(5)	$p \Rightarrow r$	MP : $3, 4$

2. $\{\vdash (p \Rightarrow p)\}$

(1)	$p \Rightarrow ((p \Rightarrow p) \Rightarrow p)$	$(A1)$
(2)	$(p \Rightarrow ((p \Rightarrow p) \Rightarrow p)) \Rightarrow ((p \Rightarrow (p \Rightarrow p)) \Rightarrow (p \Rightarrow p))$	$(A2)$
(3)	$(p \Rightarrow (p \Rightarrow p)) \Rightarrow (p \Rightarrow p)$	MP: 1,2
(4)	$p \Rightarrow (p \Rightarrow p)$	$(A1)$
(5)	$p \Rightarrow p$	MP : $3, 4$

Remark. Note that (A1)-(A3) are axiom schemes where $p, q, r \in L$, they represent a universal promise for q, p, r can be any compound/ premise propositions from L.

Also, it is a good way of thinking the synthetic entailment to prove from the bottom to top: for example example 2: at conclusion we have $(p \Rightarrow p)$, to get to conclusion, we need MP from above, the simplest guess would be $p \Rightarrow (p \Rightarrow p)$, but we cannot or very hard to get p. Hence we need the $(p \Rightarrow (p \Rightarrow p))$ part to be the premise from MP and get the conclusion $p \Rightarrow p$.

Now:

$$(p \Rightarrow p) \leftarrow (p \Rightarrow (p \Rightarrow p)) \leftarrow ((p \Rightarrow (p \Rightarrow p)) \Rightarrow (p \Rightarrow p))$$

To get the latest part, we can define $r := p, q := (p \Rightarrow p)$ and use $(A2)$:

$$(p \Rightarrow ((p \Rightarrow p) \Rightarrow p)) \Rightarrow ((p \Rightarrow (p \Rightarrow p)) \Rightarrow (p \Rightarrow p))$$

Lastly to extract the conclusion part of $A2$, we need to use MP and also define the premise using $A1$ by letting $r := p, q := (p \Rightarrow p)$.

Note. Until here, we only have limited tools like $A1 - A3$ and also MP in showing $S \vdash t$. Introducing the following result is often helpful in showing the synthetic entailment.

Proposition 1.2 (**Deduction Theorem**). *Let $S \subset L$ and $p, q \in L$. Then we have $S \vdash (p \Rightarrow q)$ if and only if $S \cup \{p\} \vdash q$.*

Remark. Note that using above deduction theorem, we can simplified steps in showing $S \vdash t$.

For example in Example 1 of showing $\{p \Rightarrow q, q \Rightarrow r\} \vdash (p \Rightarrow r)$, using the theorem, it is enough to show that $\{p \Rightarrow q, q \Rightarrow r, p\} \vdash r$, which is much easier. By observing, we can already see the steps: write down all 3 premise accordingly in order , apply MP:1,3, then MP:2,4 to get r.

Proof. Assume that $S \vdash (p \Rightarrow q)$. Write down a proof of $p \Rightarrow q$ from S and add the following two lines to obtain a proof from $S \cup \{p\}$ to p:

$$\begin{array}{ll} p & \text{(premiss)} \\ q & \text{(MP)} \end{array}$$

And we are done.

Conversely, assume that $S \cup \{p\} \vdash q$ and let $t_1, t_2, ..., t_n = q$ be a proof of q from $S \cup \{p\}$. We show S proves $p \Rightarrow t_i$ for each i by induction. Then in particular, S proves $p \Rightarrow q$ and we are done.

Case 1: If t_i is an axiom or $t_i \in S$, then

$$\begin{array}{cc} t_i \Rightarrow (p \Rightarrow t_i) & \text{(A1)} \\ t_i & \text{(axiom or premiss)} \\ p \Rightarrow t_i & \text{(MP)} \end{array}$$

is a proof of $p \Rightarrow t_i$ from S.

Case 2: If $t_i = p$, then $S \vdash (p \Rightarrow t_i)$ since $\vdash (p \Rightarrow p)$ by Example 2.

Case 3: if there exist $j, k < i$ such that $t_k = (t_j \Rightarrow t_i)$, then by induction hypothesis that $S \vdash p \Rightarrow t_r$ for all $r < i$, there are proofs of $p \Rightarrow t_j$ and $p \Rightarrow (t_j \Rightarrow t_i)$ from S i.e. $S \vdash (p \Rightarrow t_j), S \vdash (p \Rightarrow (t_j \Rightarrow t_i))$.

Adding the lines

$$\begin{array}{cc} (p \Rightarrow (t_j \Rightarrow t_i)) \Rightarrow ((p \Rightarrow t_j) \Rightarrow (p \Rightarrow t_i)) & \text{(A2)} \\ (p \Rightarrow t_j) \Rightarrow (p \Rightarrow t_i) & \text{MP} \\ p \Rightarrow t_i & \text{MP} \end{array}$$

we obtain a proof of $p \Rightarrow t_i$ from S: $S \vdash (p \Rightarrow t_i)$, hence $S \vdash (p \Rightarrow q)$. □

Remark (**Hypothesis Discharge and Conditional Proofs**). Beyond simplifying proofs, the Deduction Theorem formally justifies *working under assumptions* (or hypothesis discharge).

If we temporarily assume an unproven conjecture or auxiliary result A (such as an unproven "ABC Theorem") within a system S to derive a conclusion B, we write $S \cup \{A\} \vdash B$. The Deduction Theorem guarantees that inside the base system S alone, we have legitimately established the conditional theorem:

$$S \vdash (A \to B)$$

This provides a rigorous foundation for conditional mathematics. Furthermore, the approach is modular: if A is eventually proven within S ($S \vdash A$), applying Modus Ponens to $S \vdash A$ and $S \vdash (A \to B)$ instantly yields $S \vdash B$ without needing to revise any part of the derivation.

Note. Note that we can use Deduction Theorem to show the converse of $A2$ as well:

Let $S = \{((p \Rightarrow q) \Rightarrow (p \Rightarrow r)\}$, we want $S \vdash (p \Rightarrow (q \Rightarrow r)) \iff S \cup \{p\} \vdash q \Rightarrow r \iff S \cup \{p\} \cup \{q\} \vdash r$. So consider

$$\{(p \Rightarrow q) \Rightarrow (p \Rightarrow r), p, q\}$$

$(P1)$	$(q \Rightarrow (p \Rightarrow q))$	$(A1)$
(1)	q	premiss
(2)	$p \Rightarrow q$	MP : $P1, 1$
(3)	$(p \Rightarrow q) \Rightarrow (p \Rightarrow r)$	premiss
(4)	$(p \Rightarrow r)$	MP : $3, 2$
(5)	p	premiss
(6)	r	MP : $5, 4$

However, it takes no contribution to the system of propositional logic because of 2 reason: Note that it comes from A1,A3 and also deduction theorem, so applying the $\Leftarrow$ is just applying A1,A3. Secondly, recall that the system follows minimalism, which this redundant axiom would contradict the idea of reducing everything to minimum.

Remark. Our next goal is to prove the Completeness Theorem: $S \vdash t$ if and only if $S \models t$. Which is made up of *soundness*(if $S \vdash t$ then $S \models t$) and *adequacy*(if $S \models t$ then $S \vdash t$).

Proposition 1.3 (**Soundness Theorem**). *Let $S \subset L$, and $t \in L$. Then $S \vdash t$ implies $S \models t$.*

Note. Note that we do not need to add the premiss "If S has model v". Since the soundness theorem stated: "Let $S \subset L, t \in L$. **If** $S \vdash t$, **then** (**If** v is a model of S,

then v is also a model of t)". We assume the part "Let $S \subset L, t \in L$". If S indeed has a model, then we proof that it is also a model of v. On the other hand, if such model v does not exists, the proposition "**If** v is a model of S, **then** v is also a model of t" would be vacuously true, hence the compound proposition "**If** $S \vdash t$, **then** (**If** v is a model of S, **then** v is also a model of t)" would also be true.

Remark. Soundness says that our notion of proof is sound: it doesn't lead to absurd conclusions. Meaning that the symbol-manipulation rules are truth-preserving, we cannot have a single possibility that we proves the conclusion syntactically but turns out that the conclusion is False.

Proof. Let $t_1, t_2, ..., t_n = t$ be a proof of t from S. Let v be a model of $S(v(s) = 1 \ \forall s \in S)$. We show that $v(t_i) = 1$ for all i by induction. Then in particular, $v(t) = 1$, as required.
If t_i is an axiom, then $v(t_i) = 1$ since by example before we have shown that they are all tautologies. If t_i is a premiss, then $v(t_i) = 1$ since v is a model of S. Finally, if there exist $j, k < i$ such that $t_k = (t_j \Rightarrow t_i)$, then $v(t_k) = v(t_j \Rightarrow t_i) = 1$ by induction hypothesis, and since $v(t_j) = 1$, we have $v(t_i) = 1$ using Modus Ponen. □

Note. **System Soundness vs. Argument Soundness**

1. **Syntactic Provability ($S \vdash t$):** Reaching t from premises S via mechanical rules (e.g., Modus Ponens) without evaluating truth.

2. **Semantic Validity ($S \models t$):** The argument structure is truth-preserving: every valuation v satisfying $v(s) = 1$ for all $s \in S$ must also satisfy $v(t) = 1$.

3. **Soundness Theorem ($S \vdash t \implies S \models t$):** Guarantees that formal proofs only yield valid structures. If $S \vdash t$, then $S \models t$ is guaranteed.

4. **Argument Soundness:** Combining formal validity with real-world factuality:

$$\underbrace{\text{Semantically Valid } (S \models t)}_{\text{Guaranteed by Soundness Theorem}} + \underbrace{\text{Premises True in Reality } (v_{\text{real}}(s) = 1)}_{\text{Factually verified}} \implies v_{\text{real}}(t) = 1$$

Example Let $S = \{s_1, s_2\}$ and t be defined as:

- s_1: All humans need water to live $(h \implies w)$
- s_2: You are human (h)
- t: You need water to live (w)

1. **Derivation ($S \vdash t$):** Applying Modus Ponens to s_1 and s_2 yields $S \vdash t$.

2. **Validity ($S \models t$):** By the Soundness Theorem, $S \vdash t$ guarantees $S \models t$.

3. **Real-World Truth (v_{real}):** Factually, $v_{\text{real}}(s_1) = 1$ and $v_{\text{real}}(s_2) = 1$. Because $S \models t$, the conclusion is forced to be true in reality: $v_{\text{real}}(t) = 1$.

Remark (Understanding Semantic Entailment $S \models t$). When evaluating whether $S \models t$, we are **not** checking whether t happens to be true in the real world, nor are we checking all valuations indiscriminately.

1. **The Conditional Filter:** By definition, $S \models t$ states:

$$\forall v, \quad \Big(\big(\forall s \in S,\, v(s) = 1\big) \implies v(t) = 1\Big)$$

   The universal quantifier $\forall v$ ranges over *all* abstract valuation functions $v : P \to \{0, 1\}$, but the implication acts as a filter. We only check if $v(t) = 1$ on valuations where every premise in S evaluates to 1 (the models of S).

2. **Real-World Falsity $\neq$ Counterexample:** If a premise $s \in S$ is false in reality ($v_{\text{real}}(s) = 0$), v_{real} fails the condition $\forall s \in S, v(s) = 1$. Therefore, v_{real} is simply not a model of S and cannot serve as a counterexample to $S \models t$.

   Example Let $S = \{s_1, s_2\}$ and t be defined as:

   - s_1: All humans can fly ($h \implies w$)
   - s_2: You are human (h)
   - t: You can fly (w)

   Note that above $v_{real}(s_1) = 0$, hence we conclude that v_{real} is simply not a model of S, so it does not act as a counterexample for challenging sematic entailment and Soundness Theorem.

3. **Non-Vacuous vs. Vacuous Entailment:**

   - $S \models t$ is **vacuously true** only if S has *zero* models (e.g., $S = \{p, \neg p\}$).
   - For $S = \{p \implies q, p\}$ and $t = q$, the valuation $v(p) = 1, v(q) = 1$ satisfies all premises. Because $v(q) = 1$ under that model, $S \models t$ holds **non-vacuously**.

Valuation	$v(p)$	$v(q)$	$v(p \implies q)$	All $s \in S$ True?	$v(t) = v(q)$
v_1	0	0	1	No (filtered out)	—
v_2	0	1	1	No (filtered out)	—
v_3	1	0	0	No (filtered out)	—
v_4	**1**	**1**	**1**	**YES (Model)**	**1**

Definition 1.8. Let $S \subset L$. Say S is inconsistent if $S \vdash \bot$; otherwise S is consistent.

Note. A special case of adequacy: $S \models \bot$ implies $S \vdash \bot$, i.e., if S has no model, then S is inconsistent. Equivalently, if S is consistent, then S has model.

Example 1.7. Suppose S is inconsistent ($S \vdash \bot$), then logic breaks ($S \vdash B$ for any arbitrary $B \in L$):

$$\begin{array}{lll} (P1) & \bot \Rightarrow ((B \Rightarrow \bot) \Rightarrow \bot) & (A1) \\ (P2) & \bot & (premiss) \\ (1) & (B \Rightarrow \bot) \Rightarrow \bot = \neg\neg B & (MP : P1, P2) \\ (2) & \neg\neg B \Rightarrow B & (A3) \\ (3) & B & (MP : 1, 2) \end{array}$$

Suppose $S \vdash \bot$, we conclude that $S \vdash B$ for any arbitrary $B \in L$. Note that the system if this happens would break down since the system might show, for any p, $S \vdash p, S \vdash \neg p$ at the same time, which is contradictions, also that it establish some useless information since $B \in L$ could be some blended string, lastly and most importantly, this formal system will loss the power to distinguish between true assertions and false assertions.

Theorem 1.4 (**Model existence lemma**). *Let $S \subset L$. If S is consistent, then S has model.*

Idea. Note that if $S \vdash t$, then $S \models t$ (soundness), so $v(t) = 1$ for every model v of S. So we could try

$$v(t) = \begin{cases} 1 & \text{if } S \vdash t \\ 0 & \text{otherwise} \end{cases}$$

However, this does not work. It is possible to have $t \in L$ with $S \nvdash t$ and $S \nvdash \neg t$. So we could try to enlarge S by adding either t or $\neg t$ for every $t \in L$, while keeping S consistent.

Note. From above Idea, we notice that there might be some t existed such that the system S draws no conclusion from it: $S \nvdash t \wedge S \nvdash \neg t$ (We say that the proposition t is independent to the system S). Below, i provide both mathematical and daily life examples for reader to have a clearer sight on independency:

- **Continuum Hypothesis**:

  Let S be Zermelo-Fraenkel set theory with the Axiom of Choice (ZFC). Let t be the Continuum Hypothesis, which states that there is no set whose size is strictly between the integers ($\aleph_0$) and the real numbers ($2^{\aleph_0}$).

  The Independence Results:$S \nvdash \neg t$: In 1938, Kurt Gödel proved that you cannot disprove the Continuum Hypothesis using ZFC. He did this by constructing a

specific model of set theory called the "Constructible Universe" (L) where all the axioms of ZFC are true, and CH is also true.

$S \not\vdash t$: In 1963, Paul Cohen proved that you cannot prove the Continuum Hypothesis using ZFC. He invented a revolutionary technique called forcing to construct a completely different model of set theory where ZFC is true, but CH is false (by inserting extra sets between the integers and the reals).

- **(Euclid's Core 4 Postulates)**:

  The System S Euclid started with 4 deceptively simple, highly intuitive rules for drawing on a flat piece of paper:

  - A straight line segment can be drawn joining any two points.
  - Any straight line segment can be extended indefinitely in a straight line.
  - Given any straight line segment, a circle can be drawn having the segment as radius and one endpoint as center.
  - All right angles are equal to one another.

  The Statement t (The 5th Postulate / Parallel Postulate): Euclid's 5th rule was vastly more complicated.

  A mathematically equivalent, simpler version (known as **Playfair's Axiom**) states: Given a straight line and a point not on that line, there is exactly one straight line that passes through the point and never crosses the given line (is parallel to it).

Universe (Model)	**Description of Space**	**Status of Statement t**	**Status of Core Rules S**
Euclidean Geometry	A perfectly flat sheet of paper.	**TRUE.** Exactly one parallel line exists.	Perfectly True.
Hyperbolic Geometry	Curved space like a saddle/Pringles chip.	**FALSE.** Infinitely many parallel lines exist.	Perfectly True.
Elliptic Geometry	Curved space like a sphere (e.g., Earth).	**FALSE.** Zero parallel lines exist.	Perfectly True (with minor tweaks).

- **The System S:** The official legal traffic laws of a city.
  - Rule 1: You must drive under 50 km/h in residential zones.
  - Rule 2: You must stop completely at red lights.
  - Rule 3: You must drive on the left side of the road.
- **The Statement t:** "You must take Route A (the highway) instead of Route B (the scenic local streets) to get to the airport."
- **Why it's independent ($S \not\vdash t$ and $S \not\vdash \neg t$):** Traffic laws (S) only tell you *how* to safely operate a vehicle; they do not dictate your destination or your path choice.

  - *Model A (Driver A):* Chooses the fast highway route (t is true).
  - *Model B (Driver B):* Chooses the scenic route avoiding tolls ($\neg t$ is true).

  Both drivers can complete their trips without getting a single traffic ticket because both of their physical paths (models) obey every traffic law in the book. The choice of route t is fully independent of the legal framework.

- Or, more simply, we recall an example about the system S: "Today it is raining," and we create a statement t: "I am hungry." There is no conclusion drawn from S to t; whether I am hungry or not is entirely unaffected by the weather ($S \nvdash t$ and $S \nvdash \neg t$).

Proof. We present the proof in case the set P of primitive propositions is countable. The general case will be presented in Chapter 3. Note that if P is countable, then so is $L_1 = P\{\bot\}$. It follows inductively that L_n is also countable for every $n \in \mathbb{N}$, and hence $L = \bigcup_{n\in\mathbb{N}} L_n$ is also countable. We enumerate L as $t_1, t_2, ...$

Next observe that if $T \subset L$ is consistent and $t \in L$, then either $T \cup \{t\}$ or $T \cup \{\neg t\}$ is consistent: Suppose not, then T is consistent, and $T \cup \{t\}, T \cup \{\neg t\}$ inconsistent, then

$$\begin{cases} T \cup \{t\} \vdash \bot \\ T \cup \{\neg t\} \vdash \bot \end{cases} \overset{\text{Deduction Theorem}}{\Longrightarrow} \begin{cases} T \vdash (t \Rightarrow \bot) \\ T \vdash (\neg t \Rightarrow \bot) \end{cases} \iff \begin{cases} T \vdash (t \Rightarrow \bot) \\ T \vdash t \end{cases}$$

Since using

(1)	$t \Rightarrow \bot$	by $T \vdash (t \Rightarrow \bot)$
(2)	t	by $T \vdash t$
(3)	$\bot$	MP:1,2

and since

(1)	$\bot$	from $S \vdash \bot$
(2)	$\bot \Rightarrow (t \Rightarrow \bot)$	A1
(3)	$t \Rightarrow \bot$	MP:1,2

and also

(1)	$\bot$	from $S \vdash \bot$
(2)	$\bot \Rightarrow (\neg t \Rightarrow \bot)$	A1
(3)	$\neg t \Rightarrow \bot$	MP:1,2
(4)	$\neg\neg t$	by definition 1.4
(5)	$\neg\neg t \Rightarrow t$	$A3$
(6)	t	MP: 4,5

We arrive with $T \vdash \bot$, which is contradiction.

We now start with a consistent $S \subset L$, set $S_0 = S$ and define $S_n \subset L$ for each $n \in \mathbb{N}$ inductively as follows: Having defined S_{n-1} and assuming S_{n-1} is consistent, we let S_n be either $S_{n-1} \cup \{t_n\}$ or $S_{n-1} \cup \{\neg t_n\}$ whichever is consistent.

Finally, we let $\bar{S} = \bigcup_{n\geq 0} S_n$.

By construction, for each $t \in L$, either $t \in \bar{S}$ or $\neg t \in \bar{S}$, we cannot have both inside $\bar{S}$. We note the following two properties of $\bar{S}$.

$\bar{S}$ is consistent: if $\bar{S} \vdash \bot$, then since proofs are finite, we have $S_n \vdash \bot$ for some n, which is a contradiction since each S_n must be consistent.

$\bar{S}$ is deductively closed: if $\bar{S} \vdash t$, then $t \in \bar{S}$. Suppose that $t \notin \bar{S}$, then $\neg t \in \bar{S}$. It follows that $\bar{S} \vdash t$ and $\bar{S} \vdash \neg t$, contradicting that $\bar{S}$ is consistent, hence we must have $t \in \bar{S}$.

We now define $v : L \to \{0, 1\}$ by

$$v(t) = \begin{cases} 1 & \text{if } \bar{S} \vdash t\text{(or equivalently, if } t \in \bar{S}\text{)}, \\ 0 & \text{otherwise.} \end{cases}$$

We show that v is a valuation on L. It will then follow that v is a model of S, completing the proof.

First, $v(\bot) = 0$ since $\bot \notin \bar{S}$ as $\bar{S}$ is consistent. Next, we examine $v(p \Rightarrow q)$ for arbitrary $p, q \in L$.

Case 1: $v(p) = 1$ and $v(q) = 0$, i.e., $p \in \bar{S}$ and $q \notin \bar{S}$. We need to show that $v(p \Rightarrow q) = 0$. If not, then $(p \Rightarrow q) \in \bar{S}$. Then the sequence

$$\begin{aligned} p \quad & \text{(premiss)} \\ p \Rightarrow q \quad & \text{(premiss)} \\ q \quad & \text{(MP)} \end{aligned}$$

is a proof of q from $\bar{S}$. Since $\bar{S}$ is deductively closed, it follows that $q \in \bar{S}$, which is a contradiction.

Case 2: $v(q) = 1$, i.e., $q \in \bar{S}$. We need to show that $v(p \Rightarrow q) = 1$, i.e., that $(p \Rightarrow q) \in \bar{S}$. The sequence

$$\begin{aligned} q \quad & \text{(premiss)} \\ q \Rightarrow (p \Rightarrow q) \quad & (A1) \\ p \Rightarrow q \quad & \text{(MP)} \end{aligned}$$

is a proof of $(p \Rightarrow q)$ from $\bar{S}$. Since $\bar{S}$ is again deductively closed, it follows that $(p \Rightarrow q) \in \bar{S}$ as required.

Case 3: $v(p) = 0$, i.e., $p \notin \bar{S}$. We need to show that $v(p \Rightarrow q) = 1$, i.e., that $(p \Rightarrow q) \in \bar{S}$. As in previous cases, since $\bar{S}$ is deductively closed, it is enough to show that $\bar{S} \vdash (p \Rightarrow q)$, which is equivalent to showing that $\bar{S} \cup \{p\} \vdash q$ by the Deduction Theorem. Note that $\neg p \in \bar{S}$ since $p \notin \bar{S}$. The following then is a proof of q from

$\bar{S} \cup \{p\}$.

$$\begin{aligned} p & \quad \text{(premiss)} \\ \neg p (= p \Rightarrow \bot) & \quad \text{(premiss)} \\ \bot & \quad \text{(MP)} \\ \bot \Rightarrow (\neg q \Rightarrow \bot) & \quad (A1) \\ \neg\neg q (= \neg q \Rightarrow \bot) & \quad \text{(MP)} \\ \neg\neg q \Rightarrow q & \quad (A3) \\ q & \quad \text{(MP)} \end{aligned}$$

Since the valuation lands 1 on all 3 cases, we conclude that v is a model of $\bar{S}$ and since $S \subset \bar{S}$, it is also a model of S. □

Remark. Suppose S has a model, then S is consistent. By contrapositive, we show the statement:'Let $S \subset L$. If S is inconsistent, then S does not have a model.' Suppose not, S has a model v, then $v(s) = 1$ for all $s \in S$. Since S is inconsistent, then $\exists t \in L$ such that $S \vdash \bot$. By soundness theorem, $S \vdash \bot \implies S \models \bot$, in particular we have $v(\bot) = 1$, contradiction.

Remark. Some readers might notice that we take this result for granted: if L_n is countable for each $n \in \mathbb{N}$, then $L = \bigcup_{n\in\mathbb{N}} L_n$ will also be countable:

Some readers might notice that we take this result for granted: if L_n is countable for each $n \in \mathbb{N}$, then $L = \bigcup_{n\in\mathbb{N}} L_n$ will also be countable.

Proof. To prove that a countable union of countable sets is countable, we construct a surjective mapping from $\mathbb{N} \times \mathbb{N}$ to the union L. Since $\mathbb{N} \times \mathbb{N}$ is a known countable set, its image L must also be countable.

Step 1: Enumerate the Individual Sets

By definition, because each L_n is a countable set, its elements can be completely listed in a sequence. For every $n \in \mathbb{N}$, there exists a surjective mapping $f_n : \mathbb{N} \to L_n$. This allows us to write the elements of each L_n explicitly as:

$$L_n = \{x_{n,1}, x_{n,2}, x_{n,3}, \dots\}$$

where $x_{n,k} = f_n(k)$ represents the k-th element of the n-th set (Here we need axiom of choice to have such f: Define $S_n = \{f | f : \mathbb{N} \to L_n \text{ is surjective}\}$). By AOC, $\Phi : \mathbb{N} \to \bigcup_{n\in\mathbb{N}} S_n$, where $\forall n \in \mathbb{N}, \Phi(n) \in S_n$, such $\Phi()n := f_n$, by definition of S_n, the function $f_n : \mathbb{N} \to S_n$ is a surjective function from $\mathbb{N}$ to S_n.)

Step 2: Arrange the Elements into a 2D Array

We can visually organize all elements of the union $L = \bigcup_{n\in\mathbb{N}} L_n$ into an infinite

grid, where row n lists the elements corresponding to the set L_n:

$$\begin{array}{lccccc} L_1: & x_{1,1} & x_{1,2} & x_{1,3} & x_{1,4} & \dots \\ L_2: & x_{2,1} & x_{2,2} & x_{2,3} & x_{2,4} & \dots \\ L_3: & x_{3,1} & x_{3,2} & x_{3,3} & x_{3,4} & \dots \\ L_4: & x_{4,1} & x_{4,2} & x_{4,3} & x_{4,4} & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{array}$$

Step 3: Enumerate Diagonally (Cantor's Zig-Zag Method)

To show L is countable, we must arrange all items in this grid into a single, infinite linear list. Going row-by-row or column-by-column fails because we would get trapped in the first infinite line and never reach the second.

Instead, we count systematically along the diagonals where the sum of indices $(n + k)$ is constant:

- **Sum = 2:** $x_{1,1}$
- **Sum = 3:** $x_{1,2}, x_{2,1}$
- **Sum = 4:** $x_{3,1}, x_{2,2}, x_{1,3}$
- **Sum = 5:** $x_{1,4}, x_{2,3}, x_{3,2}, x_{4,1}$

This explicit path forms a single sequence that maps every coordinate pair $(n, k) \in \mathbb{N}\times\mathbb{N}$ directly to a natural number.

Step 4: Handle Duplicates and Conclude

We define our final listing function $g : \mathbb{N} \to L$ by tracing along this zig-zag path. If we encounter an element that has already appeared (since the sets L_n might share overlapping elements), we simply skip it and proceed to the next item on the path.

Because every element $x \in L$ belongs to at least one set L_n at some position k, it occupies the grid location (n, k) and is guaranteed to be reached by our diagonal walk. Therefore, we have successfully constructed an enumeration of L, proving that L is countable. □

Example 1.8. Let our propositional language contain the atoms M ("the steak is real beef meat") and V ("the steak is vegan"). We are given a background rule from our vocabulary definitions: $M \implies \neg V$, which states that if the steak is real beef meat, then it is not vegan.

Our original rule system S :

- s_1 (season with a suitable amount of salt)
- s_2 (add a little bit of black pepper)

- s_3 (sear both sides until they are golden brown)
- s_{meat} (the steak is real beef meat, written simply as M).

Since $S = \{s_1, s_2, s_3, M\}$ contains no logical contradictions ($S \nvdash \bot$), the **Model Existence Lemma** guarantees that a physical model—a real, juicy beef steak cooking in a pan—can exist in the universe.

Now, we expand our system into S' by adding your rule s_4, which dictates that the steak must be vegan, written as V. We can formally derive a syntactic contradiction ($S' \vdash \bot$) using standard natural deduction steps:

1. M (From premise s_{meat})
2. $M \implies \neg V$ (From background knowledge rule)
3. $\neg V$ (By Modus Ponens on lines 1 and 2)
4. V (From premise s_4)
5. $\bot$ (By $\neg$-Elimination on lines 3 and 4)

Because $S' \vdash \bot$, the expanded system is syntactically inconsistent.

To show why no model can exist under these rules, let us assume a valid semantic truth assignment (model) v satisfies S'. For v to satisfy S', it must evaluate every proposition in S' to true (1) simultaneously. Because v satisfies s_{meat}, we must have $v(M) = 1$, which forces $v(V) = 0$ via our background rule implication. However, because v must also satisfy s_4, it requires $v(V) = 1$. A propositional valuation mapping cannot assign a single atom V to be both true (1) and false (0) at the exact same time. Therefore, no such model exists. By the contrapositive of the Model Existence Lemma, we are mathematically barred from ever creating such a steak.

Corollary 1.5 (**Adequacy Theorem**). *Let $S \subset L$ and $t \in L$. If $S \models t$, then $S \vdash t$.*

Proof. If $S \models t$, suppose for contradiction that $S \cup \{\neg t\}$ has a model v. Since v is a model of $S \cup \{\neg t\}$, v must satisfy both: $v(s) = 1,\ \forall s \in S, v(\neg t) = 1$. By valuation, we know that $v(\neg t) = 1 \Rightarrow v(t) = 0$. But since $S \vdash t$, v is a model of S, the definition of entailment requires $v(t) = 1$, contradiction, so no valuation can be a model of $S \cup \{\neg t\}$. Therefore $S \cup \{\neg t\} \models \bot$, by contrapositive of Model existence lemma, since $S \cup \{\neg t\}$ has no model, $S \cup \{\neg t\}$ is inconsistent. We have $S \cup \{\neg t\} \vdash \bot$, which in turn implies $S \vdash \neg\neg t$ by Deduction Theorem. We now obtain a proof of t from S by adding the following lines to a proof of $\neg\neg t$ from S.

$$\begin{aligned} \neg\neg t \Rightarrow t \quad & (A3) \\ \neg\neg t \quad & \text{from } S \vdash \neg\neg t \\ t \quad & \text{MP} \end{aligned}$$

□

Theorem 1.6 (**Completeness Theorem**). *Let $S \subset L$ and $t \in L$. Then $S \models t$ if and only if $S \vdash t$.*

Proof. Soundness + Adequacy Theorem. □

Corollary 1.7 (**Compactness Theorem**). *Let $S \subset L$ and $t \in L$. If $S \models t$, then there is a finite subset S' of S such that $S' \models t$.*

Proof. Let $S \subset L$, suppose that $S \models t$, then by Adequacy Theorem, we have $S \vdash t$. Now note that by definition, the formal proof from S to t is constructed with only finite sequence $t_1, t_2, ..., t_n$ of propositions such that $t_n = t$. Now we define $S' = S \cap \{t_1, ..., t_n\}$. Note $t = t_n \in S$, hence $S' \vdash t$. Furthermore, because $S' \subseteq \{t_1, \ldots, t_n\}$, and since the sequence of proof $\{t_1, ..., t_n\}$ must be finite, S' must also be finite. □

Note. This is highly nontrivial if Completeness Theorem is not introduced. Below is the special case of Corollary 1.2.

Corollary 1.8. *Let $S \subset L$. If every finite subset of S has a model, then S has a model.*

Proof. If S does not have a model, then $S \models \bot$. By Corollary above, there is a finite subset S' of S such that $S' \models \bot$. □

Remark. Some people might call this corollary the Compactness Theorem. It implies the Corollary 1.2. Indeed, assume that $S \models t$. Then $S \cup \{\neg t\} \models \bot$. By Corollary 1.3, there is a finite subset S' of S such that $S' \cup \{\neg t\} \models \bot$, and thus $S' \models t$.

Note. The Compactness Theorem is seldom used in propositional logic. More frequently, readers might encounter it in model theory. It allows us not to see the whole infinite set of statements, but rather all finite subset of it, which is more helpful than considering the infinite set it's own.

Remark. The application of the Compactness Theorem is highly intuitive: suppose we have a large, potentially infinite set of premises S that logically entails a conclusion t. Because any formal syntactic proof is a finite sequence of steps, it can only ever employ a finite number of premises. The Compactness Theorem semantically mirrors this fact by guaranteeing that if $S \models t$, there exists a finite subset $S_0 \subseteq S$ such that $S_0 \models t$.

This concept of finite support inspires a broader philosophical pursuit in foundational logic: the drive toward minimality. For instance, just as compactness shows we only need a finite portion of a theory to secure a consequence, functional completeness shows we can strip away superfluous logical symbols to retain a minimal set like $\{\implies, \bot, (,)\}$ without losing expressive power.

Conversely, we often deliberately retain redundant axioms or rules within a finite framework purely for practical convenience. Consider the Reduced Row Echelon Form

(RREF) in linear algebra: although the third elementary row operation ($R_i \leftarrow R_i + cR_j$) can technically be derived from a combination of row swapping ($R_1 \leftrightarrow R_2$) and scalar multiplication (cR_1), it is explicitly maintained to simplify computations. We observe an identical pragmatic design choice in algebraic topology and category theory. Axioms such as $A2$:

$$(p \implies (q \implies r)) \implies ((p \implies q) \implies (p \implies r)) \quad \text{for } p, q, r \in L$$

are frequently preserved as core tenets of a deductive system, because immediate mathematical utility often outweighs the aesthetic ideal of strict logical independence.

Corollary 1.9 (**Decidability Theorem**)**.** *Let $S \subset L$ be a finite set and $t \in L$. Then there is an algorithm that determines in finite time whether $S \vdash t$ or not.*

Proof. Since we have $S \vdash t \iff S \models t$, let n be the total number of primitive propositions occurring in $S \cup \{t\}$ (If t contains a primitive variable outside S, checking only variables in S makes the truth table incomplete). Then writing out all the possible combinations: 2^n, there would be 2^n rows in the truth table. We recursively write out the value of $v(s) \forall s \in S, v(t)$. If there is $v(s) = 1 \forall s \in S, v(t) = 0$, terminate the algorithm and write out $S \not\models t$. Else if all 2^n rows satisfy the condition (whenever $s \in S$, we have $v(s) = 1, v(t) = 1$), then we have $S \models t$ and hence $S \vdash t$. □

Remark. If $S \vdash t$, then a proof can be found in finite time: by writing out all proofs from S, we will eventually arrive at t. However, this algorithm does not terminate if $S \nvdash t$ (no loop?).

Example 1.9 (Real-World Applications)**.** In practical software engineering, decidability marks the boundary between what computers can automate safely and what requires human intervention.

- **Cybersecurity & Malware Detection:** By Rice's Theorem: Any non-trivial semantic property of programs (or Turing machine behaviors) is undecidable.

  First, we define some terms:

  - Turing Machine (M): the ideal program. $\langle M \rangle$ represents encoded string of the machine.

    For example, M_{odd} be a Turning machine designed to accept binary strings to have an odd number of 1s. Then $\langle M_{odd} \rangle$ is the string of 0s, 1s that serializes the states and transition table of M_{odd} [1],[2],[3].

  - RE Language $L(M)$: a set of strings, which is its language. We can treat it as 'Behavior/ Output'. If M_1, M_2 is diff. Turing machine, $L(M_1) = L(M_2)$

means they have same semantic behavior even if their internal code looks completely different.

Let M_1 be a machine that loops through a string from left to right to count if the number of 1 is odd. M_2 a machine that loops through a string from right to left to count if the number of 1 is odd.

Even M_1, M_2 are completely different machine, their accepted set of strings is identical: $L(M_1) = L(M_2) = \{x \in \{0,1\| x \text{ has an odd number of 1s}\}$.[4][5]

– Semantic Property (P): Property of the Language, subset of all RE language. A property is semantic if it treats identical behaviors identically. If $L(M_1) = L(M_2)$, then either both machines possess the property P, or neither does.

Let P be the property: "The language accepted by the machine is empty $(L(M) = \emptyset)$."

If both M_1, M_2 are two different machines that both fail to accept any strings, then $L(M_1) = \emptyset, L(M_2) = \emptyset$. Since their behavioral language is identical, both satisfy property P [4][5].

For Counter-Example (Not semantic) : The property "The machine has exactly 5 states." If M_1 has 5 states and M_2 has 8 states, but they accept the exact same language, this property treats them differently. Thus, it is a syntactic property of the machine, not a semantic property of the language[4][6].

– Non-trivial: There must exist at least 1 M that has the property P, and at least one that does not.

Let's reuse the emptiness property: $P = \{\langle M \rangle \mid L(M) = \emptyset\}$.It is non-trivial because:There exists a machine M_{reject} that immediately rejects all inputs. For this machine, $L(M_{\text{reject}}) = \emptyset$, so it has the property.There exists a machine M_{accept} that immediately accepts all inputs. For this machine, $L(M_{\text{accept}}) = \Sigma^* \neq \emptyset$, so it does not have the property[5].

– Undecidable: A language L_P is undecidable if there is no decider (a M that is guaranteed to halt on all inputs and correctly output 'Accept' or 'Reject) that can determine whether an arbitrary encoded machine $\langle M \rangle$ belongs to L_P [5][4].

The language $L_P = \{\langle M \rangle \mid$ M accepts the string '101'$\}$.If you try to write a master algorithm D (a decider) to solve this, D takes an arbitrary machine string $\langle M \rangle$ as input. D must tell you if M accepts '101'.If M is a machine that gets stuck in an infinite loop when given '101', your decider D will also get stuck in a loop or fail to determine if M would have eventually accepted it. Rice's Theorem proves that a flawless, always-halting decider D cannot exist for L_P.

First, note that the Rice Theorem, in formal definition:

Let $\mathcal{R}$ be the class of all RE language over an alphabet $\sum$. Let $P \subseteq \mathcal{R}$ be a property of RE languages. We say P is non-trivial if $P \neq \emptyset$ and $P \neq \mathcal{R}$. Then, the language associated with P, defined as

$$L_P = \{\langle M \rangle | L(M) \in P\}$$

is undecidable.

We now can formulate Malware Detection as a Language Property P. If P be the property of being "Malware". We define a program as malicious based on its behavior(e.g. Does this program delete the system registry when it runs?). Hence, our language of malicious programs is

$$L_{malware} = \{\langle M \rangle | L(M) \text{ exhibits malicious behavior}\}$$

First, it is clearly a semantic property, if M_1, M_2 different program, have entirely different source code, but ultimately execute the exact same payload(Both delete the same system file under exact same conditions), their language behavior is identical ($L(M_1) = L(M_2)$). So either both are malware or neither.

Next, we show it is non-trivial (if it applies to some programs, but not all): M_{yes} exists: We can write a program that immediately formats the hard drive. This program is malware ($L(M_{yes}) \in P$). And M_{no} exists: We can write a program that simply prints 'Hi', and terminates safely. This program is not malware ($L(M_{no}) \notin P$).

By Rice's Thm., the language $L_{malware}$ is undecidable. Also, f an antivirus scanner only used the rules of your propositional logic theorem, it would look like this:

- Let S be a finite database of known malware signatures (e.g., text blocks of malicious code).
- Let t be a new file someone downloaded.
- The algorithm checks if t syntactically matches anything in S.

This is decidable and runs in finite time. But, A hacker changes the syntax of the program without changing its behavior (semantics). They can encrypt the code, scramble the variable names, or insert junk mathematical formulas.To catch this, the antivirus can no longer look at the syntax ($S \vdash t$). It must answer a semantic question: "Does this program eventually execute a malicious payload?" Because answering that question requires evaluating a non-trivial semantic property of a Turing-complete language, Rice's theorem proves it is undecidable.

How this example connects to Decidability Theorem?

The core relationship between Decidability Theorem for Propositional Logic and Rice's Theorem is a fundamental tension in computer science and logic: **Syntax (Finite, Bounded Structures) vs. Semantics (Infinite, Turing-Complete Behaviors)**.

The theorem states that if $S \subset L$ is a finite set and $t \in L$, then there is an algorithm that determines in finite time whether $S \vdash t$ or not. This problem is strictly **decidable** because Propositional Logic deals with bounded, finite search spaces. If S contains a finite number of propositional variables n (such as p, q, r), the entire truth-value universe consists of exactly 2^n combinations. Any automated proof system, such as a truth table or a semantic tree, only evaluates a **finite execution tree**. Because there are no loops, variables, or infinite paths, an algorithm can check every single branch and guarantee an absolute 'Yes' or 'No' answer in a finite number of steps.

Rice's Theorem breaks this predictability by upgrading the system from static logic formulas to **Turing-complete computer programs**. The moment a system is powerful enough to simulate loops and variables, the search space becomes **infinite**. A program does not possess a fixed, bounded truth table. It runs on an infinite number of possible inputs, and for any single input, its execution trace can run for an infinite number of steps without halting. Answering a semantic question about a program requires evaluating its runtime behavior. Because you cannot simulate an infinite execution path in a finite amount of time, the problem collapses from decidable to **undecidable**. Mathematically, the bridge connecting these concepts can be viewed as a progressive hierarchy of computational power where adding loops and variables transforms a Propositional Decidability Theorem into First-Order Logic, which via Rice's Theorem results in Undecidable Semantic Properties.

This computational boundary directly dictates why perfect malware detection is impossible. An antivirus scanner is an algorithm that takes an unknown downloaded program string $\langle M \rangle$ as input and attempts to solve a logical entailment problem. If the scanner evaluates malware purely using the mechanics of your Decidability Theorem, it restricts itself to static, finite **syntax** by checking if $S_{\text{signatures}} \vdash \langle M \rangle$. The scanner simply verifies if the text string of $\langle M \rangle$ contains a finite block of code known to be a virus. This executes in a guaranteed finite time and is fully **decidable**. However, it fails in practice because a hacker can change a single character or encrypt the file to break the syntactic match while keeping the virus functional.

To catch modern threats, the scanner must instead evaluate the **semantics**

(behavior) of the code by answering the question: "Does this program eventually execute a malicious action?" Consider a virus disguised behind an infinite logical branch where an execution of a malicious payload is nested inside an unproven mathematical statement, such as checking if the Collatz Conjecture is false. For the antivirus algorithm to accurately output a 'Yes' or 'No' in finite time without running the risk of an infinite freeze, it is forced to mathematically solve that infinite conjecture first. Rice's Theorem generalizes this mathematical trap. Because a malicious payload can be hidden inside *any* non-halting logical loop or infinite branch, no master scanner algorithm can ever exist that safely terminates in finite time for all arbitrary programs. Thus, true semantic malware detection is strictly **undecidable**.

Chapter 2

Well-Ordering and Ordinals

2.1 Introduction

Definition 2.1. A **linear order** or **total order** on a set X is a relation $<$ on X that is

1. irreflexive: $\neg(x < x)$ for all $x \in X$.
2. transitive: $((x < y) \wedge (y < z)) \Rightarrow (x < z)$ for all $x, y, z \in X$.
3. trichotomous: $(x < y) \vee (x = y) \vee (y < x)$ for all $x, y \in X$.

We say that 'X is linearly ordered by $<$' or simply 'X is a linear ordered set' .

Note. In (3), exactly one of the three possibilities hold. Suppose not, for example if $x < y$ and $y < x$, then $x < x$ by (2), which would contradict (1).

Note also that $\emptyset$ is also a linear ordered set: it is vacuously true that it is transitive. Using $\neg(x < x) = (x < x \Rightarrow \perp)$, the statement is also vacuously true, same for trichotomous.

Example 2.1. $\mathbb{N}, \mathbb{R}, \mathbb{Q}, \mathbb{Z}$ in the usual order. Note that $\mathbb{N} = \{1, 2, 3, ...\}$

Note. For a set X of size at least 2, the relation on the power set $\mathcal{P}(X)$ of X(the set of all subsets of X) defined by $a < b$ if $a \subset b$ and $a \neq b$ is not trichotomous.

Since by definition $a \subset b$ means: $(x \in a) \Rightarrow (x \in b)$ for all $x \in X$, which includes the case $a = b$, hence we have $a < b$ includes having $a = b$. For example if we set $S = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$, we have no conclusion for $\{1\}$ and $\{2\}$ (Please visit chapter 3 poset for more information).

Notation:
we write '$x > y$' for '$y < x$'
we write '$x \leq y$' for '$x < y$ or $x = y$'

Definition 2.2. The relation '$\leq$' is a relation on X that is

1. reflexive: $x \leq x$ for all $x \in X$

2. antisymmetric: $((x \leq y) \wedge (y \leq x)) \Rightarrow (x = y)$ for all $x, y \in X$

3. transitive: $((x \leq y) \wedge (y \leq z)) \Rightarrow (x \leq z)$ for all $x, y, z \in X$

4. trichotomous: $(x \leq y) \vee (y \leq x)$ for all $x, y \in X$

Note. If X is linearly ordered by $<$, then any subset Y of X is linearly ordered by $<$ (or more precisely, by the restriction of $<$ of Y).

Since we have $Y \subset X$, hence any element of Y would satisfy the 4 rules of relation.

Definition 2.3. A well-ordering of a set X is a linear order $<$ on X such that every non-empty subset of X has a least element:

$$(\forall S \subset X)(S \neq \emptyset \Rightarrow (\exists x \in S)(\forall y \in S)(x \leq y))$$

Note that the least element is unique by antisymmetry.

We will say 'X is well-ordered by $<$ or simply 'X is a well-ordered set'.

Example 2.2. $\mathbb{N}$ is well-ordered by the usual order.
$\mathbb{Z}, \mathbb{R}, \mathbb{Q}$ are not well-ordered since $(-\infty, 0] \subset \mathbb{Q}, \mathbb{R}, \mathbb{Z}$ does not possess a least element. The subset $[0, \infty)$ of $\mathbb{R}$ does have a least element 0, but it is not well-ordered since the non-empty subset $(0, \infty)$ has no least element (if i choose $b \in (0, \infty)$ to be the least element, note $b > 0$, take $a < \frac{b}{2}$, $0 < b - a < b$, contradiction).

For the case of $\mathbb{Z}$, any bounded below subset of it is well-ordered under the natural ordering ($\mathbb{Z}$), since any subset of such subset possess a least element.

Note. A subset of a well-ordered set is well-ordered. We will see that $\mathbb{Q}$ has a rich collection of well-ordered subsets.

Definition 2.4. Two linearly ordered sets X and Y are **order-isomorphic** if there is a bijection $f : X \to Y$ that is order-preserving: $x < y$ implies $f(x) < f(y)$.

We say f is an order-isomorphism. Note that f^{-1} is also an order-isomorphism, and thus $x < y \iff f(x) < f(y)$.

Suppose we have $f : X \to Y$ is an order-isomorphism, for $x, y \in X$, $x <_X y \Rightarrow f(x) <_Y f(y)$. For injectivity, it guaranteed $f^{-1}f(A) \supseteq f(A), A \subset X$, for surjectivity, it guaranteed $f(f^{-1}(B)) \subseteq B$. To show f^{-1} is an order-isomorphism from Y to X, first note that f^{-1} is already bijective. Then, if we have $a_1, a_2 \in Y, a_1 <_Y a_2$, note that $x_1 = f^{-1}(a_1), x_2 = f^{-1}(a_2)$ for some $a_1, a_2 \in X$. Suppose $x_1 \geq x_2$, then by order isomorphism of f, $f(f^{-1}(a_1)) \geq_Y f(f^{-1}(a_2)), i.e., a_1 \geq a_2$, contradiction. Hence we must have $x_1 <_X x_2$. So f^{-1} is also an order-isomorphism from the linearly ordered set Y to X.

Note. If the linearly ordered sets X and Y are order-isomorphic, and X is well-ordered, then so is Y.

Proof. Suppose that X is linearly ordered set with $(X, <_X)$ and Y with $(Y, <_Y)$. Also X, Y are order-isomorphic, so there exists a bijection map $f : X \to Y$ such that if $x <_X y$ we have $f(x) <_Y f(y)$ for $x, y \in X$. To show that the linear order on Y where all non-empty subset of Y has a least element, consider an arbitrary non-empty subset $B \subset Y$. Consider the pre-image $A = f^{-1}(B) := \{f^{-1}(y) \mid y \in B\}$. Notice that B is non-empty, we have A be non-empty subset of X. Since X is well-ordered, A has a least element, called $a_0 \in A$. Then consider $y_0 = f(a_0)$, since $a_0 \in A = f^{-1}(B)$, we have $y_0 \in B$. Take arbitrary element $y \in B$ such that $f(a) = y$ for some $a \in A$. Since X, Y are order-isomorphic, by definition we have $a_0 <_X a \implies f(a_0) <_Y f(a)$, meaning $y_0 <_Y y$. If $a_0 = a$, then $f(a_0) = f(a)$, which means that $y_0 = y$. Note that B is any arbitrary non-empty subset of Y. So the results follows.

□

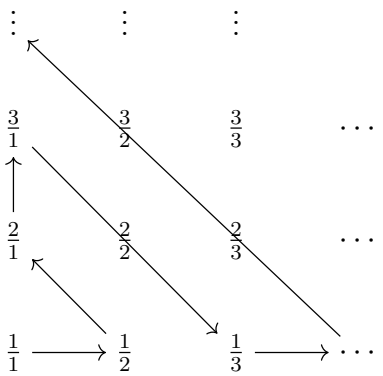
Example 2.3. 1. $\mathbb{N}$ is not order-isomorphic to $\mathbb{Q}$.

Proof. Suppose not, we assume that $\mathbb{N}$ is order-isomorphic to $\mathbb{Q}$. Consider $f(1), f(2) \in \mathbb{Q}$, and by injectivity, $f(1) \neq f(2)$. Since $\mathbb{N}, \mathbb{Q}$ are both well-ordered, we have $1 < 2 \implies f(1) < f(2)$. By density of $\mathbb{Q}$, $\exists z \in \mathbb{Q}$ such that $f(1) < z < f(2)$. By surjectivity, $\exists u \in \mathbb{N}$ such that $f(u) = z$. By order-isomorphism again, $1 < z < 2$, contradiction.

□

2. $\mathbb{Q}$ and $\mathbb{Q}\backslash\{0\}$ are order-isomorphic.

Proof. by Back-and-For method. However, we show show that $\mathbb{Q}$ are countable, dense, unbdd, linear ordered. It is trivial that it is unbounded and also linear ordered by above. To show that it is countable, consider the cantor diagram of $\mathbb{Q}$:



Where we skip the duplicated term (e.g.$\frac{1}{1}$ and $\frac{2}{2}$), this map goes through all the term in $\mathbb{Q}^+$ to $\mathbb{N}$. Hence the map from $\mathbb{N} \overset{\text{bijective}}{\to} \mathbb{Q}^+$. Hence $\mathbb{Q}^+$ is countable, we do the same for $\mathbb{N} \to \mathbb{Q}^-$. Since $\mathbb{Q} = \mathbb{Q}^+ \cup \{0\} \cup \{\mathbb{Q}^-\}$, $\mathbb{Q}$ is countable.

To show that it is dense in $\mathbb{R}$, consider $x, y \in \mathbb{R}$, we need to show that $\exists \frac{a}{b} \in \mathbb{Q}$ s.t. $x < \frac{a}{b} < y$. Note that $x < y$, then $y - x > 0$. By Archimedean Property, $\exists n \in \mathbb{Z}$ such that $n(y-x) > 1$ (since $y, x, 1 \in \mathbb{R}^+$). Hence $ny - nx > 1$, the interval (nx, ny) is larger than 1 in length, it must contain at least 1 integer, say smallest is p. So $ny > p > nx \Rightarrow y > \frac{p}{n} > x$. □

3. $A = \{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \cdots\}$ is order-isomorphic to $\mathbb{N}$ by $(m \mapsto \frac{m}{m+1})$

4. $B := A \cup \{1\}$ is well-ordered, but not order-isomorphic to $\mathbb{N}$. If we attempt to construct a bijective map by $(\frac{1}{n} \mapsto n)$, then $\mathbb{N}$ is not order preserving. However if we need to preserve order by using above mapping, the map is not surjective. Or, suppose there is an order-isomorphism $f : \mathbb{N} \to A \cup \{1\}$, meaning $\forall m, n \in \mathbb{N}$ if $n < m, f(n) < f(m)$. Since f is bijection onto $A \cup \{1\}$, $\exists k \in \mathbb{N}$ s.t. $f(k) = 1$. Consider $k < k+1 \Rightarrow f(k) < f(k+1)$, but $f(k) = 1 < f(1+k)$, which is contradiction since any element $x \in A \cup \{1\}$, $x \leq 1$.

5. $C := A \cup \{2\}$ is order-isomorphic to B by $f : \begin{cases} n \mapsto n & \text{if } n < 1 \\ 2 \mapsto 1 & \text{otherwise} \end{cases}$

6. $E := \{0\} \cup A$, then E is order-isomorphic to A.

   Proof. Note that E is order-isomorphic to $\mathbb{N} \cup \{0\}$ by $\begin{cases} 0 \mapsto 0 \\ \frac{n}{n+1} \mapsto n & \text{for } n > 0 \end{cases}$. Then E is order-isomorphic to $\mathbb{N} \cup \{0\}$. By above $\mathbb{N} \cup \{0\}$ is order-isomorphic to $\mathbb{N}$, and $\mathbb{N}$ is order-isomorphic to A. Hence E is order-isomorphic to A. □

7. $D := A \cup (A + 1)$ is well-ordered, but is not order-isomorphic to A or B

   Proof. Suppose D is order-isomorphic to A: Let $f : D \to A$ b such map. Pick $\frac{3}{2} \in D$, for any $x \in A, x < \frac{3}{2}$, we have $f(x) < f(\frac{3}{2})$. Since $f(\frac{3}{2}) \in A$, let $f(\frac{3}{2}) = \frac{n}{n+1}$ for some $n \in \mathbb{N}$. Notice that there are only $n-1$ elements smaller than $\frac{n}{n+1} : 1, 2, 3, ..., \frac{n-1}{n}$, which is finitely many such element. But since there are infinitely many such $x \in A$ s.t. $x \leq \frac{3}{2}$, contradiction. □

8. **Back and Forth Method**: Let $(X, \leq_X), (Y, \leq_Y)$ be countable, dense, unbounded, and linear-order set. Then they are order-isomorphic.

   Proof. We construct a sequence of function $(h_n)_{n \in \mathbb{N}}$, $h_n : X_n \to Y_n$ such that

   (a) $x_n \in X_n \subseteq X$, $y_n \in Y_n \subseteq Y$, $h_n \subseteq h_{n+1} (h_{n+1} \big|_{X_n} = h_n)$ for any $n \in \mathbb{N}$, where X_n, Y_n are finite.

(b) h_n is bijective and order-preserving.

Then, take $f := \bigcup_{n\in\mathbb{N}} h_n : X \to Y$. Hence $f(x) = h_n(x)$ for some n by construction of h_n is function, which is bijective and also order-preserving. To show that h_n exists, by induction:

For $n = 0$:

Define $X_0 = \{x_0\}, Y_0 = \{y_0\}$ and $h_0 : X_0 \to Y_0$ by $h(x_0) = y_0$. Clearly is bijective and also order-preserving.

Hypothesis: Suppose we have defined $h_n : X_n \to Y_n$ satisfying $(a), (b)$.

FORTH:

Let $X_n^* := X_n \cup \{x_{n+1}\}$. Since X_n is finite, X_n^* is also finite. Write $X_n = \{a_1, ..., a_k\}$, then by bijectivity of h_n, define $Y_n = \{b_1, ..., b_k\}$ where $b_i = h_n(a_i)$, also using hypothesis that both X_n, Y_n are order preserving, $a_i < a_j \iff b_i < b_j$. Now define $h_n^* : X_n^* \to Y_n \cup \{b\}$ where b is chosen later.

Case 1: If $x_{n+1} < a_1$, by unboundedness of Y, $\exists b \in Y$ such that $b < b_1$, then define $h_n^*(x_{n+1}) := b$, done. Similarly, if $x_{n+1} > a_k$, we define $h_n^*(x_{n+1}) := c$ where $\exists c(\in Y) > b_k$.

Case 2:if $x_{n+1} = a_i$ for some $i = 1, ..., k$, then we define $b = b_i$.

Case 3: if $a_i < x_{n+1} < a_j$ for some $i < j$, by density of Y, take $b_i < b < b_j$.

BACK:

We define $Y_n \cup \{b\} =: Y_n^*$. Similarly we let $h_{n+1} : Y_n^* \cup \{y_{n+1}\} \to X_n^* \cup \{a\}$, by showing all 3 cases similar to above, let $h_{n+1} : Y_n^* \cup \{y_{n+1}\} =: Y_{n+1} \to X_n^* \cup \{a\} =: X_{n+1}$, we are done.

□

Definition 2.5. A subset I of a linearly ordered set X is an initial segment of X if $x \in I$ and $y < x$ implies $y \in I$.

Example 2.4. $\{1, 2, 3, 4\}$ is initial segment of $\mathbb{N}$, but $\{2, 4, 5, 7\}, \{1, 2, 3, 5\}$ is not.

The real interval $(0, 1)$ is initial segment of $(0, \infty)$.

Generally, $\forall x \in X$, the subset $I_x = \{y \in X : y < x\}$ is a proper initial segment of X by transitivity.

Note that transitivity is used to show the downward closure where if we have $y \in I_x \implies y < x$ and if any $z < y$, then by transitivity, $z < x$ and that $z \in I_x$.

Note that every proper initial segment of X is of this form in general.(like $(-\infty, 1]$ of $\mathbb{R}$).

Note. If I is a proper initial segment of a well-ordered set X, then $I = I_x$ for x the least element of $X \backslash I$.

Proof. Let $x = \min(X \setminus I)$. If $y \in I_x$, then $y < x \implies y \notin (X \setminus I) \implies y \in I$. Conversely, if $y \in I$, then $y < x$ (otherwise $x \leq y \implies x \in I$, a contradiction), so $y \in I_x$. □

Lemma 2.1. *Let X, Y be well-ordered sets, I be an initial segment of Y, and $f : X \to I$ be an order-isomorphism. Then $f(x)$ is the least element of $Y \backslash \{f(y) : y < x\}$ for every $x \in X$.*

Proof. The set $A = Y \backslash \{f(y) : y < x\}$ is non-empty since $f(x) \in A$. Let $a \in A$ be least element. Then $a \leq f(x)$. And since $f(x) \in I$ by the f, $a \in I$ by I is initial segment of Y. Thus, $a = f(z)$ for some $z \in X$ since the map is order-isomorphic. To show that $x = z$, since $f(z) \leq f(x)$, it follows that $z \leq x$ by order-isomorphism. If $z < x$, then $f(z) \in \{f(y) : y < x\}$ and hence $f(z) \notin A$, which leads to a contradiction of the choice that $a = f(z)$ is the least element in A. Thus $x = z$. □

Proposition 2.2 (Proof by induction)**.** *Let X be a well-ordered set and $S \subset X$ such that for every $x \in X$ the following holds: if $y \in S$ for all $y < x$, then $x \in S$. Then $S = X$.*

Note. Assume that S is defined in terms of a property p : $S = \{x \in X : p(x)\}$. Then this proposition says:

$$(\forall x)\Big([(\forall y < x)p(y)] \implies p(x)\Big) \implies (\forall x)p(x)$$

The 'base case' ($p(x)$ for the least element x of X) is included in the assumption $[(\forall y < x)p(y)] \implies p(x)$'.

Proof. If $S \neq X$, then $X \backslash S$ has least element x, say. For $y < x$, we have $y \in S$ by the choice of x. It follows from the induction assumption on S that $x \in S$, which contradicts $x \in X \backslash S$. □

Remark. Note that above induction is transfinite induction: it does not require the domain to be countable, but only well-ordered set. Note also that in above proof, there is no statements or propositions that required the countability of set, but only the well-ordering of the set: which only needs $X \backslash S$ to be non-empty, hence is also a well-ordered set, and that we have a least element from it.

We summarize the table below to give a visual comparison on standard induction and transfinite induction.

Feature	**Standard Induction**	**Transfinite Induction (Proposition 2)**
Domain Set	Countable set $\mathbb{N} = \{1, 2, 3, \dots\}$	Any well-ordered set X (countable or uncountable)
Underlying Structure	Successor relation ($n \to n+1$)	Linear order where every non-empty subset has a least element
Base Case	Verified explicitly for the minimum element: $p(1)$	Included automatically in the hypothesis for x minimal: '$(y \in S \text{ s.t.} \forall y < x) \Rightarrow x \in S$' holds vacuously
Inductive Step	$(\forall n \in \mathbb{N})\,[p(n) \implies p(n+1)]$	$(\forall x \in X)\,[(\forall y < x\, p(y)) \implies p(x)]$
Proof Mechanism	Minimal counterexample in $\mathbb{N} \setminus S$	Minimal counterexample in $X \setminus S$ via well-ordering

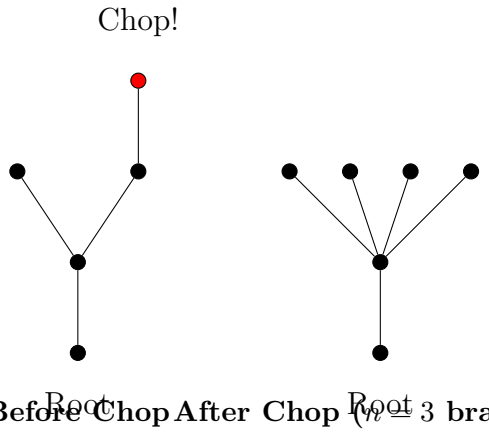
Example 2.5 (The Hydra Game — Transfinite Induction in Daily Life [7])**.** Imagine you are hired for a cleaning job, but your task is to destroy a mythological **Hydra** consisting of a root, branches, and heads. Your goal is to chop off all its heads until only the root remains.

The rules of the Hydra's regeneration are dictated by a malicious boss:

1. **The Root Move:** If you chop off a head attached directly to the root, it stays dead.

2. **The Multiplier Move:** If you chop off a head deeper in the system, the Hydra grows back n copies of the *entire branch* you just cut, attached one level lower to the structural node.

3. **The Escalation:** Every time you make a move, n increases by 1. On your 1st chop, it grows 1 copy. On your 2nd chop, 2 copies. By your millionth chop, it sprouts 1,000,000 copies of that entire branch structure!



Before Chop After Chop ($n = 3$ branches)

Why Standard Induction Fails:

If you try to prove that you will eventually finish this chore using standard mathematical induction in Peano Arithmetic (PA), you will fail. Standard induction proves a property P(n) holds for all $n \in \mathbb{N}$ by showing $P(0)$ is true and $P(n) \Rightarrow P(n+1)$.

However, the Kirby-Paris Theorem (1982) proved that the statement **"The Hydra will always eventually die"** is logically unprovable in Peano Arithmetic. The horizontal multiplication causes the number of heads to balloon at a rate that outruns standard arithmetic tools. Standard induction cannot "see" the end of the game because the numbers grow faster than any function you can describe with standard integers.

How Transfinite Induction Resolves the Scenario:

To prove that this chore must eventually end, we look past standard counting numbers and assign an **ordinal number** α to the structure of the Hydra to measure its tree height and nesting depth.

- A head (leaf node) has a value of 0.
- If a node v has children with ordinals $\alpha_1, \alpha_2, \ldots, \alpha_k$, then the ordinal value of v is modeled as a sum of powers of ω:

$$o(v) = \omega^{\alpha_1} + \omega^{\alpha_2} + \cdots + \omega^{\alpha_k}$$

The global Hydra is represented by the ordinal assigned to its root node, which quickly reaches values like ω^ω or ω^{ω^ω}.

When you chop a head off, a single branch of value $\omega^{\alpha+1}$ is replaced by a massive, finite number of branches valued at ω^α. Mathematically, because $\alpha < \alpha + 1$, we know that:

$$\underbrace{\omega^\alpha + \omega^\alpha + \cdots + \omega^\alpha}_{\text{1,000,000 times}} < \omega^{\alpha+1}$$

No matter how large n becomes, the global ordinal assigned to the root node **strictly decreases** with every single chop:

$$\Omega_{\text{step 1}} > \Omega_{\text{step 2}} > \Omega_{\text{step 3}} > \dots$$

Because ordinal numbers are **well-ordered**, they contain no infinite strictly descending sequences. This descending chain of ordinals must eventually terminate at 0, mathematically guaranteeing that you will eventually kill the Hydra, finish your shift, and go home.

Remark. Heres an example of using proof by induction.

Proposition 2.3. *Let X, Y be well-ordered , order-isomorphic set. Then there exists a unique order-isomorphism from X to Y.*

Note. Show statement holds for all preceding term. Like strong induction.

Proof. Let $f, g : X \to Y$ be order-isomorphisms. We show $(\forall x)(f(x) = g(x))$ by induction.

Let $x \in X$ and assume that $f(y) = g(y)$ for all $y < x$(the induction hypothesis). We need to show that $f(x) = g(x)$, which will then complete the induction. By Lemma 2.1, $f(x)$ is the least element of $A = Y \backslash \{f(y) : y < x\}$, and $g(x)$ is the least element of $B = Y \backslash \{g(y) : y < x\}$. By induction hypothesis, $A = B$, also by hypothesis that $f(y) = g(y)$ for any $y < x$, $\{f(y) : y < x\} = \{g(y) : y < x\}$, and the least element of A,B must also be equal, hence $f(x) = g(x)$. □

Note. This uniqueness property (Proposition 3) is false in general for linearly ordered sets.

For instance:

1. **Automorphisms on $\mathbb{Z}$:** Between the linearly ordered set $\mathbb{Z}$ and itself, the identity function $f(n) = n$ is an order-isomorphism. However, $g(n) = n + 17$ is also a valid, distinct order-isomorphism from $\mathbb{Z}$ to $\mathbb{Z}$, as $n < m \implies n + 17 < m + 17$. Because $\mathbb{Z}$ has no least element, shifting the entire structure does not disrupt the order relation.

2. **Automorphisms on $[0, \infty)$:** Between the linearly ordered set $[0, \infty)$ and itself, the identity map $f(x) = x$ is an order-isomorphism. However, $g(x) = x^2$ is also a distinct order-isomorphism from $[0, \infty)$ to $[0, \infty)$, since g is strictly increasing on $[0, \infty)$ ($x < y \implies x^2 < y^2$) and maps $[0, \infty)$ bijectively onto itself.

Takeaway: Unlike well-ordered sets—where the well-ordering property forces every element into a single, rigid, uniquely determined position via transfinite induction—general linearly ordered sets possess symmetries (automorphisms) that allow multiple distinct order-preserving bijections between the same pair of isomorphic sets.

Remark. Induction allows us to show things. And we will also need a tool to construct things: recursion. We first recall that a function from X to Y is a subset f of $X \times Y$ such that

1. for all $x \in X$, there exists $y \in Y$ such that $(x, y) \in f$.
2. for all $x \in X$, and $y, z \in Y$, if $(x, y) \in f, (x, z) \in f$, then $y = z$.

We of course write $(x, y) \in f$ as $y = f(x)$ and say that 'f maps x to y'.

Note that $f \in \mathcal{P}(X \times Y)$.

For $Z \subset X$, the restriction of f to Z is $f \restriction_Z = \{(x, y) \in f : x \in Z\}$, which is a function $Z \to Y$. Note that $f \restriction_Z$ is a subset of $Z \times Y$, so in particular, it is also an element of $\mathcal{P}(X \times Y)$.

Theorem 2.4 (Definition by recursion)**.** *Let X be a well-ordered set and Y be an arbitrary set. Then for every function $G : \mathcal{P}(X \times Y) \to Y$, there is a unique function $f : X \to Y$ such that $f(x) = G(f \restriction_{I_x})$ for all $x \in X$.*

Remark (Why Formal Recursion is Necessary). In informal mathematics, writing $F(n) = F(n-1) + F(n-2)$ appears self-evident. However, within axiomatic set theory, a function F is formally defined as a set of ordered pairs $F \subseteq X \times Y$. Simply declaring $F(n)$ in terms of F creates circularity because it evaluates a set object before that set has been shown to exist.

Theorem 4 resolves this foundational issue by:

1. **Eliminating Circularity:** Decoupling the recursive rule into a static function G that operates on already-constructed initial segments ($h|_{I_x}$).
2. **Guaranteeing Existence & Uniqueness:** Proving via transfinite induction over well-ordered sets that gluing together all valid partial attempts yields a single, non-ambiguous, non-contradictory function.

Idea:[Trace of Theorem 4 using the Fibonacci Sequence]

To construct the Fibonacci sequence strictly in set theory, we cannot simply write $F(n) = F(n-1) + F(n-2)$ because the function F does not exist prior to its definition. Instead, we apply Theorem 4 by defining a global evaluation function $G : \mathcal{P}(\mathbb{N} \times \mathbb{N}) \to \mathbb{N}$:

$$G(h|_{I_n}) = \begin{cases} 0 & \text{if } n = 0 \quad (I_0 = \emptyset) \\ 1 & \text{if } n = 1 \quad (I_1 = \{0\}) \\ h(n-1) + h(n-2) & \text{if } n \geq 2 \quad (I_n = \{0, 1, \ldots, n-1\}) \end{cases}$$

The function G evaluates each step using only the historical data stored in the restriction $h|_{I_n}$:

1. **Attempt 1 (h_1 on $I_1 = \{0\}$):**

$$h_1 = \{(0, G(\emptyset))\} = \{(0, 0)\}$$

2. **Attempt 2 (h_2 on $I_2 = \{0, 1\}$):**

$$h_2 = \{(0, 0), (1, G(h_1|_{I_1}))\} = \{(0, 0), (1, 1)\}$$

3. **Attempt 3 (h_3 on $I_3 = \{0, 1, 2\}$):**

$$h_3 = \{(0, 0), (1, 1), (2, G(h_2|_{I_2}))\} = \{(0, 0), (1, 1), (2, 1)\}$$

Uniqueness & Existence:

- *Uniqueness:* If h and h' are two attempts defined up to n, then $h|_{I_n} = h'|_{I_n}$ implies $h(n) = G(h|_{I_n}) = G(h'|_{I_n}) = h'(n)$. Any two attempts agree on their shared domain.

- *Existence:* The complete function $F = \bigcup\{h : h \text{ is an attempt}\}$ is a valid function with domain $\text{dom}(F) = \mathbb{N}$. If $\text{dom}(F)$ were a proper initial segment I_n, $F \cup \{(n, G(F|_{I_n}))\}$ would form a strictly larger attempt, contradicting maximality.

Proof. Say h is an attempt if h is a function $I \to Y$, where the domain of h, denoted by $\text{dom}(h)$, is an initial segment of X such that $h(x) = G(h \restriction_{I_x})$ for every $x \in \text{dom}(h)$ (Note that $x \in \text{dom}(h) \implies I_x \subset \text{dom}(h)$). We need to show that there is a unique attempt whose domain is X.

We first show that if $h, h'$are attempts,then $h(x) = h'(x)$ for all $x \in \text{dom}(h) \cap \text{dom}(h')$, which in particular will show the uniqueness in the statement of the theorem. We prove by induction. Fix $x \in \text{dom}(h) \cap \text{dom}(h')$ and assume that $h(y) = h'(y)$ for all $y < x$(induction hypothesis). Then $h \restriction_{I_x} = h' \restriction_{I_x}$ and thus $h(x) = G(h \restriction_{I_x}) = G(h' \restriction_{I_x}) = h'(x)$.

We complete the proof by of existence by setting $f = \bigcup\{h : h \text{ is an attempt}\}$. Then f is a function since for any $x \in X$, if there is an attempt h defined at x, then the value of $h(x)$ is independent of h by what we showed above.

The domain of f is the union $\bigcup\{h : h \text{ is an attempt}\}$, and thus $\text{dom}(f)$ is an initial segment of X.

For any $x \in \text{dom}(f)$, there is an attempt h such that $f(x) = h(x)$. It follows that $f(y) = h(y)$ for all $y < x$, and hence $f(x) = h(x) = G(h \restriction_{I_x}) = G(f \restriction_{I_x})$. Thus, f is an attempt.

Finally, assume that $\text{dom}(f) \neq X$. Then $\text{dom}(f) = I_x$ for some $x \in X$. Recall that an initial segment $I_x := \{y \in X : y < x\}$ of X. Since $\text{dom}(f) = I_x$, $x \notin \text{dom}(f)$, so there is no attempt defined at x. Consider $f' = f \cup \{(x, G(f))\}$, we show that f' is an attempt defined on $I_x \cup \{x\}$. To show f' is a function: let $a \in I_x \cup \{x\}$, if $a \in I_x$, since $f : I_x \to Y$ is an attempt, f is well-defined function on I_x. Hence there is an $y \in Y$ such that $(a, y) \in f$, and $f \subseteq f'$, it follows that $(a, y) \in f'$. If $a = x$, since we already have the ordered pair $(x, G(f))$ to f'. As $G : \mathcal{P}(X \times Y) \to Y$, then $G(f) \in Y$, hence $(x, G(f)) \in f'$.

For $a \in I_x \cup \{x\}$, $y_1, y_2 \in Y$ and $(x, y_1), (x, y_2) \in f'$. If $a \in I_x$, since f is well-defined, we must have, by above proof, that y_1, y_2 must not be $G(f)$. Hence $y_1 = y_2$ by f.

If $a = x$, then $a \notin I_x$. As $f \subseteq I_x \times Y$, there is no pair in f such that $(a, y_1), (a, y_2) \in f$. We must have only 1 pair in $f' = f \cup \{(x, G(f))\}$ with first coordinate $(a, G(f))$. Hence $y_1 = y_2 = G(f)$.

So f' is well-defined function.

Now we show that f' is an attempt that is also defined at x. We show that $\text{dom}(f') = I_x \cup \{x\}$ is an initial segment of X:given $z \in X, y \in I'_x$ such that $z < y$. If $y \in I_x$, clearly $z \in I_x \subseteq I_x \cup \{x\}$. If $y = x$, then $z < x \Rightarrow z \in I_x \subseteq I_x \cup \{x\}$. Hence it is indeed an initial segment of X . Then, consider $\text{dom}(f') = I_x \cup \{x\} =: I'_x$, which is an initial segment of X. To show that $f'(y) = G(f'|_{I'_y})$ for any $y \in \text{dom}(f')$. If $y \in I_y \subset \text{dom}(f) \subseteq \text{dom}(f')$, we have $f'|_{I_y} = f|_{I_y}$. Since f is an attempt, then $f(y) = G(f|_{I_y})$ and so $f'(y) = f(y) = G(f|_{I_y}) = G(f'|_{I_y})$. If $y \in \text{dom}(f)' \backslash \text{dom}(f') = \{x\}$ $(i.e. y = x)$, note that $f'|_{I_x} = f$, and $f'(x) = G(f)$, so $f'(x) = G(f) = G(f'|_{I_x})$. Hence $f' = f \cup \{(x, G(f))\}$ is indeed the attempt defined at x. Hence we have f is defined on the whole set X. □

Remark. Note that the above theorem is highly non-trivial, where one might not understand the meaning of existence of such theorem in advance. Consider the following question: When we are construction a recursive function, like the Fibonacci Sequence, where $F_0 := 0, F_1 := 1, F_{n+1} = f_n + f_{n-1}$. One might miss that construction of such sequence of functions might not be well-defined.

To prove that this recursive construction yields a well-defined function $f \subseteq X \times Y$ satisfying the standard functional axioms (existence and uniqueness of outputs), we must guarantee the following structural properties:

1. **Initial Segment Domains:** The domain $\text{dom}(f) \subseteq X$ (and each attempt domain $\text{dom}(h) \subseteq X$) must be an initial segment of X. This guarantees that whenever an element x is in the domain, all strictly preceding elements $y < x$ are also included.

2. **Availability of Predecessor History:** Evaluating $f(x)$ requires knowing all prior values $f(y)$ for $y < x$. Formatting the predecessor set as an initial segment

$I_x = \{y \in X : y < x\}$ ensures that the restriction $f \restriction_{I_x}$ contains complete prior history.

3. **Adherence to the Recursion Rule:** Each attempt h acts as a partial solution that strictly obeys the master rule via $h(x) = G(h \restriction_{I_x})$ for every $x \in \mathrm{dom}(h)$.

4. **Compatibility of Attempts:** Any two attempts h_1 and h_2 must agree on their overlapping domain (i.e. $h_1(y) = h_2(y)$ for all $y \in \mathrm{dom}(h_1) \cap \mathrm{dom}(h_2)$). This guarantees that prior evaluation history remains identical across all trials, preventing contradictory ordered pairs (x, y_1) and (x, y_2) which could be lead by different attempts.

5. **Assembly via Set Union:** Defining $f = \bigcup h$ glues all compatible attempts into a single relation $f \subseteq X \times Y$. Because all attempts are mutually consistent, this union satisfies functional well-definedness, with its domain $\mathrm{dom}(f) = \bigcup \mathrm{dom}(h)$ ultimately spanning all of X.

Proposition 2.5 (Subset Collapse)**.** *Let Y be a well-ordered set and $X \subset Y$. Then X is order-isomorphic to a unique initial segment of Y.*

Remark. Note that we cannot use Proposition 2.2 to show the uniqueness of the map. Since Proposition 2.3 asks us for a unique initial segment, but not a unique order isomorphism to an initial segment. So, we need to guarantee that both I, f are unique.

Proof. For uniqueness, assume that f is order-isomorphism from X to an initial segment of Y. By Lemma 2.1, we have $f(x)$ is the least element of $Y \backslash \{f(y) : y < x, y \in X\}$ for every $x \in X \subset Y$. By induction, assume that $f : X \to I_1$, $g : X \to I_2$ are 2 order-isomorphism from X to 2 initial segment of Y. We fix $x \in X$ and assume that $f(y) = g(y)$ for any $y < x$. Since $f(y) = g(y)$ for any $y < x$, then $\{f(y) : y < x, y \in X\} = \{g(y) : y < x, y \in X\}$, leading $Y \backslash \{f(y) : y < x, y \in X\} = Y \backslash \{g(y) : y < x, y \in X\}$ and hence the 2 least element $f(x) = g(x)$. Since we showed that the function $f = g$, we conclude that $\mathrm{Im}(f) = \mathrm{Im}(g)$ and hence $I_1 = I_2$.

For existence, we may assume that $Y \neq \emptyset$ and fix $y_0 \in Y$. Define $f : X \to Y$ by recursion as follows:

$$f(x) = \begin{cases} \min(Y \backslash \{f(y) : y < x, y \in X\}) & \text{if this exists} \\ y_0 & \text{otherwise} \end{cases}$$

We first show that the 'otherwise' clauses never arises by showing that $f(x) \leq x$ for all $x \in X$:. Indeed, fix $x \in X$ and assume that $f(y) \leq y$ holds for all $y \in X$ with $y < x$. Assume that $x \in \{f(y) : y < x\}$, since $x \in X \implies x \in Y$. Then we have $x = f(y')$ for some $y' < x$. Since by induction hypothesis, $f(y') \leq y'$, we have $x \leq y' < x \implies x < x$,

contradiction. Hence we have $x \in Y \backslash \{f(y) : y < x\}$. Now since $Y \backslash \{f(y) : y < x\}$ contains element, by well-ordering of Y, the subset $Y \backslash \{f(y) : y < x\}$ is also well-ordered which has least element. So by definition, $f(x) = \min(Y \backslash \{f(y) : y < x\})$. Hence $f(x) \leq x$. Note that since $f(x) \leq x$, which leads to the set $Y \backslash \{f(y) : y < x\}$ must be non-empty, hence the minimum of such set must exist, so the otherwise clause never arise.

Given $y < x$ in X, since $\{f(z) : z < y\} \subset \{f(z) : z < x\}$,

$$f(x) \in Y \backslash \{f(z) : z \in X, z < x\} \subset Y \backslash \{f(z) : z \in X, z < y\} \Rightarrow f(x) \in Y \backslash \{f(z) : z \in X, z < y\}$$

it follows that $f(y) < f(x)$. Thus, f is order-preserving.

Finally, we show that $\mathrm{Im}(f) = I$ i.e.

$$\forall a \in Y, \forall b \in Y. \text{ If } b \in \mathrm{Im}(f) \text{ and } a < b \text{ in } Y, \text{ then } a \in \mathrm{Im}(f).$$

By contrapositive take $b = f(x)$ arbitrary element in $\mathrm{Im}(f)$, then the contrapositive is ' If$a \in Y \backslash \mathrm{Im}(f)$, then $b \notin \mathrm{Im}(f) \vee b \leq a$ in Y'. Since $b = f(x) \in \mathrm{Im}(f)$, and $a \notin \mathrm{Im}(f)$, it reduces to showing that If $a \notin \mathrm{Im}(f)$, then $f(x) < a$ for all $x \in X$. Assume that $a \in Y \backslash \mathrm{im}(f)$. We show by induction that $f(x) < a$ for all $x \in X$: Fix $x \in X$ and assume that $f(y) < a$ for all $y \in X$ with $y < x$. Since $a \in Y, a \notin \mathrm{Im}(f), a$ cannot be any $f(y)$. Hence $a \in Y \backslash \{f(y) : y \in X, y < x\}$, and thus by minimality of $f(x)$, $f(x) \leq a$ and since $a \notin \mathrm{Im}(f), a \neq f(x)$, so $f(x) < a$ as required. So $f(x) < a$ holds for any $x \in X$, and a acts as an upper bound for $\mathrm{Im}(f)$ □

Remark. It follows from the above proposition that a well-ordered set X is not order-isomorphic to any proper initial segment of X.

The conceptual foundation of order-isomorphisms between well-ordered sets and the impossibility of self-compression can be summarized as follows:

1. **Canonical Relabeling ("Gapless Packing"):** An order-isomorphism $f : X \to I \subseteq Y$ onto an initial segment acts as a systematic relabeling of X using elements of Y. By mapping each $x \in X$ greedily to the least available element in Y, it "packs" X down to the very bottom of Y without leaving gaps.

2. **The Inflation Property ($f(x) \geq x$):**(By systematic $Y \backslash \{f(y) : y < x\}$ Lemma) Any order-preserving embedding $f : X \to X$ satisfies $f(x) \geq x$ for all $x \in X$. Transfinite induction guarantees that order preservation forces every element to map to a position at or strictly above itself.

   Proof. First we show that for any strictly well-ordering function $f : X \to X$ where X is well-ordered, $f(x) \geq x$ for all $x \in X$: Assume by induction that $f(y) \geq y$ for any $y < x$.

Assume on contrary that $f(x) < x$. Then there exists $y_0 \in X$ such that $f(x) = y_0 < x$, since $y_0 < x$, by induction hypothesis, $f(y_0) \geq y_0 = f(x)$, however we know that f is order-preserving, hence it is a contradiction. So we conclude that for any order-preserving function that maps from X to itself where X is well-ordered, $f(x) \geq x$ for all $x \in X$.

Now, suppose that there exists a function $f : X \to I_a$ where X is well-ordered and I_a is a proper initial segment of X where $I_a := \{x \in X : x < a\}$. Note that the function is bijective, in particular it is surjective. So every $f(X) = I_a$ and hence let $x = a$, $f(a) < a$, which contradict that $f(x) \geq x$ for all $x \in X$. □

3. **Impossibility of Self-Compression:** A well-ordered set X cannot be order-isomorphic to any proper initial segment $I_a = \{y \in X : y < a\}$. Mapping onto I_a would require $f(a) \in I_a \implies f(a) < a$, which directly contradicts $f(a) \geq a$.

4. **Order Structure vs. Cardinality:** While finite well-ordered sets cannot map onto proper initial segments due to cardinality ($|I_a| < |X|$), infinite well-ordered sets may share the same cardinality with their proper initial segments. In the infinite case, the impossibility is governed entirely by *order topology* (height/structure) rather than element count alone.

Example 2.6.

1. $\mathbb{N}$ vs $\mathbb{N} \cup \{\infty\}$: Let $X = \mathbb{N} = \{1, 2, 3, ...\}$, $Y = \mathbb{N} \cup \{\infty\}$ where $n < \infty$ for all $n \in \mathbb{N}$. First note that X is order-isomorphic to the proper initial segment $I_\infty = \{y \in Y : y < \infty\} = \mathbb{N}$ of Y. The map $f(n) = n$ is unique.

2. $\{2, 4, 6, ...\}$ vs $\{0, 1, 2, 3, ...\}$. Let $X = \{2, 4, 6, ...\}$, and $Y = \mathbb{N} \cup \{0\}$, note that $X \subset Y$ and that $f(k) = \frac{k}{2} - 1$ is an order-isomorphic map from X to Y, which it is unique. Note that above is still order-isomorphic even if the 0 is dropped. If we add ∞ at Y, then X can only be isomorphic to the initial segment I_∞ of Y.

Notation: Let X, Y be well-ordered sets, we write $X \leq Y$ if X is order-isomorphic to an initial segment of Y.

Theorem 2.6. *Let X, Y be well-ordered sets. Then either $X \leq Y$ or $Y \leq X$.*

Proof. Assume that $Y \not\leq X$. Then in particular, $Y \neq \emptyset$ else there exists the trivial order-isomorphism from $Y = \emptyset \to \emptyset$ the trivial initial segment of X. Fix $y_0 \in Y$ and define $f : X \to Y$ by recursion as follows:

$$f(x) = \begin{cases} \min(Y \backslash \{f(y) : y < x\}) & \text{if this exists} \\ y_0 & \text{otherwise} \end{cases}$$

If the 'otherwise ' clause ever arises (meaning that the set $Y \backslash \{f(y) : y < x\}$ is empty set), meaning that all element in Y is exhausted and is used to assign order for elements

in X, then let x be the least element of X for which $Y \backslash \{f(y) : y < x\}$ becomes empty. Then $f(I_x) = Y$ and for $y < x$, since x is defined to be the least element in X such that $Y \backslash \{f(y) : y < x\} = \emptyset$, so by definition for any $y < x$, there is an element in Y where y is mapped to from X, so the 'otherwise' clause does not arise in the definition of $f(y)$. It follows as in the proof of Proposition 2.3 that f is an order-isomorphism from I_x to Y meaning that$f^{-1} : Y \to I_x \subset X$ is order-isomorphism to proper initial segment of X, meaning that $Y \leq X$, which contradicts $Y \not\leq X$. Hence the 'otherwise ' clause can never arises, and then it follows again as in the proof of Proposition 2.3 that f is an order-isomorphism from X to an initial segment of Y, so we have $X \leq Y$. □

Proposition 2.7. *Let X, Y be well-ordered sets. If $X \leq Y$ and $Y \leq X$, then X and Y are order-isomorphic.*

Proof. Let $f : X \to I_Y$ and $g : Y \to I_X$ be order-isomorphisms to initial segments of Y and X, respectively. Then $g \circ f$ is an order-isomorphism from X to an initial segment of X. Note that using above results, one cannot have an order-isomorphism from X to any proper initial segment of X. So the only possible choice of initial segment of X is itself, where to restrict the isomorphism into order-preserving, the only possible map would be identity map. Hence $g \circ f = \mathrm{Id}_X$. Similarly, $f \circ g = \mathrm{Id}_Y$. □

Remark. From above, we have shown that '$\leq$' is a linear order:

- Reflexive: $X \leq X$
- Anti-symmetric: $(X \leq Y) \wedge (Y \leq X) \Rightarrow X$ order isomorphic to Y (Prop. 2.4)
- Transitive: $((X \leq Y) \wedge (Y \leq Z)) \Rightarrow (X \leq Z)$:

  Suppose $f : X \to I_1$ and $g : Y \to I_2$ are order-isomorphisms onto initial segments $I_1 \subseteq Y$ and $I_2 \subseteq Z$, respectively. Define $h = g \circ f : X \to g(I_1)$.

  Order-Preserving & Bijective onto Image: For $x, y \in X$ with $x <_X y$, order-preservation of f gives $f(x) <_Y f(y)$. Since $f(x), f(y) \in I_1$, order-preservation of g gives $g(f(x)) <_Z g(f(y))$, so h is order-preserving. Since $f : X \to I_1$ is bijective and $g|_{I_1} : I_1 \to g(I_1)$ is bijective, $h : X \to g(I_1)$ is a bijection onto its image $h(X) = g(I_1) \subseteq I_2$.

  Initial Segment Proof: We show that $g(I_1)$ is an initial segment of Z. Let $z \in Z$ and $z <_Z g(y)$ for some $g(y) \in g(I_1)$. Since $g(y) \in I_2$ and I_2 is an initial segment of Z, $z \in I_2$. By surjectivity of $g : Y \to I_2$, there exists $y' \in Y$ such that $g(y') = z$. Since g is order-isomorphic, $g(y') <_Z g(y) \implies y' <_Y y$. Because I_1 is an initial segment of Y and $y \in I_1$, we have $y' \in I_1$. Thus, $z = g(y') \in g(I_1)$.

  Hence, h is an order-isomorphism from X onto the initial segment $g(I_1)$ of Z, showing $X \leq Z$.

- Trichotomous: $(X \leq Y) \vee (Y \leq X)$ (Thm. 2.2)

Definition 2.6. For well-ordered sets X, Y, write $X < Y$ if $X \leq Y$ and X is not order-isomorphic to Y.

Example 2.7. Let $X = \omega = \{0, 1, 2, \dots\}$ and $Y = \omega + \omega = \{0, 1, 2, \dots\} \cup \{0', 1', 2', \dots\}$, where $n < m'$ for all $n, m \in \mathbb{N}$. Note that X is order-isomorphic to the proper initial segment $I_{0'} = \{y \in Y : y < 0'\}$ of Y via the identity mapping $f(n) = n$. Hence, by the Comparability Theorem (Theorem 6), Y cannot be order-isomorphic to X or to any initial segment of X. This establishes $X < Y$ in terms of order type (i.e., $\text{ord}(X) < \text{ord}(Y)$), even though their cardinalities are equal ($|\omega| = |\omega + \omega| = \aleph_0$).

Note. Equivalently, $X < Y$ if and only if X is order-isomorphic to a proper initial segment of Y.

Then, '$<$' is irreflexive, transitive, trichotomous.

Note. For any set X, there is always some z not in X, for example, no set can have a power set contained inside itself, because there is no surjection from X to the power set $\mathcal{P}(X)$ by Cantor's diagonal argument.

Proof. Let us define the map $f : X \to \mathcal{P}(X)$. We define $T := \{s \in X | x \notin f(x)\}$. Assume that the map f is surjective. Assume that $T = f(x)$ for some $x \in X$. If $x \in T$, by definition of T, $x \in T$ implies $x \notin f(x)$. But $x \in T = f(x)$, contradiction. If $x \notin T$, since $T = f(x)$, meaning that $x \notin f(x)$. But by definition of T, $x \in T$, contradiction arises.

So no such T can exist, the map simply is not surjective. □

Note. Alternatively, using Zermelo–Fraenkel set theory, the Axiom of Foundation implies that no set can contain itself as an element. Thus, setting $z = X$ directly guarantees that $z \notin X$, as X cannot be an element of itself. Following Cantor's realization that not all infinities are created equal, he proved in 1874 that $|\mathbb{N}| < |\mathbb{R}|$ (Since he showed that $\mathbb{R}$ are of the same equivalence class with $\mathcal{P}(\mathbb{N})$). Shortly after establishing that there are strictly more real numbers than integers, Cantor posed the natural follow-up question: does there exist an intermediate cardinality between the integers and the reals? Conjecture 1 (Cantor, 1878)[8]. There is no set S such that

$$|\mathbb{N}| < |S| < |\mathbb{R}|.$$

Remark (Purpose in Well-Ordering Theory). When constructing a strictly larger well-ordered set $X^+ = X \cup \{z\}$ (where $x < z$ for all $x \in X$), we must guarantee that a new, distinct element $z \notin X$ exists to extend the ordering without introducing set-theoretic paradoxes.

Note (Construction of new well-ordered set). Let X be a well-ordered set. Fix $z \notin X$, and define $X^+ := X \cup \{z\}$ well-ordered by extending the well-ordering $<$ on X by defining $x < z$ for all $x \in X$. Then X^+ is uniquely defined up to order-isomorphism (*i.e.* if we have $X \cup \{z_1\}$ and $X \cup \{z_2\}$, then $z_1 = z_2$). And $X < X^+$ since X is order-isomorphic to the proper initial segment X of X^+.

Remark.

- For well-ordered sets $(X, <_X), (Y, <_Y)$, we say Y is extend of X if $X \subset Y$, $<_X$ is a restriction of $<_Y$ to X (formally , $<_X = <_Y \cap (X \times X)$)) and X is initial segment of Y.

- We say that the set $\{X_i : i \in I\}$ is nested if for all $i, j \in I$, either X_j extends X_i or X_i extends X_j .

- From above definition, suppose that we have sequence of extension like (X_1, X_2, X_3, X_1), we must have $X_1 = X_2 = X_3$ and will not form loop inside since using the definition of extend, $X_1 \subset X_2 \subset X_3 \subset X_1$, hence we have $X_1 = X_2 = X_3$.

Remark. The reason of construction of Upper Bound and nested set of well-ordered sets lies to diversity of such in different area of mathematics: from analysis, to geometry, or algebra. For example the Jordan Holder Theorem from abstract algebra, field extension tower from Galois Theory, Baire Category Theorem from analysis, finite intersection property in topology.

Proposition 2.8 (**Upper Bound**). *If $\{X_i : i \in I\}$ is a nested set of well-ordered sets, then there exist a well-ordered set X such that $X_i \leq X$ for all $i \in I$.*

Proof. Set $X = \bigcup_{i \in I} X_i$, and for $x, y \in X$, set $x < y \iff \exists i \in I, x <_i y$ where "$<_i$" denotes the well-ordering of X_i. From the assumption that the X_i are nested, it follows that $<$ is well-defined linear order of X and each X_i is an initial segment of X.

Given a non-empty subset $S \subset X$, we have $S \cap X_i \neq \emptyset$ for some $i \in I$. Since $S \cap X_i \subset X_i$ and that each X_i is well-ordered, it follows that $S \cap X_i$ has a least element. Since X_i is an initial segment of X, take arbitrary element $s \in S$, if $s \in X_i$, $s \in S \cap X_i$, we have $x \leq s$. If $s \notin X_i$, then by X_i is initial segment of X and $S \subset X$, if $s \in X$ with $s \notin X_i$, then $s > x$. Hence in both case we must have x is least element of S. □

Remark. The same result holds without the assumption of X_i are nested, but the proof will be postponed to Chapter 5.

The well-ordered set X constructed above is in fact the least upper bound of the set $\{X_i : i \in I\}$.

Meaning that if we pick any arbitrary upper bound Y of the nested family $\{X_i : i \in I\}$, first by definition of upper bound, $X_i \leq Y$ for all $i \in I$. We show that Y is extended of X. Since each $X_i \subset Y \implies X = \bigcup_{i \in I} X_i \subset Y$. And take $y \in Y$ where

if $x \in X$ with $y <_X x$, since $x \in X$, then $x \in X_i$ for some $i \in I$. Note that X_i is initial segment of Y, then $y <_{X_i} x$, using the fact that $y \in X_i \subset X$, $y, x \in X$ and $y <_i x \implies y <_X x$ so X is initial segment of Y. Also using the fact that $y <_X x$ for $x, y \in X$, $y <_i x$ for any X_i, also the initial segment of Y for all $i \in I$.

2.2 Ordinals

Definition 2.7. An ordinal is a well-ordered set with two ordinals regarded the same if they are order-isomorphic. The order-type of a well-defined set X is the unique ordinal to which X is order-isomorphic.

Remark. The formal definition of ordinal will be given in Chapter 5. For now, we view ordinal as the equivalent class of well-ordered sets under the relation that they are order-isomorphic.

Example 2.8.

- For $k \in \{0\} \cup \mathbb{N}$, we write k for the order-type of a well-ordered set of size k.
- We let ω denote the order-type of $\mathbb{N}$ (same as order-type of $\mathbb{N} \cup \{0\}$ since $\mathbb{N} \cong \mathbb{N} \cup \{0\}$ by $f : x \mapsto x - 1$).
- Since from the beginning of this section, we have shown that $A = \{\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \cdots\}$ is order-isomorphic to $\mathbb{N}$, hence A has order-type ω.

Definition 2.8. Let α, β be ordinals and X, Y be well-ordered sets with order-type α, β, respectively. We write $\alpha \leq \beta$ if $X \leq Y$ and $\alpha < \beta$ if $X < Y$. Similarly, we write α^+ for the order-type of X^+.

Note. The notations above are well-defined. *i.e.*, they don't depend on the choice of X, Y. For arbitrary ordinals α, β, we either have $\alpha \leq \beta$ or $\beta \leq \alpha$ (Thm 2.2), and if $\alpha \leq \beta$ and $\beta \leq \alpha$, then $\alpha = \beta$ (Prop. 2.4).

Example 2.9 ($\mathbb{N}$ vs. $\mathbb{Z}$ and Their Extensions)**.** Using the definitions introduced above, we now have the necessary tools to explore the structural relationships between $\mathbb{N}$, $\mathbb{Z}$, and variations of their order types.

- $\mathbb{N}$ **vs.** $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$**:**

  As demonstrated previously, $\mathbb{N} \cong \mathbb{N}_0$ via the order-isomorphism $f : n \mapsto n - 1$. Hence, we can conclude that their order types are identical: $\mathrm{Ord}(\mathbb{N}) = \mathrm{Ord}(\mathbb{N}_0) = \omega$.

- **$\mathbb{N}$ vs. Finite Downward Extensions ($\mathbb{N}_{-n}$):**

  For any $n \in \mathbb{N}$, we can inductively define a sequence of finite downward extensions by setting $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$ and $\mathbb{N}_{-n} := \mathbb{N}_{-(n-1)} \cup \{-n\}$. By utilizing the shifting map $f(k) = k - n - 1$, it follows that $\mathrm{Ord}(\mathbb{N}_{-n}) = \mathrm{Ord}(\mathbb{N}) = \omega$ for all $n \in \mathbb{N}$. **Crucially, this preservation of order type holds because each extension is strictly finite.**

- **$\mathbb{N}$ vs. Infinite Downward Extensions ($\mathbb{Z}$):**

  The moment the downward extension becomes infinite, the construction yields the full set of integers, $\mathbb{Z}$. Unlike the finite steps, $\mathbb{Z}$ is famously not a well-ordered set under its standard ordering, as it lacks a least element. Therefore, it does not possess an ordinal order type.

- **Can we fix $\mathbb{Z}$ by adding a bottom element ($\mathbb{Z} \cup \{-\infty\}$)?**

  Suppose we adjoin a absolute minimum element to create $\mathbb{Z}^* := \mathbb{Z} \cup \{-\infty\}$, where $-\infty < k$ for all $k \in \mathbb{Z}$. Even with this adjustment, $\mathbb{Z}^*$ still fails to be well-ordered. Consider the subset $\mathbb{Z} \subset \mathbb{Z}^*$; it still contains no least element, meaning that adding a single floor at the very bottom cannot patch the infinite descending chains within the set. Thus, $\mathbb{Z}^*$ bears no structural relationship to $\mathbb{N}$ in the context of ordinals.

- **$\mathbb{N}$ vs. Finite Extensions with a Bottom Element ($\mathbb{N}_{-n} \cup \{-\infty\}$):**

  In stark contrast to the infinite case, consider a finite downward extension topped with a bottom element:

  $$\mathbb{N}_{-n} \cup \{-\infty\} = \{-\infty, -n, -(n-1), \dots, -1, 0, 1, 2, \dots\}$$

  This set successfully maintains the structure of a well-ordered set. For a fixed n, we can construct an explicit order-isomorphism $f : \mathbb{N}_{-n} \cup \{-\infty\} \to \mathbb{N}$ defined piece-wise by:

  $$f(k) = \begin{cases} 1 & \text{if } k = -\infty \\ k + n + 2 & \text{otherwise} \end{cases}$$

  Because this map is a strict bijection that preserves the order relation, we conclude that $\mathrm{Ord}(\mathbb{N}_{-n} \cup \{-\infty\}) = \mathrm{Ord}(\mathbb{N}) = \omega$.

- **The Prepending Takeaway:**

  From these observations, we establish a vital principle regarding linear orderings: infinite downward prepending completely destroys a set's well-ordered property. Conversely, finite downward adjustments—even when a formal minimum $-\infty$ is

adjoined—are entirely absorbed by the vastness of infinity, leaving the fundamental order type ω unchanged.

Theorem 2.9. *Let α be an ordinal. Then the ordinals strictly smaller than α form a well-ordered set of order-type α.*

Proof. Fix a well-ordered set X with order-type α, and form a collection of set

$$X' = \{Y \subset X : Y \text{ is a proper initial segment of } X\}$$

To show X' is linearly order by the relation $<$, from construction of X' it is irreflexive and transitive. For Trichotomous, note that each element in the collection X' is a proper initial segment of X, so any element must be well-ordered and for any $A, B \in X'$ we either have $A \subsetneq B$ or $B \subsetneq A$. So by Proposition 2.3, if A is order-isomorphic to B, we must have $A = B$. Together using Theorem 2.2, and Proposition 2.4 we either have $A < B$ or $B < A$ or $A = B$.

Now consider the map $x \mapsto I_x$ from X to X', it is bijective and also order-preserving :

Bijectivity:

Surjective from construction of initial segment of X. If $x \neq y$, then WLOG assume that $x < y$, then we have $x \in I_y$, but $x \notin I_x$, hence $I_y \neq I_x$.

Order-preserving:

Suppose $x < y$, then any element $z < x < y$, hence $I_x \subseteq I_y$, now since $x \in I_y, x \notin I_x$, so $I_x \subsetneq I_y$.

Hence, X' and X are in the same equivalent class and hence it is well-ordered with order-type α. We can now form the set

$$I = \{\text{order-type}(Y) : Y \in X'\}$$

which consists precisely of all ordinal with order-type less than α. It is linearly ordered by '$<$', and moreover, the map $Y \mapsto$ order-type(Y) is an order-isomorphism $X' \to I$:

Bijectivity:

Surjective is immediate by construction. For injectivity, if order-type(X)=order-type(Y), then $X \cong Y$, since $X, Y \in X'$, from above we have $X = Y$.

Order-preserving:

If $Y \subsetneq Z$, then Y is proper initial segment of Z, hence $Y \cong I_\beta$ for some initial segment of order type $\beta <$order-type(Z), hence order-type$(Y)<$ orde-type(Z).

and hence I is well-ordered of order-type α. □

Note. It is natural to denote the set of ordinals $\{\beta : \beta < \alpha\}$ by I_α. Using above Theorem, it is a well-ordered set of order-type α.

Proposition 2.10. *A non-empty set S of ordinals has a least element.*

Proof. Fix $\alpha \in S$. If α is not a least element of S, then $\exists \beta (\in S) < \alpha$, $\beta \in I_\alpha$, then $S \cap I_\alpha \neq \emptyset$. Hence by Thm. 2.3, $S \cap I_\alpha \subset I_\alpha$ has a least element β. Since I_α is the initial segment of S, take arbitrary element $s \in S$, if $s \in I_\alpha$ we have $s \in S \cap I_\alpha$ and $\beta < s$, if $s \notin I_\alpha$, then $s > \beta$, in both case β is the least element of S. □

Theorem 2.11 (Burali-Forti Paradox)**.** *The ordinals do not form a set.*

Proof. Assume that the ordinals do form a set X. Then X is well-ordered by Proposition 2.6. Let α be the order-type of X. Then $\alpha \in X$ and I_α is a proper initial segment of X also of order-type α. Thus, X is order-isomorphic to its proper initial segment I_α contradicting Proposition 2.3.

□

Remark. Let $S = \{\alpha_i : i \in I\}$ be a set of ordinals. Applying Proposition 2.5 to the nested set $\{I_{\alpha_i} : i \in I\}$: where $I_{\alpha_i} = \{\beta : \beta < \alpha_i\}$, first we have $I_\alpha := \bigcup_{i \in I} I_{\alpha_i}$ is well-ordered set and also an initial segment where $I_{\alpha_i} < I_\alpha$ for all $i \in I$. Since each $\beta \in S, I_\beta \subseteq I_\alpha \implies \beta \leq \alpha$, we obtain an ordinal α that is an upper bound for S, . The least element of the non-empty set $B = \{\beta \in I_\alpha \cup \alpha : \beta \text{ is an upper bound for } S\}$ of ordinals is a least upper bound for S denoted $\sup S$. Note that the set I_α is well-ordered, and using the same technique that extending the well-ordering of I_α be setting $\beta < \alpha$ for all $\beta \in I_\alpha$, B is also well-ordered. Note that $\bigcup_{i \in I} I_{\alpha_i} = I_{\sup S}$.

Example 2.10. Now we list further examples of ordinals. We have already seen $0, 1, 2, \ldots, \omega$. Note that $\omega = \sup\{0, 1, 2, \ldots\}$.

The next ordinal is ω^+ which we denote as $\omega + 1$. For now this is just notation but it is consistent with ordinal addition defined later in this section. We then have $\omega^{++} =: \omega + 2$, and then $\omega + 1, \omega + 2, \ldots$.

We denote $\sup\{\omega + 1, \omega + 2, \ldots\} =: \omega + \omega$ or $\omega \cdot 2$. Then we continue.

$$0, 1, 2, ..., \omega$$
$$\omega + 1, \omega + 2, \omega + 3, ..., \omega + \omega =: \omega \cdot 2$$
$$\omega \cdot 2 + 1, \omega \cdot 2 + 2, ..., \omega \cdot 3$$
$$\omega \cdot 3..., \omega \cdot 4..., \omega \cdot 5..., \omega \cdot \omega =: \omega^2 (\text{officially} \sup\{\omega, \omega \cdot 2, \omega \cdot 3, ...\})$$
$$\omega^2 + 1, \omega^2 + 2, ...\omega^2 + \omega$$
$$\omega^2 + \omega + 1, ..., \omega^2 + \omega \cdot 2, ..., \omega^2 + \omega \cdot 3 ... \omega^2 + \omega^2 =: \omega^2 \cdot 2$$
$$\omega^2 \cdot 2 + 1, ..., \omega^2 \cdot 2 + \omega, ..., \omega^2 \cdot 3, ..., \omega^2 \cdot 4, ..., \omega^3, ..., \omega^3 \cdot 2, ...\omega^3 \cdot 3, ...$$
$$\omega^4, ...\omega^5, ...\omega^\omega (\text{officially} \sup\{\omega, \omega^2, \omega^3, ...\})$$
$$, ...\omega^\omega \cdot 3, ..., \omega^\omega . 4, ..., \omega^\omega \cdot \omega =: \omega^{\omega+1}$$
$$\omega^{\omega+1}, ..., \omega^{\omega+2}, ..., \omega^{\omega^2} = \omega^{(\omega^2)}, ...\omega^{(\omega^2)\cdot 2}, ...\omega^{(\omega^2)\cdot 3}, ...\omega^{\omega^3}, ...\omega^{\omega^4}, ...\omega^{\omega^\omega}$$
$$, ...\omega^{\omega^{\omega^2}}, ..., \omega^{\omega^{\omega^\omega}}, ...\varepsilon_0 =: \omega^{\omega^{\omega^{\cdot^{\cdot}}}} \ (\text{officially} \ \sup\{\omega, \omega^\omega, \omega^{\omega^{\omega,\cdots}}\})$$
$$\varepsilon_0 + 1, \varepsilon_0 + 2, \varepsilon_0 + 3, ..., \varepsilon_0 + \omega, ...\varepsilon_0 \cdot 2, ...\varepsilon_0 \cdot \omega, ...\varepsilon_0 \cdot \varepsilon_0 =: \varepsilon_0^2$$
$$, ...\varepsilon_0^3, ...\varepsilon_0^\omega, ..., \varepsilon_0^{\omega^2}, ..., \varepsilon_0^{\omega^\omega}, ...\varepsilon_0^{\omega^{\omega^{\cdot^{\cdot}}}} =: \varepsilon_0^{\varepsilon_0}, ...\varepsilon_0^{\varepsilon_0^{\varepsilon_0}}, ..., \varepsilon_0^{\varepsilon_0^{\varepsilon_0^{\varepsilon_0}}}, ..., \varepsilon_1 = \varepsilon_0^{\varepsilon_0^{\varepsilon_0^{\cdot^{\cdot}}}}$$
$$, ..., \varepsilon_2, ...\varepsilon_3, ..., \varepsilon_\omega, ..., \varepsilon_{\varepsilon_0}, ...\varepsilon_{\varepsilon_{\varepsilon_0}}, ..., \varepsilon_{\varepsilon_{\cdot_{\cdot}}}, ...$$

Note. All the ordinals above are countable! By this we mean, of course, that they are order-types of countable well-ordered sets.

Theorem 2.12. *There exists an uncountable ordinal.*

Idea: If there is an uncountable ordinal (like $\mathcal{P}(\mathbb{N})$) , then there is a least one, say α is the least uncountable ordinal. Then I_α is the set of countable ordinals, *i.e.*, the set of order-types of well-orderings of subsets of $\mathbb{N}$.

Proof. Since any countable well-ordered set can be relabeled using subset of natural numbers, meaning that it is order-isomorphic to any proper initial segment M of $\mathbb{N}$ with well-ordering relation $R \subset M \times M$. We can form the set

$$A = \{(M, R) \in \mathcal{P}(\mathbb{N}) \times \mathcal{P}(\mathbb{N} \times \mathbb{N}) : R \text{ is a well-ordering of } M\}$$

of well-orderings of subsets of $\mathbb{N}$.

First, to show that A is indeed set of well-orderings of subsets of $\mathbb{N}$: For $(M, R) \in A$ Note first that $M \in \mathcal{P}(\mathbb{N})$, then $R \subset M \times M \in \mathcal{P}(\mathbb{N} \times \mathbb{N})$. Then the set

$$B = \{\text{order-type(X)}: X \in A\}$$

consists of all the countable ordinals. Set $\omega_1 = \sup B$. If ω_1 is countable, then so is ω_1^+, and hence $\omega_1^+ \in B$. It follows that $\omega_1^+ \leq \omega_1$, which is a contradiction. Hence ω_1 is the first uncountable ordinal. □

Note. The ordinal ω_1 is the least uncountable ordinal. (Indeed, if $\alpha < \omega_1$, then α must not be an upper bound of B, hence there must be an element β in B such that $\alpha < \beta$ and so α is countable.) It follows that every proper initial segment of ω_1 is countable. If $\alpha_1, \alpha_2, ...$ are countable ordinals, then so is $\sup\{\alpha_1, \alpha_2, ...\}$ being order-type of $\bigcup_{n\in\mathbb{N}} I_{\alpha_n}$ where $\sup\{\alpha_1, \alpha_2, ...\}$ is the least-upper bound of $\bigcup_{n\in\mathbb{N}} I_{\alpha_n}$.(Here we are using the fact that a countable union of countable sets is countable.)

Theorem 2.13 (Hartog's Lemma)**.** *For every set X, there is an ordinal γ which does not inject into X.*

Proof. This is a generalization of the above theorem. We form the set

$$W = \{(Y, R) \in \mathcal{P}(X) \times \mathcal{P}(X \times X) : R \text{ is a well-ordering of } Y\}$$

and construct the set of order-types of well-ordered subsets of X:

$$B = \{\text{order-type}(Y, R) : (Y, R) \in W\}$$

We let $\gamma = (\sup B)^+$. Suppose an injection $f : \gamma \to X$ exists, restricting its co-domain to its image yields a bijection $f' : \gamma \to f[\gamma] \subseteq X$.

This turns $(f[\gamma], R)$ into a well-ordered subset of X with order-type γ, so $(f[\gamma], R) \in W$.

Thus, $\gamma \in B$, which implies $\gamma \leq \sup B < (\sup B)^+ = \gamma$, a contradiction.

Which in other word, if such injection exists, then we can you can construct a specific well-ordered subset inside W whose order-type is strictly larger than B, but because it is in W, its order-type must also belong to B. This means an ordinal would have to be strictly smaller than itself, which is contradiction. □

Notation: The least ordinal that does not inject into X is denoted $\gamma(X)$.(So the function $f : \gamma \to X$ must not be injective, else it triggers the Hartog's contradiction).

Type of ordinals Let α be an ordinal. We have two cases according to whether α (or, more precisely, any well-ordered set of order-type α, *e.g.*, I_α) has a greatest element or not.

- If I_α has greatest element β, then $I_\alpha = I_\beta \cup \{\beta\}$, and hence $\alpha = \beta^+$. In this case, we say α is a *successor ordinal*. Note that in this case $\beta = \sup I_\alpha < \alpha$.
- If I_α has no greatest element, then $\alpha = \sup I_\alpha$ and we say α is a *limit ordinal*.

Example 2.11. Rather trivially, since $I_0 = \emptyset$, so the statement 'If I_0 has no greatest element , then $0 = \sup I_0$' must be vacuously true, leading 0 is a limit ordinal, whereas $1 = 0^+$ is a successor. Any ordinal n with $0 < n < \omega$ is a successor since any non-empty, finite well-ordered set has a greatest element. Since $\omega = \sup\{n : n < \omega\}$, it follows that ω is a limit. On the other hand, ω^+ is a successor.

2.3 Ordinal Arithmetics

2.3.1 Ordinal Addition

Definition 2.9 (Ordinal addition)**.** For ordinals α, β, we define $\alpha + \beta$ by recursion on β, with α fixed, as follows:

$$\begin{aligned} \alpha + 0 &= \alpha \\ \alpha + \beta^+ &= (\alpha + \beta)^+ \\ \alpha + \lambda &= \sup\{\alpha + \beta : \beta < \lambda\} \text{ for a non-zero limit ordinal } \lambda \end{aligned}$$

Remark. Technically, since the ordinals do not form a set, we need to fix an ordinal γ and define $\alpha + \beta$ for $\beta < \gamma$ by recursion in the well-ordered set I_γ. The definition is then independent of γ by uniqueness of recursion. This justifies the recursive definition above and others given below.

In a similar way, induction on ordinals works even though ordinals do not form a set.

Assume p is a property of ordinals. Then

$$(\forall\alpha)((\forall\beta < \alpha)p(\beta) \Rightarrow p(\alpha)) \implies (\forall\alpha)p(\alpha)$$

To show that above is true, assume in opposition that $(\forall\alpha)((\forall\beta < \alpha)p(\beta) \Rightarrow p(\alpha))$ holds, but for some γ, we have $\neg p(\gamma)$. In particular, we choose $\alpha = \min S$ where $S = \{\beta \leq \gamma : \neg p(\beta)\}$. Note that the set is non-empty by hypothesis that $(\exists\gamma)(\neg p(\gamma))$. Note that I_γ^+ is a set and $S \subset I_\gamma^+$, hence S is a set of ordinals. By Theorem above, S has least element, called α. Since α is the first ordinal such that $\neg p(\alpha)$, so $(\forall b < \alpha)p(\beta)$. From hypothesis, since $(\forall b < \alpha)p(\beta)$, then we have $p(\alpha)$, but $\alpha \in S \implies \neg p(\alpha)$, contradiction.

Example 2.12.

- $0, \omega$ are both limit ordinal. $0^+ = (\sup I_0)^+ = 1$
- $\alpha + 1 = \alpha + 0^+ = (\alpha + 0)^+ = \alpha^+$
- $m + 0 = m$, $m + (n + 1) = m + n^+ = (m + n)^+ = (m + n) + 1$ for all $m, n < \omega$

- $\omega + 1 = \omega^+$, $\omega + 2 = \omega + 1^+ = (\omega + 1)^+ = \omega^{++}$
- $\omega + \omega = \sup\{\omega + \beta : \beta < \omega\} = \sup\{\omega, \omega + 1, \omega + 2, ...\}$
- $1 + \omega = \sup\{1 + \beta : \beta < \omega\} = \sup\{1, 2, 3, ...\} = \omega \neq \omega + 1$, hence ordinal addition is not commutative.

Proposition 2.14. $\beta \leq \gamma \implies \alpha + \beta \leq \alpha + \gamma$

Proof. We fixed α and apply induction on γ. Since γ could be 0, a successor ordinal and also limit ordinal, then we separate into 3 cases:

$\gamma = 0$: if $\beta \leq \gamma$, then $\beta = 0$, it follows that $\alpha + \beta = \alpha + \gamma = \alpha$

$\gamma = \delta^+$ is a successor: (Recall that I_γ has greatest element δ, $\sup I_\gamma = \delta$, and $\delta^+ = \gamma$)

If $\beta \leq \gamma$, then either $\beta = \gamma$, which the result is trivial, or if $\beta < \gamma (\beta \leq \delta)$, and so by induction, (induction hypothesis that for all $\beta \leq \delta$, $\alpha + \beta \leq \alpha + \gamma$)

$$\alpha + \beta \leq \alpha + \delta < (\alpha + \delta)^+ = \alpha + \delta^+ = \alpha + \gamma$$

γ is a non-zero limit ordinal: If $\beta \leq \gamma$, we can again assume that $\beta < \gamma$, and so $\beta < \delta$ for some $\delta < \gamma$ by the definition of a limit ordinal. It follows from induction (induction hypothesis: for all $\beta < \delta, \alpha + \beta \leq \alpha + \delta$), and the definition of $\alpha + \gamma = \sup\{\alpha + \varepsilon : \varepsilon < \gamma\}$ where $\alpha + \delta \in \{\alpha + \varepsilon : \varepsilon < \gamma\}$,

$$\alpha + \beta \leq \alpha + \delta \leq \sup\{\alpha + \varepsilon : \varepsilon < \gamma\} = \alpha + \gamma$$

□

Remark. It follows directly from the result above that $\beta < \gamma \implies \alpha + \beta < \alpha + \gamma$. Indeed, we have $\beta^+ \leq \gamma$, and hence

$$\alpha + \beta < (\alpha + \beta)^+ = \alpha + \beta^+ \leq \alpha + \gamma.$$

Note, however, that $\beta < \gamma$ does not imply $\beta + \alpha < \gamma + \alpha$ in general. For example, we have $0 < 1$, and from the result above, $1 + \omega = \omega$. By induction again, we have $n + \omega = \omega = m + \omega$ for any $n, m \in \mathbb{N}$.

OTOH, $\beta \leq \gamma \implies \beta + \alpha \leq \gamma + \alpha$.

Again, we separate into 3 cases and fixed $\beta \leq \gamma$:

$\alpha = 0$: Trivial.

α is successor ordinal($\alpha = \delta^+$): By induction hypothesis, we assume for any $\delta \leq \alpha, \beta + \delta \leq \gamma + \delta$. Since $\alpha = \delta^+$,

$$\beta + \alpha = \beta + \delta^+ = (\beta + \delta)^+ = (\beta + \delta) + 1 \leq (\gamma + \delta) + 1 = (\gamma + \delta)^+ = \gamma + \delta^+ = \gamma + \alpha$$

α is a limit ordinal: Assume for induction that for any $\varepsilon < \alpha$, $\beta \leq \gamma \implies \beta + \varepsilon \leq \gamma + \varepsilon$.

By Induction hypothesis, $\beta + \varepsilon \leq \gamma + \varepsilon$ for any $\varepsilon < \alpha$. Since $\beta + \varepsilon \leq \gamma + \varepsilon \leq \sup\{\gamma + \delta : \delta < \alpha\} = \gamma + \alpha$, $\gamma + \alpha$ is upper bound of $\beta + \varepsilon$, which we must have $\gamma + \alpha \geq \sup\{\beta + \varepsilon : \varepsilon < \alpha\} = \beta + \alpha$.

Lemma 2.15. *Let S be a non-empty set of ordinals. Then*

$$\alpha + \sup S = \sup\{\alpha + \beta : \beta \in S\}$$

for any ordinal α.

Proof. Let $T = \{\alpha + \beta : \beta \in S\}$. We would like to show that $\alpha + \sup S \leq \sup T$ and $\sup T \leq \alpha + \sup S$.

For $\beta \in S$, we have $\beta \leq \sup S$. By above proposition, $\alpha + \beta \leq \alpha + \sup S$. Since it illustrates that $\alpha + \sup S$ is the upper bound of T, hence we conclude that $\sup T \leq \alpha + \sup S$.

Conversely, we separate into 2 cases:

If S has greatest element γ. Then $\gamma = \sup S$, and by proposition 2.7, for any $\beta \in S, \beta \leq \gamma$, we have $\alpha + \beta \leq \alpha + \gamma$ and $\alpha + \gamma \in T$, so $\alpha + \gamma$ is the greatest element of T. It follows that $\sup T = \alpha + \sup S$.

If S has no greatest element. Then $\lambda = \sup S$ is a non-zero limit ordinal. Indeed, $\lambda \notin S$ as S has no greatest element. So $S \subset I_\lambda$, which indicates that $\lambda = \sup S \leq \sup I_\lambda$. Note we have 2 cases: If I_λ has greatest element $\pi \implies \sup I_\lambda = \pi < \lambda \implies \sup I_\lambda < \lambda \leq \sup I_\lambda$, contradiction. Hence we must have I_λ has no greatest element and $\lambda = \sup I_\lambda$.It follows that $\lambda = \sup I_\lambda$ and that $\lambda > 0, S \neq \emptyset$, then λ is also a limit ordinal. By definition of ordinal addition,

$$\alpha + \sup S (= \lambda) = \sup\{\alpha + \gamma : \gamma < \lambda\}$$

Given that $\gamma < \lambda$, there exists $\beta \in S$ such that $\gamma < \beta$ Hence $\alpha + \gamma \leq \alpha + \beta \leq \sup T$. It follows $\alpha + \sup S \leq \sup T$. □

Proposition 2.16 (Associativity)**.** $\alpha + (\beta + \gamma) = (\alpha + \beta) + \gamma$.

Proof. We proceed by induction on γ with α, β fixed. As usual, there are 3 cases:

$\gamma = 0$: $\alpha + (\beta + 0) = \alpha + \beta = (\alpha + \beta) + 0$

$\gamma = \delta^+$ is a successor: By induction, we get

$$\alpha+(\beta+\gamma) = \alpha+(\beta+\delta^+) = \alpha+(\beta+\delta)^+ = [\alpha+(\beta+\delta)]^+ \overset{I.H.}{=} [(\alpha+\beta)+\delta]^+ = (\alpha+\beta)+\gamma$$

γ is a non-zero limit: We use induction and also Lemma 2.2.

$$\begin{aligned}\alpha + (\beta + \gamma) &= \alpha + \sup\{\beta + \delta : \delta < \gamma\} \\ &= \sup\{\alpha + \beta + \delta : \delta < \gamma\} \\ &= \sup\{\alpha + (\beta + \delta) : \delta < \gamma\} \\ &\overset{I.H.}{=} \sup\{(\alpha + \beta) + \delta : \delta < \gamma\} \\ &= (\alpha + \beta) + \gamma\end{aligned}$$

□

Note. Here we can use Lemma 2.2 since the term $\delta < \gamma$, is the same as $\delta \in I_\gamma$, hence $\delta < \gamma$ and $\delta \in I_\gamma$ are interchangeable, hence we can use Lemma 2.2.

Remark. The definition of ordinal addition we gave above is called *inductive definition*. We now give an alternative.

Definition 2.10 (**Synthetic definition of ordinal addition**)**.** For well-ordered sets X, Y, we write $X \sqcup Y$ for $X \times \{0\} \cup Y \times \{1\}$(the disjoint union of X and Y), well-ordered by the relation

$$(x, i) < (y, j) \iff \begin{cases} \text{either} & (i = j = 0) \wedge (x < y \text{ in } X) \\ \text{or} & (i = j = 1) \wedge (x < y \text{ in } Y) \\ \text{or} & i = 0, j = 1. \end{cases}$$

Informally, it is called 'X followed by Y'. For ordinals α, β, we define $\alpha + \beta$ to be the order-type of $\alpha \sqcup \beta$ or, more precisely, the order-type of $X \sqcup Y$, where X is well-ordered set of order-type α, and Y is well-ordered set of order-type β.

Example 2.13. $X = \{x_1, x_2, x_3\}, Y = \{y_1, y_2\}$, then the set

$$X \times \{0\} \cup Y \times \{1\} = \{(x_1, 0), (x_2, 0), (x_3, 0), (y_1, 1), (y_2, 1)\}$$

we have $(x_1, 0) < (x_2, 0) < (x_3, 0) < (y_1, 1) < (y_2, 1)$. Clearly, the order-type of $X \times \{0\}$ and $Y \times \{1\}$ is still 3 and 2. And from example we see that the order-type of $X \sqcup Y$ is $5 = 3 + 2$.

Remark. In a moment we show that the synthetic and inductive definitions coincide. Now we state the definition of arithmetic in synthetic definitions:

For $\alpha^+ = \alpha + 1$: Suppose that X is a set of order-type α, we need to show that X^+ is order-isomorphic to the set $X \sqcup \{*\}$. Since by definition of $\alpha + 1$, it is the order-type of $\alpha \sqcup \{1\}$, which is the order-type of $X \sqcup \{*\}$, where $X \sqcup \{*\} := X \times \{0\} \cup \{*\} \times \{1\}$

with order$(x,0) < (y,0) \iff x <_X y$,$(x,0) < (*,1)$, $\forall x \in X$. And that $X^+ := X \cup \{\infty\}$, where the order is defined to be $x <_X y, x, y \in X$, and $x < \infty$, $\forall x \in X$. We define $f : X \sqcup 1 \to X^+$ by $\begin{cases} (x,0) \mapsto x & \forall x \in X \\ (*,1) \mapsto \infty & \text{otherwise} \end{cases}$ it is clearly bijective since $X \times \{0\} \to X$ is bijective and that $(*,1)$ to ∞. Now if $u = (x,0), v = (y,0)$, then $u < v \iff x <_X y \iff f(u) < f(v)$. If $u = (x,0), v = (*,1)$,then$u < v$ and $f(u) = u < \infty = f(v)$. Lastly, note that if $v = (x,0), u = (*,1)$, we have $u < v$ and note also $f(u) = \infty \not< x = f(v)$. Hence we have $X \sqcup 1$ is order isomorphic to X^+, and we conclude $\alpha + 1 =$ order-type$(X \sqcup 1) =$ order-type$(X^+) = \alpha^+$.

For $(\alpha+\beta)+\gamma = \alpha+(\beta+\gamma)$: It suffices to show that the sets $X \sqcup (Y \sqcup Z), (X \sqcup Y) \sqcup Z$ are order-isomorphic. Consider

$$(X \sqcup Y) \sqcup Z = \Big(X \times \{0\} \cup Y \times \{1\}\Big) \times \{0\} \cup Z \times \{1\}$$

and

$$X \sqcup (Y \sqcup Z) = X \times \{0\} \cup \Big(Y \times \{0\} \cup Z \times \{1\}\Big) \times \{1\}$$

Note the element in $(X \sqcup Y) \sqcup Z$ lies into 3 category:

- Elements from X $(x \in X)$: $((x,0),0)$
- Elements from Y $(y \in Y)$: $((y,1),0)$
- Elements from Z $(z \in Z)$: $(z,1)$

With $((x,0),0) < ((y,1),0) < (z,1)$

Note the element in $X \sqcup (Y \sqcup Z)$ lies into 3 category:

- Elements from X $(x \in X)$: $(x,0)$
- Elements from Y $(y \in Y)$: $((y,0),1)$
- Elements from Z $(z \in Z)$: $((z,1),1)$

With $(x,0) < (1) < ((z,1),1)$

Define $f : X \sqcup (Y \sqcup Z) \to (X \sqcup Y) \sqcup Z$

$$f : \begin{cases} ((x,0),0) & \text{if } u = (x,0) \text{ for } x \in X \\ ((y,1),0) & \text{if } u = ((y,0),1) \text{ for } y \in Y \\ (z,1) & \text{if } u = ((z,1),1)) \text{ for } z \in Z \end{cases}$$

A moment of thinking concludes that the function is bijective and that we can see it is order-preserving from the ordering in both $X \sqcup (Y \sqcup Z), (X \sqcup Y) \sqcup Z$, it follows that $X \sqcup (Y \sqcup Z), (X \sqcup Y) \sqcup Z$ are order-isomorphic.

For $\beta \leq \gamma, \alpha+\beta \leq \alpha+\gamma$: Suppose that Y is initial segment of Z with Y is of order-type of β and Z is well-ordered set of order-type of γ. We first have $\beta \leq \gamma$. And to show that $X \sqcup Y$ is initial segment of $X \sqcup Z$, for $(x,i) \in X \sqcup Y$, for $(y,j) < (x,i)$, we can only have 3 cases: If $i = j = 0, y <_X x$ in X, then we still have $(y,j) \in X \times \{0\} \subset X \sqcup Y$. If $i = j = 1$, $y <_Y x$ in Y, which trivially that $(y,j) \in Y \times \{1\}$ hence in $X \sqcup Y$. If $i = 1, j = 0$, clearly $(y,j) \in X \times \{0\} \implies (y,j) \in X \sqcup Y$. Hence we conclude $X \sqcup Y$ is initial segment of $X \sqcup Z \implies$ order-type$(X \sqcup Y) \leq$ order-type$(X \sqcup Z) \implies \alpha+\beta \leq \alpha+\gamma$.

Proposition 2.17. *The synthetic and inductive definition of ordinal addition coincide.*

Proof. Let us temporarily define $\alpha \dot{+} \beta$ for synthetic addition and $\alpha + \beta$ for inductive addition. We prove $\alpha \dot{+} \beta = \alpha + \beta$ by induction on β wile keeping α fixed:

$\beta = 0 :$ $\alpha + 0 = \alpha =$order-type$(\alpha \sqcup 0) = \alpha \dot{+} 0$.

$\beta = \delta^+$ is a successor: by induction, we have $\alpha + \beta = (\alpha + \delta^+) = (\alpha \dot{+} \delta^+)$. The latter is the order-type of $(\alpha \sqcup \delta) \sqcup 1$, which is order-isomorphic to $\alpha \sqcup (\delta \sqcup 1)$ and, is order-isomorphic to $\alpha \sqcup \beta$. Hence $\alpha + \beta = \alpha \dot{+} \beta$.

β is a non-zero limit ordinal: By induction, we have

$$\alpha + \beta = \sup\{\alpha + \delta : \delta < \beta\} \overset{I.H.}{=} \sup\{\alpha \dot{+} \delta : \delta < \beta\}$$

For $\delta < \beta, \alpha \dot{+} \delta$ is the order-type of $\alpha \sqcup \delta$, and since the nested $\{\alpha \sqcup \delta : \delta < \beta\}$ is well-ordered set, we take the union of all such set, having $\bigcup_{\delta<\beta}(\alpha \sqcup \delta) = \alpha \sqcup \beta$. Note that $\sup\{\alpha \dot{+} \delta : \delta < \beta\} =$ order-type$(\bigcup_{\delta<\beta}(\alpha \sqcup \delta))$ = order-type$(\alpha \sqcup \beta) = \alpha \dot{+} \beta$, we conclude that $\alpha + \beta = \alpha \dot{+} \beta$. □

2.3.2 Ordinal Multiplication

As for addition, we give both the inductive and synthetic definition. We start with the former:

Definition 2.11 (Inductive Multiplication Definition)**.** Define $\alpha \cdot \beta$ by recursion on β with fixed α:

$$\begin{aligned}
&\alpha \cdot 0 = 0 \\
&\alpha \cdot \beta^+ = \alpha \cdot \beta + \alpha \\
&\alpha \cdot \lambda = \sup\{\alpha \cdot \delta : \delta < \lambda\} \text{ for non-zero limit ordinal } \lambda
\end{aligned}$$

Definition 2.12. The product of two well-ordered set $X \times Y$ is defined to be their

Cartesian Product well-ordered as follows:

$$(x,y) < (u,v) \iff \begin{cases} \text{either} & y < v \text{ in } Y \\ \text{or} & (y = v) \wedge (x < u \text{ in } X) \end{cases}$$

Definition 2.13 (Synthetic Multiplication Definition)**.** The Synthetic Multiplication $\alpha \cdot \beta$ is defined to be the order-type of $\alpha \times \beta$ or, more precisely, the order-type of $X \times Y$, where X is a well-ordered set with order-type α, Y is a well-ordered set with order-type β.

Proposition 2.18. *The synthetic and inductive definitions of ordinal multiplication coincide.*

Proof. □

Example 2.14.

- For $m, n < \omega$, $m \cdot 0 = 0$, $m \cdot (n+1) = m \cdot n^+ = m \cdot n + m$
- $\omega \cdot 2 = \omega \cdot 1^+ = \omega \cdot 1 + \omega = \omega \cdot 0^+ + \omega = (\omega \cdot 0 + \omega) + \omega = \omega + \omega$
- $2 \cdot \omega = \sup\{2 \cdot n : n < \omega\} = \omega$, hence multiplication is not commutative.

Note. we show that $\sup\{2 \cdot n : n < \omega\} = \omega$. Set $S = \{2 \cdot n : n < \omega\}$, since $S \subset \mathbb{N} \implies \sup S \leq \sup \mathbb{N} = \omega$. Suppose that $\gamma < \omega$, then γ must be finite ordinal of order-type γ. Pick $n = \gamma + 1$, also a finite ordinal, then $2n = 2\gamma + 2 > \gamma$. However $2\gamma + 2 \in S$, meaning γ will not be an upper bound of S. So $\sup S$ is the least upper bound.

Remark. The following distributivity law hold: $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$, but $(\alpha + \beta)\gamma = \alpha\gamma + \beta\gamma$ does not hold.

For $\alpha(\beta+\gamma) = \alpha\beta+\alpha\gamma$: Fixed α, β. If $\gamma = 0$, trivial. If $\gamma = \delta^+$, assume as inductive hypothesis that $\alpha(\beta + \delta) = \alpha\beta + \alpha\delta$. Since $\alpha(\beta\gamma) = \alpha(\beta + \delta^+) = \alpha((\beta + \delta)^+) = \alpha(\beta + \delta) + \alpha = \alpha\beta + \alpha\delta + \alpha = \alpha\beta + \alpha\delta^+ = \alpha\beta + \alpha\gamma$. If γ is non-zero limit ordinal: Assume that $\alpha(\beta+\delta) = \alpha\beta+\alpha\delta$ for all $\delta < \gamma$, by definition of $\beta+\gamma = \sup\{\beta+\delta : \delta < \gamma\}$, so $\alpha(\beta + \gamma) = \sup\{\alpha(\beta + \delta) : \delta < \gamma\} = \sup\{\alpha\beta + \alpha\delta : \delta < \gamma\} \overset{Lemma\ 2.2}{=} \alpha\beta + \sup\{\alpha\delta : \delta < \gamma\} = \alpha\beta + \alpha\gamma$.

Note that the $(\alpha+\beta)\gamma \neq \alpha\gamma+\beta\gamma$: Take $\alpha = \beta = 1, \gamma = \omega$. Then $(\alpha+\beta)\gamma = 2\omega = \omega$ and $\alpha\gamma + \beta\gamma = \omega + \omega \neq \omega = (\alpha + \beta)\gamma$.

Properties. The following can be verified either by induction on γ (or by induction

on α for the last implication) or by using the synthetic definition.

$$\alpha(\beta\gamma) = (\alpha\beta)\gamma$$
$$\beta \leq \gamma \Rightarrow \alpha\beta \leq \alpha\gamma$$
$$\beta < \gamma \Rightarrow \alpha\beta < \alpha\gamma \text{ provided that } \alpha > 0$$
$$\beta \leq \gamma \Rightarrow \beta\alpha \leq \gamma\alpha$$

- $\alpha(\beta\gamma) = (\alpha\beta)\gamma$:

  We fixed α, β. If $\gamma = 0$, trivial. If $\gamma = \delta^+$, by induction on γ,$\alpha(\beta\gamma) = \alpha(b\delta^+) = \alpha((\beta\delta) + \beta) = \alpha(\beta\delta) + \alpha\beta \overset{I.H.}{=} (\alpha\beta)\delta + \alpha\beta = (\alpha\beta)\delta^+ = (\alpha\beta)\gamma$. If γ is non-zero limit ordinal: Assume by induction that for any $\delta < \gamma$, $\alpha(\beta\delta) = (\alpha\beta)\delta$. Then $\alpha(\beta\gamma) = \alpha \sup\{\beta\delta : \delta < \gamma\} = \sup\{\alpha(\beta\delta) : \delta < \gamma\} = \sup\{(\alpha\beta)\delta : \delta < \gamma\} = (\alpha\beta)\gamma$.

- $\beta \leq \gamma \Rightarrow \alpha\beta \leq \alpha\gamma$:

  Fix α and β. We proceed by induction on γ.

  $\gamma = 0$: If $\beta \leq 0$, then $\beta = 0$, so $\alpha \cdot 0 \leq \alpha \cdot 0$ holds trivially. $\gamma = \delta^+$: Assume the claim holds for δ. If $\beta = \delta^+$, then $\alpha\beta = \alpha\gamma$. If $\beta < \delta^+$, then $\beta \leq \delta$. By the inductive hypothesis, $\alpha\beta \leq \alpha\delta$. Since $\alpha\delta \leq \alpha\delta + \alpha = \alpha\delta^+ = \alpha\gamma$, we obtain $\alpha\beta \leq \alpha\gamma$. γ is a non-zero limit ordinal: Assume $\alpha\beta \leq \alpha\delta$ for all $\delta < \gamma$ whenever $\beta \leq \delta$. If $\beta = \gamma$, $\alpha\beta \leq \alpha\gamma$ is trivial. If $\beta < \gamma$, we may choose $\delta = \beta < \gamma$. By I.H., $\alpha\beta \leq \alpha\beta$. Since $\beta < \gamma$, $\alpha\beta \in \{\alpha\delta : \delta < \gamma\}$, so $\alpha\beta \leq \sup\{\alpha\delta : \delta < \gamma\} = \alpha\gamma$.

- $\beta < \gamma \Rightarrow \alpha\beta < \alpha\gamma$ provided that $\alpha > 0$:

  $\gamma = \delta^+ > \beta^+$: Since $\delta > \beta$, the induction hypothesis gives $\alpha\beta < \alpha\delta$.By definition of multiplication and because $\alpha > 0$:$\alpha\delta < \alpha\delta + \alpha = \alpha\delta^+$ By transitivity, $\alpha\beta < \alpha\delta < \alpha\delta^+$. $\gamma = \lambda$, a limit ordinal with $\lambda > \beta$: Since λ is a limit ordinal and $\beta < \lambda$, the successor β^+ satisfies $\beta < \beta^+ < \lambda$. Applying the induction hypothesis to β^+, we have $\alpha\beta < \alpha(\beta^+)$. By the definition of multiplication at a limit ordinal: $\alpha\lambda = \sup_{\xi<\lambda}(\alpha\xi)$. Since $\beta^+ < \lambda$, $\alpha(\beta^+)$ is part of the set over which the supremum is taken, so $\alpha(\beta^+) \leq \alpha\lambda$. Combining these gives $\alpha\beta < \alpha(\beta^+) \leq \alpha\lambda$, so $\alpha\beta < \alpha\lambda$.

Note, however, the last inequality cannot be strengthened. For example, $1 < 2$ but $1 \cdot \omega = 2 \cdot \omega = \omega$.

Note.

2.3.3 Ordinal Exponentiation

Definition 2.14. The ordinal exponentiation is defined to be α^β by recursion on β:

$$\begin{aligned} \alpha^0 &= 1 \\ \alpha^{\beta^+} &= \alpha^\beta \cdot \alpha \\ \alpha^\lambda &= \sup\{\alpha^\delta : \delta < \lambda\} \text{ for a non-zero limit ordinal} \end{aligned}$$

Chapter 3

Posets and Zorn's Lemma

3.1 Introduction

Definition 3.1. A partial order set on a set X is a relation $\leq$ on X that is

$$
\begin{aligned}
&reflexive : x \leq x \text{ for all } x \in X;\\
&antisymmetric : \text{ if } x \leq y \text{ and } y \leq x, \text{ then } x = y \text{ for all } x, y \in X;\\
&transitive : \text{if } x \leq y \text{ and } y \leq z, \text{ then } x \leq z \text{ for all } x, y, z \in X.
\end{aligned}
$$

We then define $x < y$ to mean $x \leq y$ and $x \neq y$ for $x, y \in X$. Then $<$ is a relation on X that is irreflexive and transitive.

Definition 3.2. A partially ordered set or poset is a set with a partial order on it.

Example 3.1.

1. Any linearly ordered set is a poset.

   Since a linearly ordered set is a set with relation which must be reflexive, antisymmetric, and also transitive.

2. On $\mathbb{N}$ letting $a \leq b \iff a|b$ is a partial order.

   Since by observation, $a|a$, $\forall a \in \mathbb{N}$, if $a|b, b|a$, then $a = b$ for all $a, b \in X$, and lastly, $a|b, b|c \implies a|c$ for all $a, b, c \in X$.

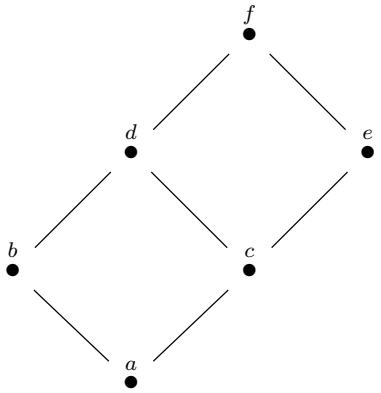
3. For any set S, the power set $X = \mathcal{P}(S)$ is a poset with $a \leq b \iff a \subset b$.

4. Any subset of a poset is a poset by restricting the partial order to the subset.

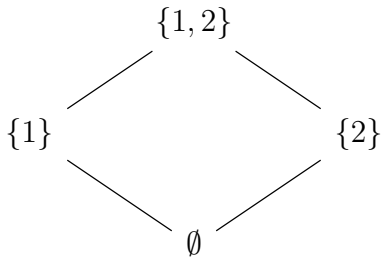
   For example if G is a group, then the subset $\{H \subset G : H \text{ a subgroup of } G\}$ of $\mathcal{P}(G)$ is a poset.

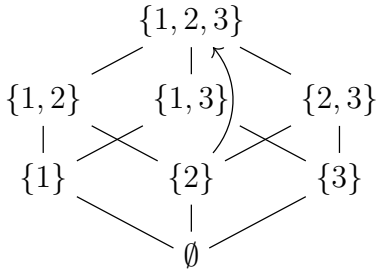
Definition 3.3. We introduce the Hasse diagram: a Hasse diagram of a poset X is a drawing of the points of X with x joined to y with an upward line if y covers x meaning : $x < y$ and there is no z with $x < z < y$.

Example 3.2. 1. For example the example of a poset of $X = \{a, b, c, d, e, f\}$ with $a \leq b, a \leq c, b \leq d, c \leq e, d \leq f, e \leq f$



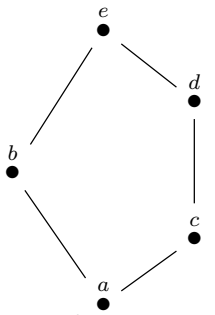
2. For example, here are the drawings of $\mathcal{P}\{1, 2\}$ and $\mathcal{P}\{1, 2, 3\}$:



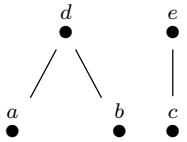


$\mathbb{Q}$ has no Hasse diagram as there is no $x, y \in \mathbb{Q}$ such that y covers x (Since $\mathbb{Q}$ is dense).

3. The following Hasse diagram shows that unlike in the previous examples, in general there is no sensible notion of 'height' or 'rank' in a Hasse diagram.



Note that we cannot identify the 'height' of b if the 'height' of a, c, d, e is $0, 1, 2, 3$, respectively.

4. 

There needs to be no relation between different parts.

5. 

There needs to be no relation at all other than what is necessary by reflexivity.

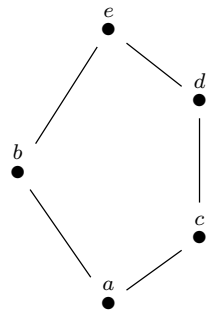
Definition 3.4. A subset S of a poset X is a chain if it is linearly ordered by the partial order of X. (Meaning the partial order is also trichotomy in S.)

Example 3.3. 1. Every subset of a linearly ordered set is a chain.

2. Every subset of a chain in a poset is a chain.

3. In $\mathbb{N}$ with $a \leq b \iff a|b$, the set $\{1, 2, 4, 8, ...\}$ of powers of 2 is a chain.

4. The set $\{\emptyset, \{1\}, \{1, 2\}, \{1, 2, 3\}\}$ is a chain in $\mathcal{P}(\{1, 2, 3\})$. Besides, we also have $\{\emptyset, \{2\}, \{2, 3\}, \{1, 2, 3\}\}$ a chain,$\{\emptyset, \{3\}, \{2, 3\}, \{1, 2, 3\}\}$,$\{\emptyset, \{3\}, \{1, 3\}, \{1, 2, 3\}\}$,... a chain of $\mathcal{P}(\{1, 2, 3\})$. There are total of 6 chains for $\mathcal{P}(\{1, 2, 3\})$.

5. In the poset 

,$\{a, c, d, e\}$ and $\{a, b, e\}$ is a chain.

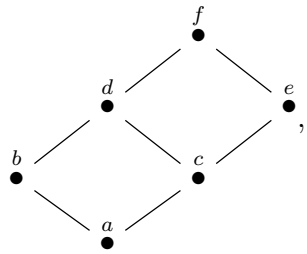
6. The set $\{(\infty, x) \cap \mathbb{Q} : x \in \mathbb{R}\}$ is an uncountable chain in $\mathcal{P}(\mathbb{Q})$.

Definition 3.5. A subset S of a poset X is an antichain if no two distinct members of S are related: for all $x, y \in S$, if $x \leq y$, then $x = y$.

Example 3.4. 1. In a linearly ordered set, since each elements obey the trichotomy of linear order, so each element is comparable. Hence there is no antichain of size > 1.

2. In $\mathbb{N}$ with $a \leq b \iff a|b$, the set of primes is an antichain. For any prime, if $a|p$, we either have $a = 1$ or $a = p$. Since $a = 1 \notin$the set of prime, we conclude that if there exists $a|p$, $a = p$.

3. In $\mathcal{P}(\{1, 2, 3, ..., n\})$, the family $\mathcal{F}_k = \{A \subset \{1, 2, 3, ..., n\} : |A| = k\}$ is an antichain for each $k = 0, 1, 2, ..., n$.

4. In 

, $\{b, c\}, \{d, e\}$ are antichains.

5. In 

the whole set $\{a, b, c, d\}$ is an antichain.

Definition 3.6. Let X be a poset, $S \subset X$ and $x \in X$.

1. x is an upper bound for S if $y \leq x$ for all $y \in S$.

2. x is a least upper bound or supremum for S if x is an upper bound for S and $x \leq y$ for every upper bound y for S.

3. If a supremum for S exists, it is unique and denote $\sup S$ or $\bigvee S$.

Example 3.5. 1. If $S \subset \mathcal{P}(X)$, then $\sup S = \bigcup\{A : A \in S\}$.

Define $x \leq y \iff x \subset y$. Clearly, $\bigcup\{A : A \in S\}$ is an upper bound of S, as if we have $H \in S$, $H \subset \bigcup\{A : A \in S\} \implies H \leq \bigcup\{A : A \in S\}$. Suppose we have another upper bound U of S, since we must have $A \subset U$ for all $A \in S$, then $\bigcup\{A : A \in S\} \subset U \implies \bigcup\{A : A \in S\} \leq U$. We conclude that $\sup S = \bigcup\{A : A \in S\}$.

2. In $\mathbb{R}$, $\sup(0, 1) = 1$ and $\sup[0, 1] = 1$.

3. In $\mathbb{Q}$, the set $S = \{x : x^2 < 2\}$ has an upper bound, *e.g.*, 2, but it has no supremum.

Definition 3.7. A poset is complete if every subset of X has a supremum.

Example 3.6. 1. The real interval $[0, 1]$ in the usual order is complete.

2. $\mathbb{R}$ is not complete since it has not upper bound.

3. $[0,2] \cap \mathbb{Q}$ is not complete since the subset $\{x : x^2 < 2\}$ has no supremum.

4. For any set S, the power set $X = \mathcal{P}(S)$ is complete.

Note. If X is complete: then X has greatest element $\sup X$ and least element $\sup \emptyset$. In particular, X is non-empty.

Note that by completeness, the supremum of X must exists and by definition of supremum, $\sup X \in X$. To show that $\sup \emptyset$ is the least element, note that $u \in X$ is upper bound of $\emptyset$ if $\forall x \in \emptyset, x \leq u$, which holds vacuously for any element in X. And by definition, the smallest element of all $x \in X$ is denoted by $\sup \emptyset$.

Definition 3.8. A function $f : X \to Y$ between posets X, Y is order-preserving if $f(x) \leq_Y f(y)$ whenever $x \leq_X y$.

Example 3.7. 1. $f : \mathbb{N} \to \mathbb{N}, f(n) = n + 1$.

2. $f : \mathcal{P}(S) \to \mathcal{P}(S), f(A) = A \cup B$, where $B \subseteq S$ is fixed. We use $A_1 \leq A_2 \iff A_1 \subseteq A_2, A_1, A_2 \in \mathcal{P}(S)$

   Above is also order=preserving if we change the $f(A) = A \cup B$ into $f(A) = A \cap B$ for fixed $B \subset S$.

Note. If $f : X \to Y$ is order-preserving and injective, then $x <_X y \implies f(x) <_Y f(y)$. The converse holds if X is linearly ordered.

Note that by order-preserving of f, and since $x <_X y$, we have $x \leq_X y \implies f(x) \leq_Y f(y)$. Note since $x \neq y \implies f(x) \neq f(y)$. Hence we conclude $f(x) <_Y f(y)$.

Conversely, assume that X is linearly ordered and $x < y \implies f(x) < f(y)$, let $x, y \in X$ with $x \leq y$, if $x = y, f(x) = f(y)$ so $f(x) \leq f(y)$. If $x < y \implies f(x) < f(y) \implies f(x) \leq f(y)$. To show injectivity, let $x, y \in X$ with $x \neq y$. WLOG assume $x < y$, then $f(x) < f(y) \implies f(x) \neq f(y)$.

Theorem 3.1 (Knaster-Tarski fixed point theorem)**.** *Need Present* *Let X be a complete poset and $f : X \to X$ is order-preserving. Then f has a fixed point.*

Proof. Set $S = \{x \in X : x \leq f(x)\}$ and let $z = \sup S$. We show that $f(z) = z$. For $x \in S$, we have $x \leq z$, and by ordedr-preserving of f, $x \leq f(x) \leq f(z)$. Thus, $f(z)$ is an upper bound for S, and so $z \leq f(z)$. It follows that $f(z) \leq f(f(z))$, and so $f(z) \in S$ and $f(z) \leq z$. □

Corollary 3.2 (Schröder-Bernstein theorem)**.** *Need Present* *Suppose that A, B are sets such that there are injections $f : A \to B$ and $g : B \to A$. Then there is a bijection $h : A \to B$.*

Proof. We seek partitions $A = P \cup Q$ of A and $B = R \cup S$ of B such that $f(P) = R, g(S) = Q$. If we define $h : A \to B$ by

$$h(x) = \begin{cases} f(x) & \text{if } x \in P \\ g^{-1}(x) & \text{if } x \in Q \end{cases}$$

then h is a bijection: if $x, y \in A$ such that $x \neq y$, if $x, y \in P$, then by injectivity of $f|_P : P \to R$, $h(x) = f(x) \neq f(y) = h(y)$. If $x \in P, y \in Q$, then $h(x) = f(x) \in R$ and $h(y) = g^{-1}(y) \in S$. Since $R \cap S = \emptyset$, $h(x) \neq h(y)$. If $x, y \in Q$, $h(x) = g^{-1}(x), h(y) = g^{-1}(y) \in S$. Note that the restriction of g^{-1} is also injective, we conclude $h(x) \neq h(y)$. To show it is surjective, suppose $y \in R$, since $f(P) = R$, then there exists $x \in P$ such that $f(x) = y$, since $x \in P$, $h(x) = f(x) = y$. If $y \in S$, then there exists $x \in Q$ $g(x) = y$, since $x \in Q$, then $h(x) = g^{-1}(x) = g^{-1}(g(y)) = y$.

To show that such partitions exists, define $H : \mathcal{P}(A) \to \mathcal{P}(A)$ by $H(A) = A \backslash g(B \backslash f(P))$. Then H is an order-preserving map on the complete poset $\mathcal{P}(A)$: Since if we have $P_1 \leq P_2$ by $P_1 \subseteq P_2$, then $f(P_1) \subseteq f(P_2) \implies B \backslash f(P_2) \subseteq B \backslash f(P_1) \implies g(B \backslash f(P_2)) \subseteq g(B \backslash f(P_1)) \implies A \backslash g(B \backslash f(P_2)) \subseteq A \backslash g(B \backslash f(P_1))$. Hence by Knaster-Tarski fixed point theorem, H has fixed poitn P. Setting $Q = A \backslash P$, $R = f(P)$ and $S = B \backslash R$, we get the desired partitions.

□

3.2 Zorn's Lemma

Definition 3.9. Let X be a poset. Say $x \in X$ is a maximal element of X if $x \leq y$ implies $x = y$ for every $y \in X$, *i.e.*, there is no y in X such that $x < y$.

Remark. Note there are 3 concepts: Let $(X, \leq_X)$ be a poset and $A \subseteq X$

$$\begin{cases} x \in A \text{ is maximum(greatest element) of } A \text{ if } y \leq x \ \forall y \in A \\ x \in A \text{ is a maximal element of } A \text{ if } x \leq y \implies x = y \ \forall y \in A \\ x \in X \text{ is supremum of } A \text{ if } x \text{ is the least upper bound of } A \text{ in } X \end{cases}$$

A greatest element of A is automatically maximal if it exists and it is also supremum of A. But maximal element might not be a greatest element of A.

Example 3.8.

1. For a set S, the poset $X = \mathcal{P}(S)$ has maximal element X. In fact, X is a maximum element.

2. In general, a greatest element is maximal, but the converse is false in general. For example we have the poset $A = \{a, b, c, d\}$ with $a \leq c, a \leq d, b \leq c, b \leq d$.

Note that we have c, d the maximal since there is no $y \in A$ such that $c, d < y$. But A has no greatest element since $c \not\leq d, d \not\leq c$.

3. In $\mathbb{N}$ with $a \leq b \iff a|b$, there are no maximal elements.

Theorem 3.3 (Zorn's Lemma)**.** *Let X be a (non-empty) poset in which every chain has an upper bound. Then X has a maximal element.*

Remark. The empty set is a chain in X, so it has upper bound by assumption. Thus $X \neq \emptyset$. However, we sometimes verify the conditions of Zorn's lemma by checking that $X \neq \emptyset$ and that every non-empty chain has an upper bound.

Proof. Assume that X has no maximal elements. Then for every $x \in X$ we can fix an element $x' \in X$ with $x < x'$. Let us also fix, for every chain C, an upper bound $u(C)$ for C in X. Let $\gamma = \gamma(X)$ from Hartogs' Lemma and define $f : \gamma \to X$ by recursion as follows:

$$\begin{aligned} f(0) &= u(\emptyset) \\ f(\alpha + 1) &= f(\alpha)' \\ f(\lambda) &= u(\{f(\alpha) : \alpha < \lambda\}) \ , \lambda \neq 0 \text{ limit ordinal} \end{aligned}$$

An easy induction (on β with α fixed) shows that $f(\alpha) < f(\beta)$ for all $\alpha < \beta$. It follows that f is injective contradicting the choice of γ. □

Theorem 3.4. *Every vector space V has a basis.*

Proof. We seek a maximal(w.r.t. inclusion) linearly independent set B. Any such set is a basis, *i..e*, spans V otherwise for any $x \in V \backslash \text{span}(B)$, the set $B \cup \{x\}$ is linearly independent contradicting the maximality of B.

Let $X = \{A \subset V : A \text{ is linearly independent}\}$ partially ordered by inclusion. Let $C = \{A_i : i \in I\}$ be a chain in X. We show that $A = \bigcup_{i \in I} A_i$ is linearly independent. Then A is an upper bound for C. So assume that $\sum_{j=1}^{n} \lambda_j x_j = 0$ is a linear relation on A. For each $j = 1, ..., n$, we have $x_j \in A_{i_j}$ for some $i_j \in I$. Since C is a chain, there is a $k, 1 \leq k \leq n$, such that $A_{i_j} \subset A_{i_k}$ for each $j = 1, 2, ..., n$. Since A_{i_k} is linearly independent, it follows that $\lambda_j = 0$ for all j, and hence A is linearly independent.

We showed that every chain in X has an upper bound. By Zorn's Lemma, X has a maximal element. □

Remark.

1. A similar argument shows that every linearly independent subset B_0 of V is contained in a basis B of V.

2. $\mathbb{R}$ is a vector space over the field $\mathbb{Q}$. A basis of $\mathbb{R}$ over $\mathbb{Q}$ is called a *Hamel Basis*.

3. The real vector space $\mathbb{R}^{\mathbb{N}}$ of all real sequences has no countable basis. By Theorem 3.3, a basis of $\mathbb{R}^{\mathbb{N}}$ does exist.

4. There are many examples for the use of Zorn's Lemma in different areas of mathematics. *E.g.*, the existence of maximal ideals in rings with 1, the existence of continuous linear functionals in normed spaces, the compactness of a product of compact topological spaces(Tychonov's theorem).

Theorem 3.5 (Model Existence Lemma (countable+uncountable))**.** *Let P be any set of primitive propositions. Let $S \subset L = L(P)$ be consistent. Then there exists $\bar{S} \subset L$ such that $S \subset \bar{S}$ and for all $t \in L$, either $t \in \bar{S}$ or $\neg t \in \bar{S}$.*

Proof. We seek a maximal (w.r.t. inclusion) consistent set $\bar{S} \supset S$. We then complete the proof as follows. Given $t \in L$, one of $\bar{S} \cup \{t\}$ and $\bar{S} \cup \{\neg t\}$ is consistent, else $\bar{S} \cup \{t\} \vdash \bot, \bar{S} \cup \{\neg t\} \vdash \bot \implies \bar{S} \vdash \neg t, \bar{S} \vdash \neg\neg t \implies \bar{S} \vdash \bot$. It follows by maximality of $\bar{S}$ that either $t \in \bar{S}$ or $\neg t \in \bar{S}$.

Let $X = \{T \subset L : S \subset T, T \text{ is consistent}\}$ partially ordered by inclusion. Note that $X \neq \emptyset$ since $S \subset X$. Next, let $C = \{T_i : i \in I\}$ be a non-empty chain in X. We show that $T = \bigcup_{i \in I} T_i$ is an upper bound of C. We have $S \subset T$ (as $C \neq \emptyset$) and $T_i \subset T$ for all $i \in I$. So we just need to show that $T \in X, i.e.$, T is consistent. If $T \vdash \bot$, then since proofs are finite, in particular, we can only use finitely many members in T in chain of proof, where if we list out all T_k's in T which involves in the proof, there exists $i_1, ..., i_n$ in I such that $\bigcup_{j=1}^{n} T_{i_j} \vdash \bot$. Since C is a chain, for some $k = 1, ..., n$, we have $\bigcup_{j=1}^{n} T_{i_j} = T_{i_k} \vdash \bot$, contradicting T_{i_k} is consistent. Hence T is an upper bound of C.

By Zorn's Lemma, X has a maximal element required. Since we find such $\bar{S}$, following the similar argument from chapter 1 Theorem 1.1, we can construct a model of $\bar{S}$, which is also a model of S. □

Theorem 3.6 (Well-ordering Principle (WO))**.** *Every set A can be well-ordered.*

Proof. Let $X = \{(B, R) \in \mathcal{P}(A) \times \mathcal{P}(A \times A) : R \text{ is a well-ordering on } B\}$. We partially order X by $(B_1, R_1) \leq (B_2, R_2) \iff B_2$ extends $B_1 (i.e., B_1 \subset B_2, R_1 = R_2 \cap (B_1 \times B_1)$ and B_1 is an initial segment of $B_2)$.

Note that $X \neq \emptyset$ as $(\emptyset, \emptyset) \in X$. Assume that $\{(B_i, R_i) : i \in I\}$ is a nonempty chain in X, *i.e.*, a nested set of well-ordered sets. Then $(\bigcup_{i \in I} B_i, \bigcup_{i \in I} R_i)$ is an upper bound (from chapter 2 Proposition 2.5).

By Zorn's Lemma, there is a maximal element (B, R) in X. If $B \neq A$, then for any $x \in A \backslash B$, the extension $(B, R)^+ = (B \cup \{x\}, R \cup \{(b, x) : b \in B\})$ of (B, R) contradicts the maximality of (B, R). This, $A = B$ is well-ordered by R. □

Example 3.9. $\mathbb{R}$ can be well-ordered.

Remark. In applications of Zorn's Lemma, like the previous example, the maximal object whose existence it asserts cannot be described explicitly.

3.3 The Axiom of Choice (AC)

Note. In the proof of Zorn's Lemma we used two functions: $' : X \to X, x \mapsto x' \in \{y \in X : x < y\}$, and $u : \{C \subset X : C \text{ is a chain}\} \to X$ with $u(C) \in \{x \in X : x \text{ is an upper bound for } C\}$. These are examples of choice functions. Another example from IA Numbers and Sets: to show that $\bigcup_{n\in\mathbb{N}} A_n$ is countable if each A_n is countable, we chose, for each $n \in \mathbb{N}$ an injection $f_n : A_n \to \mathbb{N}$.

Axiom of Choice (AC). This is the assertion for every set X of non-empty sets, $X = \{A_i : i \in I\}$, there is a function $f : I \to \bigcup_{i\in I} A_i$ with $f(i) \in A_i$ for all $i \in I$, called a *choice function* for X.

Note. Note. This rule differs in character from other rules for building sets (e.g., union, power set) in that the object whose existence it asserts is not unique. It is therefore often of interest whether a result whose proof uses **AC** can be proved without **AC**. We show in a moment that **ZL** and **WO** both need **AC**.

Theorem 3.7. ***AC*** $\iff$ ***OW*** $\iff$ ***ZL***

Proof. From Theorem 3.2, we have shown **AC** $\implies$ **ZL** and from Theorem 3.5, we have shown **ZL** $\implies$ **WO**. To show **WO** $\implies$ **AC** , let $X = \{A_i : i \in I\}$ be a set of non-empty sets. Let $Y = \bigcup_{i\in I} A_i$ and fix a well-ordering of Y. Define $f : I \to Y$ by letting $f(i)$ be the least element of A_i. Then f is a choice function of X. □

Chapter 4

First-Order Predicate Logic

4.1 Introduction

In Propositional Logic, we had a set P of primitive propositions and we combined them using $\bot, \Rightarrow$ (and later $\wedge, \vee, \neg, \top, \iff$) to build the language $L = L(P)$ of all (compound) propositions. The primitive propositions, however, had no meanings attached to them.

Our aim now is to describe a wide variety of mathematical theories. We replace primitive propositions with statements like

$$m(x, m(y, z)) = m(m(x, y), z), \quad m(x, i(x)) = e$$

in the language of groups, or $x \leq y$ in the language of posets. These statements are built using variables (like $x, y, z, ...$ above), operation symbols (like the binary symbol m, the unary symbol i and the nullary symbol, *i.e.*, constant, e) and predicate symbols (like the binary predicate $\leq$).

We then combine these statements into formulae. For example, in the language of groups we have

$$(\forall x)(\forall y)(\forall z)(m(x, m(y, z)) = m(m(x, y), z))$$

to describe associativity, or

$$(\forall x)(m(x, x) = e) \Rightarrow (\forall x)(\forall y)(m(x, y) = m(y, x))$$

to describe the statement you proved in IA Groups that if every non-identity element has order 2, then the group is abelian. An example from the language of posets is the sentence

$$(\forall x)(\forall y)(\forall z)(((x \leq y) \wedge (y \leq z)) \Rightarrow (x \leq z))$$

to describe transitivity.

In Propositional Logic we had valuations. A valuation v can be thought of as a choice of constant $v(p)$, for every proposition p, from the set $\{0, 1\}$. In turn, a choice of constant from $\{0, 1\}$ is a function from a singleton set to $\{0, 1\}$. In first-order logic

we will have a structure which will be a set A together with a choice of function $p_A : A^n \to \{0, 1\}$ for every formula p, where n is the number of variables in p. For a set S of formulae, we will define the notion of model for S and we will define semantic and syntactic consequences of S in a way very similar to what we did in Chapter 1. It is now time to give the formal definitions.

4.2 The Language

Definition 4.1 (The Language)**.** The language is specified by a disjoint pair of sets: the set Ω of operation symbols and the set Π of predicate symbols together with an arity function $\alpha : \Omega \cup \Pi \to \mathbb{N}_0 = \mathbb{N} \cup \{0\}$. The language $L = L(\Omega, \Pi)$ then consists of the following:

Variables. This is a countable set disjoint from Ω and Π. We usually denote variables as $x_1, x_2, ...$ or $x, y, z, ...$

Terms. Also called $\Omega-$terms, these are defined inductively as follows:

1. Every variable is a term.

2. If $\omega \in \Omega, n = \alpha(\omega)$ and $t_1, ..., t_n$ are terms, then $\omega t_1 t_2 \cdots t_n$ is also a term.

Example 4.1. the language of groups consists of $\Omega = \{m, i, e\}$ and $\Pi = \emptyset$ with arities $\alpha(m) = 2, \alpha(i) = 1, \alpha(e) = 0$. Here where $'m'$ stands for the multiplication on group elements, $'i'$ stands for the inverse of element in group and $'e'$ stands for identity of group. The reason why $\alpha(m) = 2, \alpha(i) = 1, \alpha(e) = 0$ is that $'m'$ takes 2 elements from group to form an operation, while $'i'$ and $'e'$ takes 1 and 0, respectively. The following then are examples of terms in the language of groups.

$$mxmyz, \quad mmxyz, \quad mxe, \quad mxix, \quad mee$$

Note that brackets are not needed as their positions are uniquely determined by the order of operation symbols and variables. The following strings of operation symbols and variables are not terms: $mxymz, emx, mxyz$.

Note that for example the first one: $mxmyz$, since $\alpha(m) = 2$, and y, z are terms, we have myz is also a term. And note that since x, myz are terms, $m\ x\ myz$ is a term. So in a cleaner version $mxmyz$ stands for $m(x, m(y, z))$.

While $mxymz$ is not a term since $\alpha(m) = 2$, it takes at least 2 terms for m to construct a new term, hence mz is not a term. Same issue for $mxyz$.

Note. Notice that $mxyz$ is not a term but $mmxyz$ is a term, since m takes only 2 terms, so $mxyz$ along cannot be a term, but for $mmxyz$, the latter m takes only the terms x, y, while the first m takes the terms mxy, z.

Definition 4.2 (**Atomic formulae**)**.** These are two kinds:

1. If s, t are terms, then $(s = t)$ is an atomic formula.
2. If $\phi \in \Pi, \alpha(\phi) = n$ and $t_1, t_2, ..., t_n$ are terms, then $\phi t_1 t_2 \cdots t_n$ is an atomic formula.

Example 4.2. the language of posets consists of $\Omega = \emptyset$ and $\Pi = \{\leq\}$ with arity $\alpha(\leq) = 2$. The following are then atomic formulae in the language of posets.

$$(x_1 = x_2), \quad (x_1 \leq x_2)$$

The latter example should really be written as $\leq x_1 x_2$. However, here and below we shall often revert to the more conventional way of writing algebraic expressions which then may require the insertion of brackets to avoid ambiguity.

Remark. Note that the Atomic formulae are Not defined to be a term. Hence we cannot construct a compound atomic formula from 2 atomic formulae like $(x_1 \leq x_2) \leq x_3 : (\leq\leq x_1 x_2 x_3)$.

Example 4.3. We combine the above atomic formulae and terms together to give a more informative example.

Let $\Omega = \{f, m, a\}, \Pi = \{\leq\}$, and $\alpha(f) = 1, \alpha(m) = \alpha(a) = 2, \alpha(\leq) = 2$, and $\mathcal{V} = \{x_1, x_2, ...\}$ be set of variables. Then the following are terms and atomic formulae:

1. $amx_1x_2x_3 : a(m(x_1, x_2), x_3)$ is a term
2. $famx_1x_2x_3 : f(a(m(x_1, x_2), x_3))$ is a term
3. $\leq famx_1x_2x_3x_4 : f(a(m(x_1, x_2), x_3)) \leq x_4$ is a atomic formula

Note that the usage of () are not needed. For example we have $(x_1 \cdot x_2) \cdot x_3$ vs $x_1 \cdot (x_2 \cdot x_3)$, since $(x_1 \cdot x_2) \cdot x_3 \to mmx_1x_2x_3$ and $x_1 \cdot (x_2 \cdot x_3) \to mx_1mx_2x_3$.

Definition 4.3 (**Formulae**)**.** These are defined inductively as follows:

1. Atomic formulae are formulae.
2. $\bot$ is a formula.
3. If p, q are formulae, the $(p \Rightarrow q)$ is a formula. As before, we introduce $\wedge, \vee, \neg, \top, \iff$ in order to abbreviate certain formulae.
4. If p is a formula and x is a free variable in p, then $(\forall x)p$ is a formula. We also introduce $(\exists x)p$ as an abbreviation for $\neg(\forall x)\neg p$.

To explain clause 4, note that a formula is a string of symbols from $\Omega \cup \Pi$, from variables and from the set $\{\bot, \Rightarrow, (,), =, \forall\}$ (and other symbols introduced as abbreviations). An occurrence of a variable inside a term or inside a term as part of formula is either *free* or *bound*, defined inductively as follows:

Definition 4.4 (Free and bounded variable)**.** 1. Any occurrence of a variable inside any term or in any atomic formula is free.

2. Given formulae p, q, an occurrence of a variable in p or q remains an occurrence of the same type in the formula $(p \Rightarrow q)$.

3. If q is a formula containing free occurrences of a variable x, then those occurrences become bound in the formula $(\forall x)q$ (all other occurrences of a variables in q remain unchanged in $(\forall x)q$).

4. For a formula p, we denote by $FV(p)$, the set of free variables of p, *i.e.*, the set of variables that have at least one free occurrence in p.

Example 4.4. 1. $P(x) \vee (\exists x)Q(x)$, the occurrence of x in $P(x)$ is free (since $P(x), Q(x)$ is atomic formulae) , while the occurrence of x in $(\exists x)Q(x)$ is bound (by 3). While if we have $R(x) \Rightarrow (P(x) \vee (\exists x)Q(x))$, the occurrence of x in $R(x), P(x)$ are sill free by 2, and the occurrence of x in $Q(x)$ are still bound (again by 2).

2. The following are formulae in the language of groups. In each case we indicate for every occurrence of the variable x whether it is a free or bound occurrence.

$$\underset{\text{free}}{(mxix} = e) \Rightarrow (\underset{\text{free}}{mixx} = e)$$

$$(\forall x)(\underset{\text{free}}{mxx} = e) \Rightarrow (\forall x)(\forall y)(\underset{\text{bound}}{mxy} = \underset{\text{bound}}{myx})$$

$$\underset{\text{free}}{(mxx} = y) \Rightarrow \neg(\exists x)(mm\underset{\text{bound}}{xxx} = y)$$

Note that in the last example, the variable x has both free and bound occurrences. Although such formulae are technically allowed, it is usual mathematical practice to avoid them. In the second example, there are no free variables: both x and y only have bound occurrences. Such formulae have a special name.

Definition 4.5. A formula with no free variables is called a sentence.

Example 4.5. Here below are some examples of sentence.

1. $(\forall x)(mxx = e) \Rightarrow (\forall x)(\forall y)(mxy = myx)$
2. $(\forall x)R(x)(\Rightarrow (\forall y)P(y) \vee (\exists x)Q(x))$
3. $(\forall x_1)(\forall x_2)(\exists x_3)(\forall x_4)(\leq famx_1x_2x_3x_4)$

4.3 Structures

Definition 4.6. A structure in a language $L = L(\Omega, \Pi)$, or $L-$structure, is a non-empty set A together with functions

$$\omega_A : A^n \to A, \quad (\omega \in \Omega, n = \alpha(\omega))$$

and subsets $\varphi_A \subset A^n$ or equivalently, identifying a subset with its indicator function, functions

$$\varphi_A : A^n \to \{0, 1\}, \quad (\varphi \in \Pi, n = \alpha(\omega))$$

Note. An operation symbol ω of arity 0 is called a constant. Its interpretation in a structure A is a function ω_A from A^0, a singleton set, to A, *i.e.*, ω_A is simply an element of A.

Remark. Note that A^0 is defined to be $A^\emptyset := \{f | f : \emptyset \to A\}$. Since a function $\emptyset$ is defined to be the only subset of $\emptyset \times A$ such that

- If $(x \in \emptyset)$, then $(\exists a \in A)$ such that $(x, y) \in \emptyset$
- If $(x, a_1) \in \emptyset, (x, a_2) \in \emptyset$, then $a_1 = a_2$

Note above are vacuously true as there cannot have element in empty set. Hence we conclude that $A^0 = \{\emptyset\}$. So $\omega_A : A^0 \to A$ is a map $\emptyset \mapsto a$ has only 1 input $\emptyset$. So $\omega_A(\emptyset) = a$ for any input. Hence it is an element in A.

Note. In normal mathematical practice, we allow the empty set to be a structure. Its exclusion here is a simplifying assumption (see later for an explanation).

Example 4.6.

1. If L is the language of groups, then an $L-$ structure is a set A together with functions

   $$m_A : A \times A \to A, \quad i_A : A \to A$$

   and an element $e_A \in A$. Note that A is not a group yet.

2. If L is the language of posets, then an $L-$structure is a non-empty set A together with a subset $\leq_A$ of $A \times A$, *i.e.*, a relation on A. Again, note that A is not yet a poset.

3. If L is the language described in Example 4.3, then an $L-$structure is a non-empty set A together with functions

$$f_A : A \to A, \quad m_A : A^2 \to A \quad a_A : A^2 \to A, \quad \leq_A \subset A^2 : A \times A \to \{0,1\}$$

and it have arity 0 symbol, so no constant in the structure A.

Motivation. Let $L = L(\Omega, \Pi)$ be a first-order language, A be an $L-$structure. Given a formula p in the language L, we want to define the interpretation of p in A and what it means that 'p is satisfied in A'.

For example, if p is the formula $(mxix = e)$ in the language of groups, then we let p_A be the subset $p_A := \{a \in A : m_A(a, i_A(a)) = e_A\} \subset A$ of A, and then say that p is satisfied in A if $p_A = A$. Equivalently, identifying p_A with its indicator function, we say p is satisfied in A if $p_A : A \to \{0,1\}$ is the constant function with value 1, where $p_A(a) = 1$ if $m_A(a, i_A(a)) = e_A$, and $p_A(a) = 0$ otherwise.

Meaning that

$\Big($ for a formula $p_A \subset A$, we say p is satisfied in A if $p_A = A$, $\Big) \iff \forall a \in A, p_A(a) = 1$.

If q now is the sentence $(\forall x)p$, then its interpretation in A should be a function $A^0 \to \{0,1\}, i.e., q_A$ is simply an element of $\{0,1\}$. We set

$$q_A = \begin{cases} 1 & \text{if } p_A(a) = 1, \ \forall a \in A \text{ (,i.e., if } p \text{ holds in } A) \\ 0 & \text{otherwise} \end{cases}$$

We now give the formal description of how to interpret formulae in a structure. This is rather dry, and it is best to let these motivating examples be the guide.

Definition 4.7. Let $L = L(\Omega, \Pi)$ be a first-order language and A an $L-$structure. Let t be a term in L and p be a formula in L both with free variables contained in the set $\{x_1, ..., x_n\}$. We define interpretations on terms $t_A : A^n \to A$ of t in A and formulae $p_A : A^n \to \{0,1\}$ (equivalently, $p_A \subset A^n$) of p in A by induction on the language L as follows.

- If t is x_i for some $i = 1, ..., n$, then

$$t_A : A^n \to A, \quad t_A(a_1, ..., a_n) = a_i$$

- If t is $\omega t_1 \cdots t_m$ for some $\omega \in \Omega$ with arity m and terms $t_1, ..., t_m$, then

$$t_A : A^n \to A, \ t_A(a_1, ..., a_n) = \omega_A\big((t_1)_A(a_1, ..., a_n), ..., (t_m)_A(a_1, ..., a_n)\big)$$

[Note: if $\omega \in \Omega$ with $\alpha(\omega) = n$ and t is the term $\omega x_1 \cdots x_n$, then $t_A = \omega_A$.]

- If p is $(u = v)$ for terms u, v, then

$$p_A = \{(a_1, ..., a_n) \in A^n : u_A(a_1, ..., a_n) = v_A(a_1, ..., a_n)\}.$$

- If p is $\varphi t_1 \cdots t_m$ for some $\varphi \in \Pi$ with arity m and terms $t_1, ..., t_m$, then

$$p_A = \{(a_1, ..., a_n) \in A^n : \big((t_1)_A(a_1, ..., a_n), ..., (t_m)_A(a_1, ..., a_n)\big) \in \varphi_A\}$$

or, identifying $\varphi_A \subset A^m$ with its indicator function, $p_A : A^n \to \{0, 1\}$ is given by

$$p_A(a_1, ..., a_n) = \varphi_A\big((t_1)_A(a_1, ..., a_n), ..., (t_m)_A(a_1, ..., a_n)\big)$$

[Note: if $\varphi \in \Pi$ with $n = \alpha(\varphi)$ and p is $\varphi x_1 \cdots x_n$, then $p_A = \varphi_A$.]

- $\bot_A$ is the constant function $A^n \to \{0, 1\}$ with value 0.

- If p is $(q \Rightarrow r)$, then

$$p_A = \{(a_1, ..., a_n) \in A^n : q_A(a_1, ..., a_n) = 0 \text{ or } r_A(a_1, ..., a_n) = 1\}.$$

- If P is $(\forall x_{n+1})q$ with $FV(q) \subset \{x_1, ..., x_n, x_{n+1}\}$, then

$$p_A = \{(a_1, ..., a_n) \in A^n : (a_1, ..., a_n, a_{n+1}) \in q_A \text{ for all } \ a_{n+1} \in A\}.$$

Example 4.7. 1. Suppose we have the language and structure described in Example 4.3 and Example 4.6-3,

Let's set A, and we use the variables x_1, x_2, x_3. To find the interpretation of $t_A = amx_1x_2x_3$ in A, first note that using (1):

$$(x_1)_A(a_1, a_2, a_3) = a_1 \ , \ (x_2)_A(a_1, a_2, a_3) = a_2 \ , \ (x_3)_A(a_1, a_2, a_3) = a_3$$

and using (2) that the term

$$(mx_1x_2)_A(a_1, a_2, a_3) = m_A\big((x_1)_A(a_1, a_2, a_3), (x_2)_A(a_1, a_2, a_3)\big) = m_A\big(a_1, a_2\big)$$

then, by 2 again,

$$(amx_1x_2x_3)_A(a_1, a_2, a_3) = a_A\big((mx_1x_2)_A(a_1, a_2, a_3), (x_3)_A(a_1, a_2, a_3)\big) = a_A\big(m_A(a_1, a_2), a_3\big)$$

Hence the interpretation of $amx_1x_2x_3$ in A is $t_A(a_1, a_2, a_3) = a_A(m_A(a_1, a_2), a_3)$.

2. For the formula $(\forall x_1)(\forall x_2)(\exists x_3)(\forall x_4)(\leq famx_1x_2x_3x_4)$, to find its interpretation in A,

   again we first partite the formula into smaller terms,

   Since we already got $(amx_1x_2x_3)_A(a_1, a_2, a_3) = a_A(m_A(a_1, a_2), a_3)$, so

$$\begin{aligned}(famx_1x_2x_3)_A(a_1, a_2, a_3) &= f_A\Big(\big(amx_1x_2x_3\big)_A(a_1, a_2, a_3)\Big)\\ &= f_A(a_A(m_A(a_1, a_2), a_3))\end{aligned}$$

   And for the term $\leq famx_1x_2x_3x_4$,

$$\begin{aligned}(\leq famx_1x_2x_3x_4)_A(a_1, a_2, a_3) &=\leq_A \Big(\big(famx_1x_2x_3\big)_A(a_1, a_2, a_3), (x_4)_A(a_1, a_2, a_3)\Big)\\ &=\leq_A \big(f_A(a_A(m_A(a_1, a_2), a_3)), a_4\big)\end{aligned}$$

   Or we can use 4,

$$\begin{aligned}(\leq famx_1x_2x_3x_4)_A(a_1, a_2, a_3) &= \{(a_1, a_2, a_3, a_4) \in A^4 :\\ &\quad \big((famx_1x_2x_3)_A(a_1, a_2, a_3), (x_4)_A(a_1, a_2, a_3)\big) \in\leq_A\}\\ &= \{(a_1, a_2, a_3, a_4) \in A^4 : \big(f_A(a_A(m_A(a_1, a_2), a_3)), a_4\big) \in\leq_A\}\end{aligned}$$

   First, let $q =\leq famx_1x_2x_3x_4$,

   $q_A = \{(a_1, a_2, a_3) \in A^3 : \big(f_A(a_A(m_A(a_1, a_2), a_3)), a_4\big) \in\leq_A\}$ or $=\leq_A (f_A(a_A(m_A(a_1, a_2), a_3)), a_4)$

   Now using 7, we define $p = (\forall x_4)q$, since $FV(q) = \{x_1, x_2, x_3, x_4\} \subset \{x_1, x_2, x_3, x_4\}$,
   then

   $p_A = \{(a_1, a_2, a_3) \in A^3 : (a_1, ..., a_4) \in\leq_A \big(f_A(a_A(m_A(a_1, a_2), a_3)), a_4\big)$ for all $a_4 \in A\}$

   Similarly, define $q^{(1)} := p, q_A^{(1)} := p_A$, and $q^{(2)} = (\exists x_3)q^{(1)}$, note that $q^{(2)} = (\exists x_3)q^{(1)} = \neg(\forall x_3)\neg q^{(1)} = \neg(\forall x_3)(q^{(1)} \Rightarrow \bot)$.

   For the term $(q^{(1)} \Rightarrow \bot)$, note that $\bot_A$ is a constant function which always maps

to 0, hence

$$\begin{aligned}(q^{(1)} \Rightarrow \bot)_A &= \{(a_1, a_2, a_3) \in A^3 : q_A^{(1)}(a_1, a_2, a_3) = 0 \text{ or } \bot_A(a_1, a_2, a_3) = 1\} \\ &= \{(a_1, a_2, a_3) \in A^3 : p_A(a_1, a_2, a_3) = 0\}\end{aligned}$$

To interpret $(\forall x_3)(q^{(1)} \Rightarrow \bot)$, note that $FV((q^{(1)} \Rightarrow \bot)) = \{x_1, x_2, x_3\} \subset \{x_1, x_2, x_3\}$, then

$$\begin{aligned}\big((\forall x_3)(q^{(1)} \Rightarrow \bot)\big)_A &= \{(a_1, a_2) \in A^2 : (a_1, a_2, a_3) \in (q^{(1)} \Rightarrow \bot)_A \text{ for all } a_3 \in A\} \\ &= \{(a_1, a_2) \in A^2 : p_A(a_1, a_2, a_3) = 0 \text{ for all } a_3 \in A\}\end{aligned}$$

Hence $q^{(2)} = \neg(\forall x_3)(q^{(1)} \Rightarrow \bot) = \big((\forall x_3)(q^{(1)} \Rightarrow \bot) \Rightarrow \bot\big)$, which is

$$\begin{aligned}q_A^{(2)} &= \{(a_1, a_2) \in A^2 : \big((\forall x_3)(q^{(1)} \Rightarrow \bot)\big)_A(a_1, a_2) = 0 \text{ or } \bot_A(a_1, a_2) = 1\} \\ &= \{(a_1, a_2) \in A^2 : \big((\forall x_3)(q^{(1)} \Rightarrow \bot)\big)_A(a_1, a_2) = 0\}\end{aligned}$$

Now, set $q^{(3)} = (\forall x_2)q^{(2)}$, since $FV(q^{(2)}) = \{x_1, x_2\} \subset \{x_1, x_2\}$, then

$$\begin{aligned}q_A^{(3)} &= \{a_1 \in A : (a_1, a_2) \in q_A^{(2)} \text{ for all } a_2 \in A\} \\ &= \{a_1 \in A : \big((\forall x_3)(q^{(1)} \Rightarrow \bot)\big)_A(a_1, a_2) = 0 \text{ for all } a_2 \in A\}\end{aligned}$$

Finally, set $q^{(4)} = (\forall x_1)q^{(3)}$, since $FV(q^{(3)}) = \{x_1\} \subset \{x_1\}$, then

$$\begin{aligned}q_A^{(4)} &= \{\langle\rangle \in A^0 : a_1 \in q_A^{(3)} \text{ for all } a_1 \in A\} \\ &= \begin{cases} 1 & \text{if } q_A^{(3)}(a_1) = 1 \text{ , } \forall a_1 \in A \\ 0 & \text{otherwise} \end{cases}\end{aligned}$$

But what is $q_A^{(3)}(a_1) = 1$, $\forall a_1 \in A$?

$$\begin{aligned}\forall a_1 \in A,\ q_A^{(3)}(a_1) = 1 &\iff \forall a_1 \in A,\ \forall a_2 \in A,\ q_A^{(2)}(a_1, a_2) = 1 \\ &\iff \forall a_1 \in A,\ \forall a_2 \in A,\ \big((\forall x_3)(q^{(1)} \Rightarrow \bot)\big)_A(a_1, a_2) = 0 \\ &\iff \forall a_1 \in A,\ \forall a_2 \in A,\ \exists a_3 \in A,\ p_A(a_1, a_2, a_3) = 1 \\ &\iff \forall a_1 \in A,\ \forall a_2 \in A,\ \exists a_3 \in A,\ \forall a_4 \in A, \\ &\quad \big(f_A(a_A(m_A(a_1, a_2), a_3)), a_4\big) \in \leq_A\end{aligned}$$

Example 4.8. Suppose L is the language of groups, with an $L-$structure A, recall that the functions

$$m_A : A^2 \to A, \;\; i_A : A \to A$$

and an element $e_A \in A$. Suppose we have the set of free variables $\{x, y\}$ and the term $t = mmymxixe$. To find the interpretations of the term t on A, we will follow the similar process in above example:

First, note that for $a_1, a_2 \in A$

$$x_A(a_1, a_2) = a_1, \;\; y_A(a_1, a_2) = a_2, \;\; e_A(a_1, a_2) = e_A$$

The term $(ix)_A$ are interpret as follows:

$$(ix)_A(a_1, a_2) = i_A\big(x_A(a_1, a_2)\big) = i_A(a_1)$$

The term $mxix$ are interpret as follows:

$$(mxix)_A(a_1, a_2) = m_A\big(x_A(a_1, a_2), (ix)_A(a_1, a_2)\big) = m_A(a_1, i_A(a_1))$$

The term $mymxix$ are then interpret as follows:

$$\begin{aligned}(mymxix)_A &= m_A\big(y_A(a_1, a_2), (mxix)_A((a_1, a_2))\big) \\ &= m_A(a_2, m_A(a_1, i_A(a_1)))\end{aligned}$$

Note that since the term e_A is an element in A. Hence for the term t, yields

$$\begin{aligned}t_A &= m_A\Big(\big(mymxix\big)_A(a_1, a_2), e_A(a_1, a_2)\Big) \\ &= m_A\big(m_A(a_2, m_A(a_1, i_A(a_1))), e_A\big)\end{aligned}$$

If we let $A = \mathbb{Z}$, the interpretation function mapping:

1. $m_{\mathbb{Z}} : \mathbb{Z}^2 \to \mathbb{Z}$ by $m_{\mathbb{Z}}(a_1, a_2) = a_1 + a_2$
2. $i_{\mathbb{Z}} : \mathbb{Z} \to \mathbb{Z}$ by $i_{\mathbb{Z}}(a) = -a$
3. $e_{\mathbb{Z}} = 0$

Then the above term $t = mmymxixe$ becomes

$$\begin{aligned} t_{\mathbb{Z}} &= m_{\mathbb{Z}}\Big(m_{\mathbb{Z}}\big(a_2, m_{\mathbb{Z}}(a_1, i_{\mathbb{Z}}(a_1))\big), e_{\mathbb{Z}}\Big) \\ &= m_{\mathbb{Z}}\Big(m_{\mathbb{Z}}\big(a_2, a_1 - a_1\big), 0\Big) \\ &= m_{\mathbb{Z}}\Big(m_{\mathbb{Z}}\big(a_2, 0\big), 0\Big) \\ &= m_{\mathbb{Z}}(a_2, 0) \\ &= a_2 \end{aligned}$$

Remark. Crucially, a language consists purely of formal symbols and syntactic rules devoid of intrinsic semantic meaning. To endow a language L with meaning, we define an L-structure $\mathcal{A}$ comprised of a underlying domain of actual mathematical objects. Through the structure's interpretation mapping, these abstract symbols and syntactic rules are instantiated as concrete operations and relations, thereby constructing a well-defined mathematical structure.

4.4 Theories and Models

Definition 4.8. Let $L = L(\Omega, \Pi)$ be a first-order language and A be an $L-$structure. Given a formula p in the language L, we say p is satisfied in A (or p holds in A, or p is true in A or A is a model of p) if $p_A = A^n$ or ,equivalently, $p_A : A^n \to \{0, 1\}$ is a constant function with value 1, where n is the **number of free variables** in p.

Note. In other words, p is satisfied in A if $p_A(a_1, \ldots, a_k) = 1$ for all $(a_1, \ldots, a_k) \in A^k$, where k is the number of free variables in p.

Note that k counts only the free variables, not the total number of variables appearing in p. Bound variables are quantified within the formula and evaluated internally; thus, satisfaction requires evaluating whether p_A holds constantly as 1 across all possible assignments to the remaining k free variables.

Note. When p is a sentence, then p_A is a function $A^0 \to \{0, 1\}$, *i.e.*, it can only be either 0 or 1. Thus, p holds in A if and only if $p_A = 1$.

Definition 4.9. A theory in L is a set of sentences in L. A model for a theory T is an $L-$structure in which each sentence in T is satisfied.

Remark. For example if $T = \{p_1, p_2, ..., p_n\}$, where p_i are sentence for all $i = 1, ..., n$. If p_i satisfied in A for all $i = 1, ..., n$: meaning that $(p_i)_A = 1$ for all i, then we say A is a model of T, and A is an $L-$structure in which each sentence in T is satisfied.

Example 4.9. 1. **Theory of groups**

The language of groups has been defined above: it consists of $\Omega = \{m, i, e\}$ and $\Pi = \emptyset$ with the arities of m, i, e being $2, 1, 0$, respectively. The theory of groups is the following set of sentences. (Recall that the group has 4 axioms: if $a, b \in G$, then $a \cdot b \in G$. For $a, b, c \in G$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$. There exists $e \in G$ such that for all $a \in G, a \cdot e = a, e \cdot a = a$. For any $a \in G$, there exists $a^{-1} \in G$ such that $a \cdot a^{-1} = e, a^{-1} \cdot a = e$).

$$\begin{aligned} T = \{&(\forall x)(\forall y)(\forall z)(mxmyz = mmxyz) \ , \\ &(\forall x)(mxe = x \wedge mex = x) \ , \\ &(\forall x)(mxix = e \wedge mixx = e)\} \end{aligned}$$

Here a group is an $L-$structure. Then every model of T is a group (Since the model must satisfy all the axioms of group) and every group is a model of T (Since any group must have group axiom, which is the 3 sentence in the theory, in its language). So groups can be axiomatized as first-order theory

Meaning every $L-$structure that satisfy the above 3 sentences in T are called groups.

For example the $L-$structure $\mathbb{N}$ is not a group, and hence $\mathbb{N}$ is not a model of the theory of groups T (Since it lacks inverse axiom).

2. **Theory of posets**

Recall that a definition of posets is a relation $\leq$ on a set X that is $x \leq x, \ \forall x \in X.(x \leq y) \wedge (y \leq x) \Rightarrow x = y, \ \forall x, y \in X.(x \leq y) \wedge (y \leq z) \Rightarrow x \leq z, \ \forall x, y, z \in X$.

The language is as defined above: $\Omega = \emptyset, \Pi = \{\leq\}$ and the arity of $\leq$ is 2. The theory of posets is then the following set of sentences.

$$\begin{aligned} T = \{&(\forall x)(x \leq x) \ , \\ &(\forall x)(\forall y)\big((x \leq y \wedge y \leq x) \Rightarrow x = y\big) \ , \\ &(\forall x)(\forall y)(\forall z)\big((x \leq y \wedge y \leq z) \Rightarrow x \leq z\big)\} \end{aligned}$$

This axiomatizes posets: every (non-empty) poset is a model of T and every model of T is a poset.

3. **Theory of rings with 1**

   The language consists of $\Omega = \{+, \times, -, 0, 1\}$ with arities $2, 2, 1, 0, 0$, respectively, and $\Pi = \emptyset$. The theory is as follows.

$$\begin{aligned} T = \{&(\forall x)(\forall y)(\forall z)\big((x + (y + z)) = ((x + y) + z)\big) \ , \\ &(\forall x)(\forall y)(x + y = y + x) \ , \\ &(\forall x)(x + 0 = x) \ , \\ &(\forall x)(x + (-x) = 0) \ , \\ &(\forall x(\forall y)(\forall z)\big(x(yz) = (xy)z\big) \ , \\ &(\forall x)(\forall y)(\forall z)\big((x(y + z) = xy + xz) \wedge ((x + y)z = xz + yz)\big) \ , \\ &(\forall x)\big((x1 = x) \wedge (1x = x)\big)\} \end{aligned}$$

   The models of T are precisely the rings with 1. Note that here we reverted to writing $x + y$ instead of $+xy$, xy instead of $x \times y$, etc. This and the theory of group are examples of algebraic theories: the sentences only involve equations; well, they don't actually, but for example the sentence $(\forall x)((x1 = x) \wedge (1x = x))$ can be replaced with the two sentences $(\forall x)(x1 = x)$ and $(\forall x)(1x = x)$, etc.

4. **Theory of fields**

   The language is the same as for rings with 1. The theory is the union of the theory of rings with 1 and the following set of three sentences.

$$\{(\forall x)(\forall y)(\forall z)(xy = yx), \neg(0 = 1), (\forall x)(\neg(x = 0) \Rightarrow (\exists y)(xy = 1))\}$$

   This axiomatizes fields. Note that this is not algebraic theory, and indeed fields cannot be axiomatized as an algebraic theory. This is because every field has at least 2 elements, and it is easy to see that the singleton set is a model for every algebraic theory.

   Note that every field is a model of T_{ring1}, but not every ring with 1 is a field (e.g. $\mathbb{Z}$ is a ring with 1, but not a field since no mult. inverse for $1 \neq 0$).

   But in here, the models of this theory T are precisely all fields.

5. **Graph theory**

   The language consists of $\Omega = \emptyset, \Pi = \{a\}$ with a having arity 2 (a is adjacency predicate). The theory is

$$\{(\forall x)\neg(axx), (\forall x)(\forall y)(axy \Rightarrow ayx)\}$$

So graphs can also be axiomatized as a first-order theory.

6. **Propositional theories**

   This example shows that propositional logic is a special case of predicate logic. Let P be a set of primitive propositions. Set $\Omega = \emptyset, \Pi = P$ with each primitive proposition having arity 0. In the language $L = L(\Omega, \Pi)$ every primitive proposition is an atomic formula, and thus every proposition in $L(P)$ (as defined in Chapter 1) is a formula. An $L-$structure is a nonempty set A together with a function $v : P \to \{0, 1\}$ that maps $p \in P$ to $p_A \in \{0, 1\}$ (as usual we identify a function $A^0 \to \{0, 1\}$ with the value it takes in $\{0, 1\}$). Given a (compound) proposition $t \in L(P)$, its interpretation t_A in A is an element of $\{0, 1\}$ (i.e. a function $A^0 \to \{0, 1\}$), and the map $t \mapsto t_A$ is precisely the extension $v : L(P) \to \{0, 1\}$ of v to $L(P)$ as defined in Chapter 1. Given $S \subset L(P)$, a model of S is a structure A with a function $v : P \to \{0, 1\}$ such that for every $s \in S$, we have $s_A = v(s) = 1$. Thus, the meaning of model coincides with the definition of model in Chapter 1. Note that the underlying set A here is irrelevant.

Summery.

The Language $L = L(\Omega, \Pi)$ consists purely of formal symbols and syntactic rules. Based on that, we construct the $L-$ structure A, which defines mathematical objects, like Groups, Posets, Rings, Propositions based on the Language we build up.

Given a formula $p \in L$, we say that the $L-$ structure A is a model of p if p_A, the interpretation of p at A, if $p_A = A^n$ (Equivalently, $\forall(a_1, ..., a_n) \in A^n,\ p_A(a_1, ..., a_n) = 1$). When p is a sentence, meaning no free variable, A is a model of p if p_A is constant 1 (Since $p_A : A^0 \to \{0, 1\}$).

Note that for a fixed language L, there could be infinitely many $L-$structure A of the language L.

Furthermore, if set S is set of sentences in L, and a sentence $t \in L$, we say $S \models t$ if any model of any sentence in S is also a model of t. Mathematically, $\forall A\big((\forall s \in S, s_A = 1) \Rightarrow t_A = 1\big)$.

4.5 Semantic entailment

Definition 4.10. Given a first-order language $L = L(\Omega, \Pi)$, a set S of sentences in L and a sentence t in L, we say S (semantically) entails t, written $S \models t$, if t holds in every model of S.

Note. Mathematically, the above statement is equivalent to $\forall A((\forall s \in S, s_A = 1) \Rightarrow t_A = 1)$ (Since s, t do not have any free variables).

Example 4.10. 1. **(Abelian Group)** Let T be the theory of groups (in the language of groups). Then

$$T \models \big((\forall x)(xx = e) \Rightarrow (\forall x)(\forall y)(xy = yx)\big)$$

In order to show that for any $L-$structure that is a model of T, is also a model of t. Notice that a $L-$structure that is a model of T is automatically a group G from above example. Hence we need to show that a group satisfying $(\forall x)(xx = e)$, then $(\forall x)(\forall y)(xy = yx)$ (Meaning a group is abelian). Since for any $x \in G, m(ix)e = ix$, by assumption that $mxx = e$, $m(ix)m(xx) = ix \Rightarrow mmixxx = ix \Rightarrow mex = ix \Rightarrow ix = x$. Note that if $a, b \in G$, $a \cdot b \in G$, hence $i(ab) \in A$, now we show $i(mab) = m(ib)(ia) \iff$ show $m(mab)(mibia) = e$. Note that $mmabmibia = mmambibia = mmaeia = maia = e$, and note also that $mibia = me(mibia) = m(mi(mab)(mab))(mibia) = m(imab)m(mab)(mibia) = mimabe = imab$. By above, we conclude that $\forall a, b \in G,\ mab = imab = mibia = mba$ $(i.e.(\forall x)(\forall y)(xy = yx))$.

Or simply, since we have shown that in above, that the model of T are precisely all groups. So to show $T \models t$, we verify t holds in all group. Take arbitrary group G (arbitrary model of T), if we have $xx = e$ in G, then $x = x^{-1}$, hence for $xy \in G$, $(xy) = (xy^{-1}) = y^{-1}x^{-1} = yx$.

2. **(Cancellation Law)** Let T be the theory of fields (in the language of rings with 1). Then

$$T \models (\forall x)\big(\neg(x = 0) \Rightarrow (\forall y)(\forall z)(xy = xz \Rightarrow y = z)\big)$$

Note. We have defined the semantic entailment in the context of set of sentences, in order to enlarge the set that includes free variable also, we shall need some motivations.

Remark. Notice that our basic definition of semantic entailment $S \models t$ applies when S consists of sentences and t is a sentence. If S or t contains free variables, truth in an $\mathcal{L}$-structure $\mathcal{F}$ is evaluated at elements $(a_1, \ldots, a_n) \in F^n$ (as in Definition 4.7).

For example, consider the existence of multiplicative inverses in fields: Let T be the theory of fields (in the language of rings with 1). Let p be the formula $\neg(x = 0)$, let t be the formula $(\exists y)(xy = 1)$, and let $S = T \cup \{p\}$.

From above example we know that model of the theory T is precisely all the fields. So for any field $\mathcal{F} \models T$ and any element $a \in F$:

- By the field axiom, since the sentence $q = (\forall x)(p \Rightarrow t) \in T$, hence q is true in $\mathcal{F}$, which means $q_{\mathcal{F}} = 1$ (here note the sentence has no free variable). Hence, for every $a \in F$, $(p \Rightarrow t)_{\mathcal{F}}(a) = 1$.

- By the truth rule for implication in Definition 4.7:

$$(p \implies t)_{\mathcal{F}}(a) = 1 \iff \big(p_{\mathcal{F}}(a) = 0 \quad \text{or} \quad t_{\mathcal{F}}(a) = 1\big)$$

- If $a \in F$ satisfies $p \in S$ (meaning $p_{\mathcal{F}}(a) = 1$, i.e., $a \neq 0$), then $p_{\mathcal{F}}(a) = 0$ is false. Thus, $t_{\mathcal{F}}(a)$ is forced to be 1.

Thus, any element $a \in F$ that satisfies S in $\mathcal{F}$ also satisfies t.

To formalize this notion of entailment without explicitly evaluating elements $a \in F$ each time, we introduce Definition 4.11, which converts free variables into new constant symbols.

Definition 4.11. Let $L = L(\Omega, \Pi)$ be the first-order language, S be a set of formulae in L and t a formula in L. Introduce a new constant to L for each free variable occurring in $S \cup \{t\}$. For a formula $u \in S \cup \{t\}$, let u' be the sentence in the new language L' obtained from u by replacing each free occurrence of a variable with the corresponding constant and set $S' = \{s' : s \in S\}$. We then say S (semantically) entails t, written $S \models t$, if $S' \models t'$.

Example 4.11. Suppose we have $L = L_{ring1}$. $S = T \cup \{\neg(x = 0)\}, t = (\exists y)(xy = 1)$. Consider $S \cup \{t\} = T \cup \{\neg(x = 0)\} \cup \{(\exists y)(xy = 1)\}$. Here x is free in $S \cup \{t\}$. So we introduce $L' = L \cup \{c_x\}$ where $c_x \neq 0$. For the formula $u = \neg(x = 0)$, We replace x with the constant c_x to obtain $u' = \neg(c_x = 0)$, and $t' = (\exists y)(c_x u = 1)$, $S' = T \cup \{\neg(c_x = 0)\}$. We now show that $S' \models t'$: In every $L'-$structure F' where $c_x^{F'} \neq 0$, the sentence $(\exists y)(c_x^{F'} y = 1)$ must hold true. Hence $S' \models t'$, then $S \models t$.

Definition 4.12. A formula t in a first-order language L is a tautology if $\emptyset \models t$, *i.e.*, if t holds in every $L-$structure.

Note. Meaning that t is true disregarding any arbitrary $L-$structure A.

Note. In the definition of semantic entailment, we substituted constants into formulae. Later we will need more general notion of substitution.

Definition 4.13. Let p be a formula in a first-order language L. If x is a free variable in p and t is a term in L whose variables do not occur bound in p, then $p[t/x]$ is the formula obtained from p by replacing each free occurrence of x with t.

Remark. We can replace x with constant c to get $p[c/x]$.

Definition 4.14. Let p be a formula in a first-order language L. Let $x_1, \ldots, x_n$ be distinct variables, and let $t_1, \ldots, t_n$ be terms in L.

If for every $i \in \{1, \ldots, n\}$, no variable occurring in the term t_i occurs bound in p, then the simultaneous substitution

$$p[t_1/x_1, \ldots, t_n/x_n]$$

is the formula obtained from p by replacing each free occurrence of each variable x_i with its corresponding term t_i in parallel.

Example 4.12. Let p be the formula $(\forall y)(mmxyx = mmyxy)$ in the language of groups.

1. If t is the term mzz, then $p[t/x]$ is $(\forall y)(mmmzzymzz = mmymzzy)$
2. If t is the term mxx, then $p[t/x]$ is $(\forall y)(mmmxxymxx = mmymxxy)$
3. If t is the term myy, then $p[t/x]$ is not defined as y occurs bound in p.

4.6 Syntactic entailment

4.6.1 Axiom

(A1) $p \Rightarrow (q \Rightarrow p)$ (p, q any formulae)

(A2) $(p \Rightarrow (q \Rightarrow r)) \Rightarrow ((p \Rightarrow q) \Rightarrow (p \Rightarrow r))$ (p, q, r any formulae)

(A3) $\neg\neg p \Rightarrow p$ (p any formula)

(A4) $(\forall x)(x = x)$

(A5) $(\forall x)(\forall y)((x = y) \Rightarrow (p \Rightarrow p[y/x]))$ (x, y distinct variables, p a formula, y does not occur bound in p)

(A6) $((\forall x)\ p) \Rightarrow p[t/x]$ (p a formula, x a free variable of p, t a term no variable of which occurs bound in p)

(A7) $(\forall x)(p \Rightarrow q) \Rightarrow (p \Rightarrow (\forall x)q)$ (p, q formulae, x is a free variable in q that does not occur free in p)

Note. All axiom is a tautology.

4.6.2 Rules of deduction

Modus Ponens (MP): from p and $p \Rightarrow q$, we can deduce q.

Generalization (Gen): from p with free variable x, we can deduce $(\forall x)p$ provided that x does not occur free in any premiss used in the proof of p (Meaning x has to occur bound in all premises used to proof p).

Definition 4.15. Let S be a set of formulae and p be a formula. A proof of p from S is a finite sequence $t_1, ..., t_n$ of formulae such that $t_n = p$ and for every i,

1. either t_i is an axiom, or
2. t_i is a premiss (member of S), or
3. t_i follows by modus ponen (MP): $\exists j, k < i$ such that $t_k = (t_j \Rightarrow t_i)$, or
4. t_i follows by generalization (Gen): $\exists j < i$ such that $t_i = (\forall x)t_j$, where x is free variable in t_j but does not occur free in any premiss $t_k, k \leq j$.

We say S proves p, and write $S \vdash p$, if there is a proof from S to p. If S is a theory, p is a sentence and $S \vdash p$, we say p is a theorem of S.

Remark. From the definition of Generalization, we only need to ensure that x occurs bounded in all premiss $s \in S$ **that is used in the finite formal proof** leading to p. In other words, if there is an $s' \in S$ which x occurs free in it, but s' is never used in the proof from S to p, x is still valid of using Generalization.

Remark. We can now explain why a structure in a language L must be non-empty. Suppose we allowed the empty set $\emptyset$ as a valid structure.

Consider the sentence $(\forall x)\neg(x = x)$:

1. For the subformula $q(x) = \neg(x = x)$, its arity is 1, so $q_\emptyset \subseteq \emptyset^1 = \emptyset$. The complement of $\emptyset$ in $\emptyset$ is $\emptyset$, so $q_\emptyset = \emptyset$.
2. Applying Definition 4.7 for the universal quantifier with $n = 0$ free variables:

$$((\forall x)\neg(x = x))_\emptyset = \{() \in \emptyset^0 : (a) \in q_\emptyset \text{ for all } a \in \emptyset\} = \{()\} = 1$$

Thus, $(\forall x)\neg(x = x)$ holds true in $\emptyset$ (Meaning $\emptyset$ is a model of $(\forall x)\neg(x = x)$). Set $S = \{(\forall x)\neg(x = x)\}$, we have $\emptyset$ is a model of the set S of sentence. We show $S \not\models t = \bot$, meaning there is a model A of S such that $\bot_A = 0$. Consider $A = \emptyset$, by definition 4.7 $\bot_\emptyset = 0$. Hence $S \not\models t$. On the other hand, we show $\{(\forall x)\neg(x = x)\} \vdash \bot$:

$$
\begin{aligned}
(\forall x)\neg(x=x) &\quad \text{(premiss)}\\
(\forall x)\neg(x=x)\Rightarrow\neg(x=x) &\quad \text{(A6), set } t=x\\
\neg(x=x) &\quad \text{(MP)}\\
(\forall x)(x=x) &\quad (A4)\\
(\forall x)(x=x)\Rightarrow(x=x) &\quad \text{(A6), set t=x}\\
(x=x) &\quad (MP)\\
(x=x)\Rightarrow\bot &\quad \text{(from } \neg(x=x))\\
\bot &\quad \text{(MP)}
\end{aligned}
$$

Example 4.13. $\{(x=y)\}\vdash(y=x)$

$$
\begin{aligned}
(\forall x)(\forall y)\big((x=y)\Rightarrow((x=z)\Rightarrow(y=z))\big) &\quad \text{(A5), take } p:x=z\\
(\forall x)\Big((\forall y)((x=y)\Rightarrow((x=z)\Rightarrow(y=z)))\Big) &\quad \text{(A6), set t=x}\\
\Rightarrow(\forall y)((x=y)\Rightarrow((x=z)\Rightarrow(y=z))) &\\
(\forall y)((x=y)\Rightarrow((x=z)\Rightarrow(y=z))) &\quad \text{(MP)}\\
(\forall y)\Big((x=y)\Rightarrow((x=z)\Rightarrow(y=z))\Big)\Rightarrow\big((x=y)\Rightarrow((x=z)\Rightarrow(y=z))\big) &\quad \text{(A6), take t=y}\\
\big((x=y)\Rightarrow((x=z)\Rightarrow(y=z))\big) &\quad \text{(MP)}\\
(x=y) &\quad \text{(premiss)}\\
(x=z)\Rightarrow(y=z) &\quad \text{(MP)}\\
(\forall z)\big((x=z)\Rightarrow(y=z)\big) &\quad \text{(Gen)}\\
(\forall z)\big((x=z)\Rightarrow(y=z)\big)\Rightarrow((x=x)\Rightarrow(y=x)) &\quad \text{(A6), take } t=x\\
((x=x)\Rightarrow(y=x)) &\quad \text{(MP)}\\
(\forall x)(x=x) &\quad \text{(A4)}\\
(\forall x)(x=x)\Rightarrow(x=x) &\quad \text{(A6)}\\
(x=x) &\quad \text{(MP)}\\
(y=x) &\quad \text{(MP)}
\end{aligned}
$$

4.7 Relationship between $S\vdash p$ and $S\models p$

Before dive into this topic, i would like to give a useful lemma, which we will encounter it many times in the proves in this topic.

Lemma 4.1. *Let L be a language and S be the set of formulas, and p be a formula. If $S\vdash p$, then there exist a finite subset $S'\subset S$ such that $S'\vdash p$.*

Proof. Suppose that $t_1,...,t_{n-1},t_n=p$ is a finite formal proof sequence from S to p, where for each t_i for $1\leq i\leq n$, it is either axiom or premisses or it follows by MP, or it follows by Gen. Consider $S'=S\cap\{t_1,...,t_n\}$, clearly, $S'\subseteq\{t_1,...,t_n\}$ is finite, and also $S'\vdash t_n=p$ as t_n is a premiss in S'. □

Remark. The above is stated as the Compactness Theorem, stating that only finite premisses in a countable/ uncountable set S are used to contribute to the formal proof.

Lemma 4.2 (Variable Substitution Inversion)**.** *Let p be a formula in a first-order language L, and let x, y be variables in L. If x does not occur in p (neither free nor bound), then:*

$$(p[x/y])[y/x] = p$$

In particular, $(p[x/y])[y/x] \equiv p$. Furthermore, x is free in $p[x/y]$ if and only if y is free in p.

Proof. We first show by structural induction that for any term t in L in which x does not occur, $(t[x/y])[y/x] = t$:

- **Variable $t = z$ ($z \neq y$):** Since x does not occur in t, $z \neq x$. Thus $(z[x/y])[y/x] = z[y/x] = z$.
- **Variable $t = y$:** Since $y[x/y] = x$ and x is a variable, $(y[x/y])[y/x] = x[y/x] = y$.
- **Function Term $t = \omega t_1 \cdots t_m$:** Since x does not occur in t, x does not occur in any subterm t_i. By the induction hypothesis, $(t_i[x/y])[y/x] = t_i$ for each $1 \leq i \leq m$. Thus:

$$((\omega t_1 \cdots t_m)[x/y])[y/x] = \omega((t_1[x/y])[y/x] \cdots (t_m[x/y])[y/x]) = \omega t_1 \cdots t_m = t$$

Now we proceed by structural induction on the formula p:

- **Case 1 (Atomic Formulas):** Let p be $\phi t_1 \cdots t_m$ for an n-ary predicate symbol or equality $\phi \in \Pi \cup \{=\}$. Since x does not occur in p, x does not occur in any term t_i. Applying the term result to each t_i:

$$(p[x/y])[y/x] = \phi((t_1[x/y])[y/x] \cdots (t_n[x/y])[y/x]) = \phi t_1 \cdots t_m = p$$

- **Case 2 (Absurdity):** Let p be $\bot$. Since $\bot$ contains no variables, $(\bot[x/y])[y/x] = \bot = p$.
- **Case 3 (Implication):** Let p be $(q \Rightarrow r)$. Since x does not occur in p, x does not occur in q or r. By the induction hypothesis, $(q[x/y])[y/x] = q$ and $(r[x/y])[y/x] = r$. Thus:

$$((q \Rightarrow r)[x/y])[y/x] = (q[x/y])[y/x] \Rightarrow (r[x/y])[y/x] = q \Rightarrow r = p$$

- **Case 4 (Universal Quantification):** Let p be $(\forall z)q$. Since x does not occur in p, $x \neq z$ and x does not occur in q.

– *Subcase $z = y$:* Since y is bound in p, y has no free occurrences in p. Thus $p[x/y] = p$. Since x does not occur in p, $p[y/x] = p$. Hence $(p[x/y])[y/x] = p$.

– *Subcase $z \neq y$:* Since $z \neq x$ and $z \neq y$, substitution commutes with the quantifier. By the induction hypothesis on q:

$$((\forall z)q)[x/y][y/x] = (\forall z)((q[x/y])[y/x]) = (\forall z)q = p$$

By structural induction, $(p[x/y])[y/x] = p$ holds syntactically for all formulas p, which implies logical equivalence $p[x/y][y/x] \equiv p$.

To show the next result, we first establish an auxiliary result for terms: for any term t in L in which x does not occur, $x \in FV(t[x/y]) \iff y \in FV(t)$. We proceed by structural induction on t:

- **Variable $t = z$ ($z \neq y$):** Since x does not occur in t, $z \neq x$. Then $z[x/y] = z$. We have $FV(z[x/y]) = \{z\}$ and $FV(z) = \{z\}$. Since $z \neq x$ and $z \neq y$, both $x \in FV(z[x/y])$ and $y \in FV(z)$ are false. Hence, $x \in FV(z[x/y]) \iff y \in FV(z)$ holds.

- **Variable $t = y$:** Here $y[x/y] = x$. Thus $FV(y[x/y]) = \{x\}$ and $FV(y) = \{y\}$. Since $x \in \{x\}$ and $y \in \{y\}$, both sides are true, so the equivalence holds.

- **Compound Term $t = \omega t_1 \cdots t_m$:** Since x does not occur in t, x does not occur in any subterm t_i. By definition:

$$x \in FV(\omega t_1 \cdots t_m[x/y]) \iff \exists i \in \{1, \ldots, m\},\ x \in FV(t_i[x/y])$$

By the induction hypothesis on each subterm t_i, we have $x \in FV(t_i[x/y]) \iff y \in FV(t_i)$. Therefore:

$$\begin{aligned} \exists i \in \{1, \ldots, m\},\ x \in FV(t_i[x/y]) &\iff \exists i \in \{1, \ldots, m\},\ y \in FV(t_i) \\ &\iff y \in FV(\omega t_1 \cdots t_m) \end{aligned}$$

Now we proceed by structural induction on the formula p:

1. **Case 1 (Atomic Formulas):** Let p be $\phi t_1 \cdots t_n$ for an n-ary predicate symbol or equality $\phi \in \Pi \cup \{=\}$. Since x does not occur in p, x does not occur in any term t_i. Applying the term result to each t_i:

$$\begin{aligned} x \in FV(\phi t_1 \cdots t_n[x/y]) &\iff \exists i \in \{1, \ldots, n\},\ x \in FV(t_i[x/y]) \\ &\iff \exists i \in \{1, \ldots, n\},\ y \in FV(t_i) \\ &\iff y \in FV(\phi t_1 \cdots t_n) \end{aligned}$$

2. **Case 2 (Absurdity):** Let p be $\bot$. We have $FV(\bot) = \emptyset$ and $FV(\bot[x/y]) = \emptyset$. Thus, $x \in FV(\bot[x/y])$ and $y \in FV(\bot)$ are both false, and the equivalence holds trivially.

3. **Case 3 (Implication):** Let p be $(q \Rightarrow r)$. Since x does not occur in p, x does not occur in q or r. By the induction hypothesis on q and r:

$$\begin{aligned} x \in FV((q \Rightarrow r)[x/y]) &\iff x \in FV(q[x/y]) \lor x \in FV(r[x/y]) \\ &\iff y \in FV(q) \lor y \in FV(r) \\ &\iff y \in FV(q \Rightarrow r) \end{aligned}$$

4. **Case 4 (Universal Quantification):** Let p be $(\forall z)q$. Since x does not occur in p, $z \neq x$ and x does not occur in q.

   - *Subcase $z = y$:* Since y is bound in p, $y \notin FV((\forall y)q)$. By definition of substitution, $((\forall y)q)[x/y] = (\forall y)q$. Since x does not occur in $(\forall y)q$, $x \notin FV((\forall y)q)$. Thus both $x \in FV(p[x/y])$ and $y \in FV(p)$ are false, making the equivalence true.
   - *Subcase $z \neq y$:* Since $z \neq x$ and $z \neq y$, substitution commutes with quantification:

     $$((\forall z)q)[x/y] = (\forall z)(q[x/y])$$

     By definition of free variables in quantified formulas:

     $$x \in FV((\forall z)(q[x/y])) \iff x \in FV(q[x/y]) \setminus \{z\}$$

     Since $x \neq z$, $x \in FV(q[x/y]) \setminus \{z\} \iff x \in FV(q[x/y])$. By the induction hypothesis on q, we have $x \in FV(q[x/y]) \iff y \in FV(q)$. Finally, since $z \neq y$, $y \in FV(q) \iff y \in FV(q) \setminus \{z\} = FV((\forall z)q)$. Combining this sequence of bi-implications yields:

     $$x \in FV(((\forall z)q)[x/y]) \iff y \in FV((\forall z)q)$$

By structural induction, $x \in FV(p[x/y]) \iff y \in FV(p)$ holds for all formulas p.

□

Lemma 4.3 (Substitution Composition for Fresh Constants)**.** *Let A be a formula (or term) in a language L. Let $x_1, \ldots, x_n$ be distinct variables, let $c_1, \ldots, c_n$ be distinct fresh constant symbols not occurring in A, and let $t_1, \ldots, t_n$ be terms. Then:*

$$(A[c_1/x_1, \ldots, c_n/x_n])[t_1/c_1, \ldots, t_n/c_n] = A[t_1/x_1, \ldots, t_n/x_n]$$

In particular, setting $t_i = y_i$ gives:

$$(q[c_1/x_1, \ldots, c_n/x_n])[y_1/c_1, \ldots, y_n/c_n] = q[y_1/x_1, \ldots, y_n/x_n]$$

Proof. Let $\sigma' = [c_1/x_1, \ldots, c_n/x_n]$, $\sigma = [t_1/c_1, \ldots, t_n/c_n]$, and $\rho = [t_1/x_1, \ldots, t_n/x_n]$. We prove $(A\sigma')\sigma = A\rho$ by structural induction.

1. Base Case for Terms:

- **Variable $u = x_i$ for some i:** We have $x_i\sigma' = c_i$. Then $(x_i\sigma')\sigma = c_i[t_1/c_1, \ldots, t_n/c_n] = t_i$. On the right-hand side, $x_i\rho = t_i$. Thus, $(x_i\sigma')\sigma = x_i\rho$.
- **Variable $u = z$ where $z \neq x_i$ for all i:** Since $z \neq x_i$, $z\sigma' = z$. Because z is a variable (and not one of the constants c_j), $z\sigma = z$. On the right-hand side, $z\rho = z$. Thus, $(z\sigma')\sigma = z\rho$.
- **Constant symbol $u = k$ in L:** Since $c_1, \ldots, c_n \notin L$, $k \neq c_j$ for all j. Thus $k\sigma' = k$, $k\sigma = k$, and $k\rho = k$.

2. Induction Step for Compound Terms: Let $u = \omega u_1 \cdots u_m$ for an m-ary function symbol f. By the induction hypothesis, $(u_j\sigma')\sigma = u_j\rho$ for each subterm u_j. Thus:

$$(\omega(u_1 \cdots u_m)\sigma')\sigma = \omega((u_1\sigma')\sigma \cdots (u_m\sigma')\sigma) = \omega(u_1\rho \cdots u_m\rho) = \omega u_1 \cdots u_m\rho$$

3. Induction Step for Formulas:

- **Atomic Formulas $\phi u_1 \cdots u_m$:** Follows directly from the term result on each u_j.
- **Absurdity $\bot$:** $\bot\sigma'\sigma = \bot = \bot\rho$.
- **Implication $(B \Rightarrow C)$:** Follows by applying the induction hypothesis to B and C and using the distributivity of term substitution over logical connectives.
- **Universal Quantification $(\forall z)B$:** Because $c_1, \ldots, c_n$ are constant symbols, quantifiers cannot bind c_i. Furthermore, variable substitution renames bound variables if necessary to avoid capture. By definition of quantifier substitution:

  $$((\forall z)B)\sigma' = (\forall z')(B[z'/z]\sigma')$$

  Applying σ commutes directly past the quantifier because no constant c_i is a variable:

  $$(((\forall z)B)\sigma')\sigma = (\forall z')((B[z'/z]\sigma')\sigma)$$

  By the induction hypothesis on $B[z'/z]$:

  $$(\forall z')((B[z'/z]\sigma')\sigma) = (\forall z')(B[z'/z]\rho) = ((\forall z)B)\rho$$

By structural induction, $(A\sigma')\sigma = A\rho$ holds for all terms and formulas A. □

Lemma 4.4 (Constant Re-substitution)**.** *Let L be a first-order language, S a set of formulas in L, and p a formula in L. Let $\{x_1, \ldots, x_n\}$ be a finite set of distinct variables in L containing all free variables occurring in $S \cup \{p\}$.*

Let $L' = L \cup \{c_1, \ldots, c_n\}$ be an expansion of L by fresh, distinct constant symbols $c_i \notin L$. Define the substituted set of premises $S' = \{q[c_1/x_1, \ldots, c_n/x_n] \mid q \in S\}$ and the target formula $p' = p[c_1/x_1, \ldots, c_n/x_n]$.

Suppose there exists a formal deduction $\pi' = (\psi'_1, \psi'_2, \ldots, \psi'_m)$ of p' from S' in L'. If $y_1, \ldots, y_n$ are distinct variables in L that do not occur (either free or bound) in S, p, or anywhere in the deduction sequence π', then substituting y_i for c_i across every step of π' yields a valid formal deduction $\pi = (\psi_1, \psi_2, \ldots, \psi_m)$ of $p[y_1/x_1, \ldots, y_n/x_n]$ from $S[y_1/x_1, \ldots, y_n/x_n]$ in L.

Proof. Let $\sigma' = [c_1/x_1, \ldots, c_n/x_n]$ and $\sigma = [y_1/c_1, \ldots, y_n/c_n]$. We define $\psi_k = \psi'_k\sigma$ for each $1 \leq k \leq m$. We proceed by induction on the line number k to show that each formula ψ_k in the transformed sequence π is legally justified in L:

Case 1: Premise from S'. If $\psi'_k \in S'$, then $\psi'_k = q\sigma'$ for some premise $q \in S$. Since the constant symbols $c_1, \ldots, c_n$ do not occur in q, applying σ replaces c_i with y_i in the exact positions where x_i originally occurred in q. Thus:

$$\psi_k = (q\sigma')\sigma = q[y_1/x_1, \ldots, y_n/x_n] \in S[y_1/x_1, \ldots, y_n/x_n]$$

Hence, ψ_k is a valid hypothesis in $S[y_1/x_1, \ldots, y_n/x_n]$.

Case 2: Logical Axiom. If ψ'_k is an instance of a first-order axiom schema in L', applying the term substitution σ maps L'-terms to L-terms and L'-formulas to L-formulas. Because the variables $y_1, \ldots, y_n$ are fresh and do not occur in π', no variable y_i is quantified anywhere in π'. Consequently, replacing c_i with y_i cannot cause variable capture in any subterm substitution (such as in universal instantiation axioms $(\forall v\phi \Rightarrow \phi[t/v])$). Thus, $\psi_k = \psi'_k\sigma$ remains a valid instance of the same axiom schema in L.

Case 3: Modus Ponens. If ψ'_k is derived from ψ'_i and $\psi'_j = (\psi'_i \Rightarrow \psi'_k)$ for $i, j < k$, then by the distributive property of substitution over conditional connectives:

$$\psi'_j\sigma = (\psi'_i\sigma \Rightarrow \psi'_k\sigma) = (\psi_i \Rightarrow \psi_k)$$

By the induction hypothesis, ψ_i and $\psi_i \Rightarrow \psi_k$ are valid steps in π. Hence, ψ_k follows legally by Modus Ponens.

Case 4: Generalization. If $\psi'_k = (\forall z)\theta'$ was derived from θ' via Generalization, where z is not free in any premise of S' used up to line k, then applying σ yields:

$$\psi_k = \psi'_k\sigma = (\forall z)(\theta'\sigma)$$

Since $y_1, \ldots, y_n$ are completely fresh to π', the bound variable z cannot be any y_i. Furthermore, because z is not free in the active premises of S', z is not free in the corresponding transformed premises of $S[y_1/x_1, \ldots, y_n/x_n]$. Thus, the side-conditions for Generalization are preserved, and ψ_k is legally derived from $\theta'\sigma$.

Since $\psi'_m = p'$, the final formula of π is $\psi_m = p'\sigma = p[y_1/x_1, \ldots, y_n/x_n]$. Therefore, $\pi = (\psi_1, \ldots, \psi_m)$ is a valid formal deduction of $p[y_1/x_1, \ldots, y_n/x_n]$ from $S[y_1/x_1, \ldots, y_n/x_n]$ in L. □

Proposition 4.5 (Deduction Theorem)**.** *Let S be a set of formulae and p, q be formulae. Then $S \vdash (p \Rightarrow q)$ if and only if $S \cup \{p\} \vdash q$.*

Proof. Assume that $S \vdash (p \Rightarrow q)$. Write down a proof of $(p \Rightarrow q)$ from S and append the lines

$$\begin{aligned} p \quad & \text{(premiss)} \\ p \Rightarrow q \quad & \text{(premiss)} \\ q \quad & \text{(MP)} \end{aligned}$$

o obtain a proof of q from $S \cup \{p\}$.

Now assume that $S \cup \{p\} \vdash q$ and let $t_1, ..., t_n = q$ be a proof of q from $S \cup \{p\}$.

We show that $S \vdash (p \Rightarrow t_i)$ for all i by induction.

Induction hypothesis before the ith step: for each $j < i$, we have $S \vdash (p \Rightarrow t_j)$ in such a way that if a variable x did not occur free in any premiss used in the proof $t_1, ..., t_j$ of t_j from $S \cup \{p\}$, then x does not occur free in any premiss used in the proof of $(p \Rightarrow t_j)$ from S.

We now show $S \vdash (p \Rightarrow t_i)$ by considering a number of cases (The 4 cases).

Case 1: If t_i is an axiom or $t_i \in S$, then

$$\begin{aligned} t_i \Rightarrow (p \Rightarrow t_i) \quad & \text{(A1)} \\ t_i \quad & \text{(axiom or premiss)} \\ p \Rightarrow t_i \quad & \text{(MP)} \end{aligned}$$

is a proof of $p \Rightarrow t_i$ from S.

Case 2: If $t_i = p$, then $S \vdash (p \Rightarrow t_i)$ since $\vdash (p \Rightarrow p)$ (Here $(p \Rightarrow p)$ is a).

Case 3: If there exist $j, k < i$ such that $t_k = (t_j \Rightarrow t_i)$, then by induction hypothesis,

thee are proofs of $p \Rightarrow t_j, p \Rightarrow (t_j \Rightarrow t_i)$ from S. Adding the lines

$$\begin{aligned} \big(p \Rightarrow (t_j \Rightarrow t_i)\big) \Rightarrow \big((p \Rightarrow t_j) \Rightarrow (p \Rightarrow t_i)\big) \quad & \text{(A2)} \\ (p \Rightarrow t_j) \Rightarrow (p \Rightarrow t_i) \quad & \text{(MP)} \\ p \Rightarrow t_i \quad & \text{(MP)} \end{aligned}$$

we obtain a proof of $p \Rightarrow t_i$ from S.

Case 4: Finally, if for some $j < i$ we have $t_i = (\forall x)t_j$, then x occurs free in t_j and x does not occur free in any premiss used in the proof of t_j from $S \cup \{p\}$.

We now have 2 further cases.

Case 4a: If x occurs free in p, then p was not used in the proof of t_j from $S \cup \{p\}$, so instead we have $S \vdash t_j$ from the fact of Case 4, x does not occur free in any premiss used in the proof from S to t_j. We write down this proof and append the lines

$$\begin{aligned} (\forall x)t_j \quad & \text{(Gen): Since } x \text{ occurs bdd in all premiss} \\ (\forall x)t_j \Rightarrow (p \Rightarrow (\forall x)t_j) \quad & \text{(A1)} \\ p \Rightarrow (\forall x)t_j \quad & \text{(MP)} \end{aligned}$$

to obtain a proof of $p \Rightarrow t_i$ from S.

Case 4b: If x does not occur free in p, then by induction hypothesis, we write down a proof of $S \vdash (p \Rightarrow t_j)$ in which x does not occur free in any premiss. We then append the following lines

$$\begin{aligned} (\forall x)(p \Rightarrow t_j) \quad & \text{(Gen): As } x \text{ occurs bound} \\ & \quad \text{in any premiss in proof of } S \vdash (p \Rightarrow t_j) \\ (\forall x)(p \Rightarrow t_j) \Rightarrow (p \Rightarrow (\forall x)t_j) \quad & \text{(A7)} \\ p \Rightarrow (\forall x)t_j \quad & \text{(MP)} \end{aligned}$$

Hence we obtain a proof of $p \Rightarrow t_i$ from S. In all cases, it is easy to check that the induction hypothesis is valid before the $(i+1)^{th}$ step. □

Aim. We now embark on the proof of the Completeness Theorem that states that, for first-order logic, $\vdash, \models$ coincide.

Proposition 4.6 (Soundness Theorem)**.** *Let S be a set of formulae in a first-order language L and p be a formula in L. If $S \vdash p$, then $S \models p$.*

Proof. We proceed by induction on the length of a proof of p from S. Induction hypothesis: For any line $i \leq n$, if $S \vdash t_i$, then $S \models t_i$.

Case 1: If p is an axiom, then it is a tautology, automatically $\emptyset \models p \Rightarrow S \models p$.

Case 2: If $p \in S$ (p is a premiss), if A be arbitrary model of S, then $S \models p$ directly.

Case 3: If p follows by Modus Ponens, where $t_k = (t_j \Rightarrow p)$ where $j, k < i$. Since from induction hypothesis, we have $S \models (t_j \Rightarrow p), S \models t_j$, since by definition 4.7,

$S \models (t_j \Rightarrow p)$, meaning either $(t_j)_A = 0$ or $p_A = 1$. However we have $S \models t_j$, meaning $(t_j)_A = 1$. So we must have $p_A = 1$, concluding that $S \models p$.

Case 4: Note that if p follows by Generalization from an earlier line q in the proof, *i.e.*, if $p = (\forall x)q$, suppose that the formal proof from S to q is $t_1, ..., t_n = q$, since the sequence is finite, then it uses only finite number of premisses from S. Let $S_1 \subset S$ be the set of all premisses that actually occurs in the proof from S to q. Note that if p follows from generalization of q, by definition, x must not occur free in all the premisses that is actually used in the proof from S to q, and such proof exists by the construction of p. Hence the set S_1 is not empty set and $S_1 \vdash q$. Note that q comes from poof sequence that is before p. Hence by induction hypothesis, $S_1 \vdash q \Rightarrow S_1 \models q$. Since x does not occur free in S_1 we show that for arbitrary model A of S_1, $\Big(\forall s \in S_1, s_A = 1 \Rightarrow q_A = 1\Big) \Rightarrow \Big(\forall s \in S_1, s_A = 1 \Rightarrow ((\forall x)q)_A = 1\Big)$. By using definition 4.7 again, the formula p appears true in A if $p_A = ((\forall x)q)_A = 1 \iff q_A(a) = 1,\ \forall a \in A$. Note that since the sentence q has no free variable, the free variable x is basically vacuous free variable on q. $q_A(a)$ simply does not depends on a at all, meaning that $q_A(a) = q_A = 1$ for all $a \in A$, then $p_A = 1$. Hence we have $S_1 \models (\forall x)q$. Take arbitrary model A of S. if A is the model of S, it will automatically be the model of S_1. Since we know that $S_1 \models (\forall x)q$, meaning every model of S_1 satisfies $(\forall x)q$, in particular, A also satisfies $(\forall x)q$. Hence $S \models (\forall x)q$.

□

Note. For the converse, we first consider the special case when S is a theory and p is the formula $\perp$.

Definition 4.16. A theory S in a first-order language is inconsistent if $S \vdash \perp$, and otherwise S is consistent.

Theorem 4.7 (Model existence lemma)**.** *If S is a consistent theory in a first-order language $L = L(\Omega, \Pi)$, then S has a model.*

Idea of proof. We will build a model from the language L itself. We initially choose our structure to be the set A of all closed terms of L, *i.e.*, terms not involving variables (Meaning not even free/bound occur variables and terms). Examples of closed terms in the language of commutative rings with 1:

$$1 + 1,\ \ (1 + 0) + 1,\ \ 1 \cdot 1,\ \ (1 + 0) \cdot 0 + 1,\ \ etc.$$

We turn A into a structure by interpreting the operation symbols in the obvious way. In the example above, we would have $(1 + 0) +_A (1 + 1) = (1 + 0) + (1 + 1)$, or $0_A = 0, 1_A = 1$.

However, if S is the theory of fields, for example, then A is not a model of S. E.g., we have $S \vdash (1 + 0 = 1)$ (by using the axiom $(\forall x)(x + 0 = x)$ and set $x := 1.$),

but $1 + 0 = 1$ is not satisfied in A since the closed terms $1 + 0$ and 1 are different terms (since $(1 + 0)_A = +_A(1, 0) \neq 1_A$). There is an easy remedy. Introduce the equivalence relation $S \sim t \iff S \vdash (s = t)$ and replace A by its quotient $A/\sim$. Operation symbols are now interpreted on representatives of equivalence classes. E.g., $[1] +_A [1] = [1 + 1]$ in our example and also for example $([1 + 0] = [1])$. Two issues remain.

For an example of the first issue, consider the theory S of fields with characteristic 2, or 3, which consists of the theory of fields together with the sentence $(1 + 1 = 0) \vee (1 + 1 + 1 = 0)$. Then $S \nvdash (1 + 1 = 0)$ since S has models that are fields of characteristics 3. Similarly, $S \nvdash (1 + 1 + 1 = 0)$. It follows that in our structure A, we have $[1] +_A [1] = [1 + 1] \neq 0$ and $[1] +_A [1] +_A [1] = [1 + 1 + 1] \neq [0]$, and thus A is not a model (since they are not in the same equivalence class). As for propositional logic, we will first extend the theory S to a consistent theory that is complete. In general, we say that a theory S in a first-order language L is complete if for every sentence p, either $S \vdash p$ or $S \vdash \neg p$.

For an example of the second issue, consider the theory S of fields in which 2 has a root. This consists of the theory of fields together with the sentence $(\exists x)(xx = 1 + 1)$. Then the structure A defined above consisting of $\sim$- equivalence classes of closed terms is not a model, since there is no closed term t such that $[tt] = [1 + 1]$. In other words, we lack a witness to the sentence $(\exists x)(xx = 1 + 1)$, *i.e.*, we lack a closed term t such that $S \vdash p[t/x]$ where p is the formula $(xx = 1 + 1)$ (i.e. lack a constant t such that $tt = 1 + 1$). The solution is to add a new constant c to our language and the new sentence $(cc = 1 + 1)$ to our theory.

The problem is that the two processes of adding witnesses and of completion pull in different directions. When we add witnesses to a complete theory, the new theory may no longer be complete. When we complete a theory which has witnesses, the new theory may lack witnesses.

Note. Goal: Show if S is consistent theory, either $S \cup \{p\}$ or $S \cup \{\neg p\}$ is consistent. $\rightarrow$ there is a maximal consistent, complete, deductively closed $\bar{S} \rightarrow$ if S consistent and $S \vdash (\neg x)p$ where p is formula with 1 free variable x, then $S \cup \{p[c/x]\}$ is consistent where c is new constant to $L \rightarrow$ construct $\bar{L} = L(\Omega \cup \mathcal{C}, \Pi)$ and $\bar{S}$: for all sentence of form $(\exists x)p$ s.t. $S \vdash (\exists x)p$, there is a closed term t in $\bar{L}$ s.t. $\bar{S} \vdash p[t/x]$. $\rightarrow$ use above to construct each S_i from language L_{i-1}

Proof. We begin with the observation that if S is a consistent theory, then for any sentence p, one of $S \cup \{p\}$ and $S \cup \{\neg p\}$ is consistent. Indeed, otherwise, by Deduction Theorem, we have $S \vdash \neg p$ and $S \vdash \neg\neg p$, and hence an application of modus ponens yields $S \vdash \bot$, which is a contradiction.

Since we have a consistent theory S in a fixed language L, let

$\mathcal{P} := \{T \subseteq L' : S \subseteq T, T \text{ is consistent, and } L' \text{ is the set of all sentence in } L\}$. We first show that every chain $\mathcal{C} \subseteq \mathcal{P}$, there exists an upper bound U such that $T \subseteq U$ for all $T \in \mathcal{C}$. Consider

$$U = \bigcup_{T \in \mathcal{C}} T$$

We show that U is upper bound of chain $\mathcal{C}$. Since $\mathcal{C}$ is non-empty, pick any $T \in \mathcal{C}$. By definition of $\mathcal{P}$, $S \subseteq T \subseteq U$, then $S \subseteq U$. And that by definition of set union, for every $T \in \mathcal{C}$, $T \subseteq U$. We now show that U is consistent: Suppose not, then $U \vdash \bot$. Let $\{p_1, ..., p_k\} \subset U$ be the finite set of premisses used to show $U \vdash \bot$. Then, each $p_i \in T_i$ for some $T_i \in \mathcal{C}$. Since $\mathcal{C}$ is a chain, the finite collection $\{T_1, ..., T_k\}$ has a maximum element $T_{max} \in \mathcal{C}$, and each $T_i \subseteq T_{max}$. Hence all premisses $\{p_1, ..., p_k\}$ are contained in the consistent theory T_{max}. Meaning we can prove $\bot$ from T_{max} also. Which is a contradiction. Hence we must have $U \in \mathcal{P}$.

By Zorn's Lemma, $\mathcal{P}$ has maximal element $\bar{S}$ where $S \subseteq \bar{S}$ such that $\bar{S}$ is consistent.

We now show that $\bar{S}$ is deductively closed: if $\bar{S} \vdash q$, then $q \in \bar{S}$. Suppose not, then $q \notin \bar{S}$, consider $\bar{S} \cup \{p\}$, then $\bar{S} \cup \{p\} \supsetneq \bar{S}$, then we must not have $\bar{S} \cup \{p\}$ consistent, else it contradict the maximality of $\bar{S}$, hence $\bar{S} \cup \{p\} \vdash \bot$. By Deduction Theorem, $\bar{S} \vdash \neg q$. Since in the assumption, we have $\bar{S} \vdash q$, combining with $\bar{S} \vdash \neg q$, using modus ponen, we have $\bar{S} \vdash \bot$, contradiction. Hence $\bar{S}$ is deductively closed.

Now we show that either $p \in \bar{S}$ or $\neg p \in \bar{S}$. Suppose $p \notin \bar{S}$: Suppose that $p \notin \bar{S}$, the set $\bar{S} \cup \{p\}$ is a strictly larger set of sentences than $\bar{S}$, by maximality of $\bar{S}$, we must have $\bar{S} \cup \{p\} \vdash \bot \Rightarrow \bar{S} \vdash \neg p \Rightarrow \neg p \in \bar{S}$. If $\neg p \in \bar{S}$, then $\bar{S} \cup \{\neg p\} \supsetneq \bar{S} \Rightarrow \bar{S} \cup \{\neg p\} \vdash \bot \Rightarrow \bar{S} \vdash \neg\neg p \Rightarrow \bar{S} \vdash p \Rightarrow p \in \bar{S}$. Hence we have $\bar{S}$ is complete and also deductively closed.

Next assume that S is a consistent theory and $S \vdash (\exists x)p$ where p is a formula with one free variable x. We add a **new** constant c to the language L and show that $S \cup \{p[c/x]\}$ is consistent: Indeed, otherwise if $S \cup \{p[c/x]\} \vdash \bot$, by the Deduction Theorem, we get $S \vdash \neg p[c/x]$. Since c does not occur in S, if we replace every occurrence of c in the proof of $\neg p[c/x]$ from S with a new variable not used in the proof: let $t_1, t_2, ..., t_{n-1}, t_n = \neg p[c/x]$ be the sequence of finite proof from S to $\neg p[c/x]$. And define, for $1 \leq i \leq n$,

$$t_i = \begin{cases} t_i & \text{if } c \text{ does not occurs in } t_i \\ t_i[y/c] \text{ where } y \text{ is not used in the proof and not in } S & \text{if } c \text{ does occur in } t_i \end{cases}$$

. Then the new sequence of proof from S to $\neg p[c/x] : t'_1, t'_2, ..., t'_{n-1}, t_n = \neg p[y/x]$. Then,

$\neg p[y/x]$ has free variable y and by applying the axioms,

$$\begin{array}{rll} & (\forall y)(\neg p[y/x]) & \text{(Gen.) since } y \text{ is not in } S \\ ((\forall y)\neg p[y/x]) \Rightarrow (\neg p[y/x])[x/y] \overset{\text{Lemma 4.3}}{=} & \neg p & \text{(A6) since } x \text{ is free in } \neg p[y/x] \\ & \neg p & \text{(MP)} \\ & (\forall x)\neg p & \text{(Gen.) since } x \text{ is free in } x \text{ and not free in theory } S \end{array}$$

We obtained $S \vdash (\forall x)\neg p$. Note from beginning we assume that $S \vdash (\exists x)p$ and that S is consistent, combining the result that $S \vdash \neg\big((\exists x)p\big)(= (\forall x)\neg p)$ and $S \vdash (\exists x)p$, by modus ponen, $S \vdash \perp$, contradict the assumption on S is consistent.

Apply the above to all theorems of S of the form $(\exists x)p$, we obtain a new language $\bar{L} = L(\Omega \cup \mathcal{C}, \Pi)$ where $\mathcal{C}$ is a set of new constants, and a consistent theory $\bar{S} \supset S$ such that for every sentence of the form $(\exists x)p$ in the language L that is deducible from S, there is a closed term t (terms not involving variables) in $\bar{L}$ such that $\bar{S} \vdash p[t/x]$: Note that since S consists of sentence and also note that if we pick a closed term $t \in \mathcal{C}$, $p[t/x]$ has no free variables. So $S \cup \{p[t/x]\} \subseteq L'$, the set of all sentence in L. Since in above, we show that $S \cup \{p[t/x]\}$ is consistent (as $t \in \mathcal{C}$, which is not in S), $S \cup \{p[t/x]\} \in \mathcal{P}$, hence $S \cup \{p[t/x]\} \subseteq \bar{S}$, then we must have $p[t/x] \in \bar{S}$, by the property that $\bar{S}$ is deductively closed, we have $\bar{S} \vdash p[t/x]$, and since $t \in \mathcal{C}$, $\bar{S} \vdash p[t/x]$ where t is closed term in $\bar{L}$ (Note that this $\bar{S}$ is only the maximal, complete, consistent, deductively closed theory **IN** language L).

If we use all of above information, note that

1. For a language L, $\mathcal{P} := \{T \subseteq L' : S \subseteq T, T \text{ is consistent, and } L' \text{ is the set of all sentence in } L\}$ is non-empty

2. for consistent $S \in \mathcal{P}$, for arbitrary $p \in L'$, either $S \cup \{p\}$ or $S \cup \{\neg p\}$ is consistent

3. for each language L, there is a maximal, complete, consistent, deductively closed $\bar{S}$ such that if all sentence of the form $(\exists x)p(p$ is formula with 1 free variable $x)$ in the language L such that $S \vdash (\exists x)p$, then there exists a closed term $t \in \bar{L}$ such that $\bar{S} \vdash p[t/x]$ where $\bar{L} := L(\Omega \cup \mathcal{C}, \Pi)$, $\mathcal{C}$ is set of new constants.

Let us now start with a consistent theory S in a first-order logic $L = L(\Omega, \Pi)$. Put $S_0 = S, L_0 = L$ and inductively define, using the two processes described above:

For each $n \in \mathbb{N}$, define $S_0 := S, L_0 := L$,

- $S_0 \subset S_1 \subset T_1 \subset S_2 \subset T_2 \subset ...$

- $L_n = L(\Omega \cup \mathcal{C}_1 \cup \cdots \cup \mathcal{C}_n, \Pi)$, where $\mathcal{C}_1, \mathcal{C}_2, \cdots$ are pairwise disjoint sets of constants each disjoint from Ω, Π.

- S_n is consistent, complete theory in L_{n-1}

- T_n is a consistent theory in L_n which has witnesses for S_n

By using last paragraph conclusion, if we set $S_0 = S, L_0 = L$, first, L_0 has a complete consistent theory $S_1 \supset S_0$ (S_1 is $\bar{S}$ of L_0) where $\bar{S} \vdash p[t/x]$ for $t \in L_1$. And we define $L_1 = \bar{L} = L(\Omega \cup \mathcal{C}_1, \Pi)$. Note that $p[t/x] \notin S_1$ even if S_1 is deductively closed and also $S_1 \vdash p[t/x]$ as $p[t/x] \notin L_0 (\in L_1)$.So we define $T_1 := S_1 \cup \{(\exists x)p \to p[t_p/x] \mid S_1 \vdash (\exists x)p\}$, where each distinct sentence $(\exists x)p$ is assigned a distinct constant $c_p \in \mathcal{C}_1$. Note that for each such sentence, $T_1 \vdash p[c_p/x]$ by Modus Ponens (As $S_1 \vdash (\exists x)p$). We now show that T_1 is consistent theory in L_1: Suppose not, then $T_1 \vdash \bot$. Since a formal proof is a finite sequence of formulas, any derivation of $\bot$ from T_1 can only use finitely many premises from $T_1 \setminus S_1$. Thus, there exist $a_1, \ldots, a_k \in T_1 \setminus S_1$ such that $S_1 \cup \{a_1, \ldots, a_k\} \vdash \bot$, where each $a_i := (\exists x_i)p_i \to p_i[t_i/x_i]$ with $S_1 \vdash (\exists x_i)p_i$ and fresh constants $t_1, .., t_k \in \mathcal{C}_1$. By using Deduction Theorem, $S_1 \cup \{a_1, ..., a_k\} \vdash \bot \iff S_1 \cup \{a_1, ..., a_{k-1}\} \vdash (a_k \Rightarrow \bot) \iff S_1 \cup \{a_1, ..., a_{k-1}\} \vdash \neg a_k$. Meaning we have

$$S_1 \cup \{a_1, ..., a_{k-1}\} \vdash \neg\big((\exists x_k)p_k \to p_k[t_k/x_k]\big)$$

Note that $\neg\big((\exists x_k)p_k \to p_k[t_k/x_k]\big) \iff (\exists x_k)p_k \wedge \neg(p_k[t_k/x_k])$. Thus, we have, simultaneously that

- $S_1 \cup \{a_1, ..., a_{k-1}\} \vdash (\exists x_k)p_k$
- $S_1 \cup \{a_1, ..., a_{k-1}\} \vdash \neg(p_k[t_k/x_k])$

Note that the fresh constant t_k: Is not a member of S_1, it does not occur in $a_1, ..., a_{k-1}$, it does not occur in $(\exists x_k)p_k$, hence it is not a member of T_1. Hence by (Gen), (A6: replace it with y-fresh variable), and apply (Gen) again (As y is again fresh free variable in T_1), we got

$$S_1\{a_1, ..., a_{k-1}\} \vdash (\forall y)\neg p_k[y/x_k]$$

Applying the axiom,

$$\begin{aligned}(\forall y)\neg p_k[y/x_k] \Rightarrow (\neg p_k[y/x_k])[x_k/y] = \neg p_k \quad & \text{(A6) since } y \text{ is free in } \neg p_k[y/x_k] \\ \neg p_k \quad & \text{(MP)} \\ (\forall x_k)\neg p_k \quad & \text{(Gen)}\end{aligned}$$

Which

$$(\forall x_k)\neg p_k \equiv \neg(x_k)p_k$$

Hence, we have both

- $S_1 \cup \{a_1, ..., a_{k-1}\} \vdash \neg(\exists x_k)p_k$
- $S_1 \cup \{a_1, ..., a_{k-1}\} \vdash (\exists x_k)p_k$

, hence $S_1 \cup \{a_1, ..., a_{k-1}\} \vdash \bot$. By inductively applying the process k times, we eventually yield $S_1 \vdash \bot$, which contradict the assumption that S_1 is consistent. Hence we must have T_1 is consistent. Inductively, T_n is consistent with similar proof as above.

Inductively define all S_n, T_n, L_n, put $L^* = \bigcup_{n\in\mathbb{N}} L_n$, $S^* = \bigcup_{n\in\mathbb{N}} S_n$. We show S^* is a consistent theory in the language L^* that is complete and has witnesses:

- S^* is consistent in L^*: Suppose not, then $S^* \vdash \bot$ in L^*. Since formal proof is a finite sequence of formulas, a proof of $\bot$ from S^* can only use finitely many premisses from S^*. Let such finite set of premisses be $\{t_1, ..., t_k\} \subseteq S^*$. Since $S^* = \bigcup_{n\in\mathbb{N}} S_n$ and that $S_0 \subseteq S_1 \subseteq S_2 \subseteq \cdots$, each formula $t_i \in S_{n_i}$ for some S_{n_i} for $n_i \in \mathbb{N}$. Take $n' = \max\{n_1, n_2, ..., n_k\}$, then $t_i \in S_{n'}$ for all i by $S_n \subseteq S_{n+1}, \forall n \in \mathbb{N}$. Hence $S_{n'} \vdash \bot$. But $S_{n'}$ is constructed and consistent in $L_{n'-1}$ which is consistent in L^*, contradiction. Hence S^* must be consistent.

- S^* is deductively closed: for any sentence $p \in L^*$, $S^* \vdash p \Rightarrow p \in S^*$. Suppose again that $\{t_1, ..., t_k\}$ be the formal proof from S^* to p. Take $n' = \max\{n_1, ..., n_k\}$ of which $S^* = \bigcup_{n\in\mathbb{N}} S_n$ again where $\{t_1, ..., t_k\} \subseteq S_{n'}$, then $S_{n'} \vdash p$. Since p is a finite formula in L^*, $p \in L_m$ for some $m \in \mathbb{N}$. Take $K = \max\{n', m+1\}$, then $S_K \vdash p$. As S_K is constructed to be the maximal complete consistent theory in L_{K-1}, it is deductively closed w.r.t. sentence in L_{K-1}. As $p \in L_{K-1}, S_K \vdash p, p \in S_K$. As $S_K \subseteq S^* \Rightarrow p \in S^*$.

- S^* is complete: for any sentence $p \in L^*$, either $S^* \vdash p$ or $S \vdash \neg p$: Since S^* is deductively closed, then for any sentence $p \in L^*$, we either have $p \in S^*$ or $\neg p \in S^*$. Since p contains only finitely many constant symbols, $p \in L_m$ for some $m \in \mathbb{N}$. Consider S_{m+2}, which is a complete theory w.r.t. the language $L_{m+1} \supseteq L_m$ and $p \in L_m$, we must have $p \in S_{m+2}$ or $\neg p \in S_{m+2}$, and $S_{m+2} \subseteq S^*$, it follows that either $p \in S^*$, or $\neg p \in S^*$.(As S_{m+2} is maximal complete in L_{m+1}, and $p \in L_n \subset L_{n+1}$, S_{m+2} is the first $S - step$ that guarenteed to be complete for the entire language L_{m+1} and hence for $p \in L_m$, if we choose S_{m+1}, it is complete for L_m, but it lives prior to the witness expansion. Using S_{m+2} ensures p contains witness constants added in $\mathcal{C}_{m+1}$ from failing to be decided.)

- S^* has witnesses meaning If $(\exists x)p$ is a sentence in L^* and $S^* \vdash (\exists x)p$, then there exists a constant symbol $c \in L^*$ such that $S^* \vdash p[c/x]$: Suppose $S^* \vdash (\exists x)p$. Since S^* is complete and also consistent, $(\exists x)p \in S^*$. Since $(\exists x)p$ is a finite formula in L^*, then there is some $n \in \mathbb{N}$ such that $(\exists x)p \in L_n$. As S_{n+1} is complete in L_n and contains all provable formulas of S_n, we have $S_{n+1} \vdash (\exists x)p$. Now we define T_{n+1} as to be S_{n+1} union all the existential sentences in L_n proven by S_{n+1}. In particular, we have $(\exists x)p \Rightarrow p[c/x] \in T_{n+1}$ where $c \in \mathcal{C}_{n+1} \subset L_{n+1}$ is a new

constant. As $T_{n+1} \subset S^*$, both $(\exists x)p, (\exists x)p \Rightarrow p[c/x] \in S^*$, by (MP), we conclude that $S^* \vdash p[c/x]$ and hence $p[c/x] \in S^*$ with $c \in L^*$.

Note that every model of S^* is also a model of S, to complete the proof, we may assume that S is complete and has witnesses.

We let A be the set of equivalence classes of closed terms in L under the equivalence relation $s \sim t \iff S \vdash (s = t)$. We turn A into an $L-$structure as follows. For $\omega \in \Omega$ with arity n, we define $\omega_A : A^n \to A$ be setting

$$\omega_A([t_1], ..., [t_n]) = [\omega t_1 \cdots t_n]$$

and for $\varphi \in \Pi$ with arity n, we let

$$\varphi_A([t_1], ..., [t_n]) = 1 \iff S \vdash (\varphi t_1 \cdots t_n)$$

An 'easy' induction shows that if s is a term with variables in $\{x_1, ..., x_n\}$, then $s_A([t_1], ..., [t_n]) = [s[t_1/x, ..., t_n/x_n]]$:

- Base Cases:

  1. Case 1: s is a variable $x_i (1 \leq i \leq n)$: By definition of variable valuation in A:
  $$(x_i)_A([t_1], ..., [t_n]) = [t_i]$$
  putting t_i for x_i in s yields $x_i[t_1/x_1, ..., t_n/x_n] = t_i$, hence $[s[t_1/x_1, ..., t_n/x_n]] = [t_i]$.

  2. Case 2: s is a constant symbol $c \in \Omega$(0-arity function): By definition of constant interpretation in A :
  $$c_A([t_1], ..., [t_n]) = [c]$$
  Since c contains no variables, $c[t_1/x_1, ..., t_n/x_n] = c$, hence $c_A([t_1], ..., [t_n]) = [c[t_1/x_1, ..., t_n/x_n]]$

- Suppose that $s = \omega(u_1, ..., u_m)$ for an $m - ary$ function symbol $\omega \in \Omega$, where $u_1, ..., u_m$ are terms satisfying the induction hypothesis:

$$(u_j)_A([t_1], ..., [t_n]) = [u_j[t_1/x_1, ..., t_n/x_n]]$$

Evaluating s_A :

$$\begin{aligned} s_A([t_1],\ldots,[t_n]) &= \omega_A\Big((u_1)_A([t_1],\ldots,[t_n]),\ldots,(u_m)_A([t_1],\ldots,[t_n])\Big) \\ &= \omega_A\Big([u_1[t_1/x_1,\ldots,t_n/x_n]],\ldots,[u_m[t_1/x_1,\ldots,t_n/x_n]]\Big) \quad \text{(by I.H.)} \\ &= [\omega(u_1[t_1/x_1,\ldots,t_n/x_n],\ldots,u_m[t_1/x_1,\ldots,t_n/x_n])] \quad \text{(by definition of } \omega_A) \\ &= [(\omega(u_1,\ldots,u_m))[t_1/x_1,\ldots,t_n/x_n]] \quad \text{(by substitution properties)} \\ &= [s[t_1/x_1,\ldots,t_n/x_n]] \end{aligned}$$

In particular, $s_A = [s]$ for any closed term s.

Another induction on the language then shows that for any formula p with $FV(p) \subset \{x_1, ..., x_n\}$ and for closed terms $t_1, ..., t_n$, we have $S \vdash p[t_1/x_1, ..., t_n/x_n] \iff p_A([t_1], ..., [t_n]) = 1$:

Let $\sigma := [t_1/x_1, ..., t_n/x_n]$

- 1. Case 1: Equality $p \equiv (u = v)$

$$\begin{aligned} (u=v)_A([t_1],\ldots,[t_n]) = 1 &\iff u_A([t_1],\ldots,[t_n]) = v_A([t_1],\ldots,[t_n]) \\ &\iff [u\sigma] = [v\sigma] \quad \text{(by above proof)} \\ &\iff u\sigma \sim v\sigma \\ &\iff S \vdash (u\sigma = v\sigma) \\ &\iff S \vdash (u=v)\sigma \end{aligned}$$

  2. Case 2: Predicate $p \equiv \phi(u_1, ..., u_m)$ for $\phi \in \Pi$.

$$\begin{aligned} \phi(u_1,\ldots,u_m)_A([t_1],\ldots,[t_n]) = 1 &\iff \phi_A\Big((u_1)_A([t_1],\ldots,[t_n]),\ldots,(u_m)_A([t_1],\ldots,[t_n])\Big) \\ &= 1 \\ &\iff \phi_A([u_1\sigma],\ldots,[u_m\sigma]) = 1 \quad \text{(by above proof)} \\ &\iff S \vdash \phi(u_1\sigma,\ldots,u_m\sigma) \quad \text{(by definition of } \phi_A) \\ &\iff S \vdash (\phi(u_1,\ldots,u_m))\sigma \end{aligned}$$

- Assume the equivalence holds for subformulas q, r.

  1. Negation ($p \equiv \neg q$):

$$\begin{aligned}
(\neg q)_A([t_1],\ldots,[t_n]) = 1 &\iff q_A([t_1],\ldots,[t_n]) = 0 \\
&\iff q_A([t_1],\ldots,[t_n]) \neq 1 \\
&\iff S \nvdash q\sigma && \text{(by I.H.)} \\
&\iff S \vdash \neg q\sigma && \text{(since } S \text{ is complete and consistent)} \\
&\iff S \vdash (\neg q)\sigma
\end{aligned}$$

2. Implication ($p \equiv q \Rightarrow r$):

$$\begin{aligned}
(q \to r)_A([t_1],\ldots,[t_n]) = 1 &\iff q_A([t_1],\ldots,[t_n]) = 0 \text{ or } r_A([t_1],\ldots,[t_n]) = 1 \\
&\iff S \nvdash q\sigma \text{ or } S \vdash r\sigma \quad \text{(by I.H.)} \\
&\iff S \vdash \neg q\sigma \text{ or } S \vdash r\sigma \quad \text{(by completeness of } S) \\
&\iff S \vdash (q\sigma \to r\sigma) \quad \text{(by completeness/classical logic in } S) \\
&\iff S \vdash (q \to r)\sigma
\end{aligned}$$

3. Existential Quantifier ($p \equiv (\exists y)q(y, x_1, \ldots, x_n)$):

$$\begin{aligned}
((\exists y)q)_A([t_1],\ldots,[t_n]) = 1 &\iff \text{there exists } [c] \in A \text{ such that } q_A([c],[t_1],\ldots,[t_n]) = 1 \\
&\iff \text{there exists a closed term } c \in L \text{ such that} \\
&\qquad S \vdash q[c/y, t_1/x_1, \ldots, t_n/x_n] \text{ (by I.H.)}
\end{aligned}$$

- We now show this last condition is equivalent to $S \vdash ((\exists y)q)\sigma$:
  - ($\Longrightarrow$): If $S \vdash q[c/y]\sigma$, then by existential generalization ($q[c/y] \Rightarrow \exists y q$), $S \vdash (\exists y q)\sigma$.
  - ($\Longleftarrow$): If $S \vdash (\exists y q)\sigma$, since S is assumed/constructed to have witnesses, there exists a witness constant symbol $c \in L$ such that $S \vdash q[c/y]\sigma$.

In particular, if $p \in S$, then $p_A = 1$(*i.e.* p holds in A), and thus A is a model of S. □

Corollary 4.8 (Adequacy Theorem)**.** *Let S be a set of formulae in a first-order language L and p be a formula in L. If $S \models p$, then $S \vdash p$.*

Proof. To prove that $S \models p \implies S \vdash p$, it suffices to reduce to the case where S is a theory (a set of sentences) and p is a sentence.

Let $\mathcal{V}_{\text{free}}(S, p) = \{x_1, \ldots, x_n\}$ be the set of all variables occurring free in p or in any formula of S. Expand the language L to $L' = L \cup \{c_1, \ldots, c_n\}$ by introducing n fresh,

distinct constant symbols $c_1, \ldots, c_n \notin L$. Define:

$$p' := p[c_1/x_1, \ldots, c_n/x_n] \quad \text{and} \quad S' := \{q[c_1/x_1, \ldots, c_n/x_n] \mid q \in S\}$$

By the definition of semantic entailment for open formulas, $S \models p$ holds in L if and only if $S' \models p'$ holds in L', where S' is a set of sentences (a theory) and p' is a sentence in L'.

Suppose we have established the theorem for theories and sentences, yielding a formal deduction $S' \vdash p'$ in L'. Let $\pi' = (\phi'_1, \phi'_2, \ldots, \phi'_m)$ be a valid deduction sequence of p' from S', where $\phi'_m = p'$.

Choose n fresh variables $y_1, \ldots, y_n$ that do not appear anywhere in S, p, or the proof sequence π'. Replace every occurrence of c_i with y_i across the entire sequence π' to obtain a sequence $\pi = (\phi_1, \phi_2, \ldots, \phi_m)$ in L, where $\phi_i = \phi'_i[y_1/c_1, \ldots, y_n/c_n]$.

Since $c_1, \ldots, c_n$ are fresh constants not present in L, replacing them with fresh variables $y_1, \ldots, y_n$ preserves the logical validity of every axiom instance and rule application (Modus Ponens and Generalization). Thus, π is a valid formal deduction of $p[y_1/x_1, \ldots, y_n/x_n]$ from S.

To complete the deduction $S \vdash p$, append the following steps to π for each variable y_i (illustrated below for a single free variable x and fresh variable y):

1.	$S \vdash p[y/x]$	(from sequence π)
2.	$S \vdash (\forall y)\, p[y/x]$	(Gen, valid as y is not free in S)
3.	$\vdash (\forall y)\, p[y/x] \implies (p[y/x])[x/y]$	(Axiom A6, since x is free for y in $p[y/x]$)
4.	$S \vdash (p[y/x])[x/y]$	(MP on 2 and 3)

Since $(p[y/x])[x/y] \equiv p$, this yields $S \vdash p$. Iterating this process for all $y_1, \ldots, y_n$ converts $S' \vdash p'$ into a rigorous deduction $S \vdash p$ in L.

Now since $S \models p$, the theory $S \cup \{\neg p\}$ has no models, and so $S \cup \{\neg p\} \vdash \bot$ by Model existing Theorem. By Deduction Theorem, we have $S \vdash \neg\neg p$, and hence $S \vdash p$ using the axiom $\neg\neg p \Rightarrow p$ and modus ponens. □

Theorem 4.9 (Godel's Completeness Theorem for first-order logic)**.** *Let S be a set of formulae in a first-order language L and p be a formula in L. Then*

$$S \models p \iff S \vdash p$$

Proof. By Soundness Theorem and Adequacy Theorem. □

Corollary 4.10 (Compactness Theorem)**.** *Let S be a theory in a first-order language L. If every finite subset of S has a model, then S has a model.*

Proof. If S has no model, then $S \vdash \bot$ by Theorem 3. As proofs are finite, let $t_1, ..., t_k =$

$\perp$ be the formal finite proof from S to a $\perp$, there is a finite set $S' := S \cap \{t_1, ..., t_k\}$ of S such that $S' \vdash \perp$. By Soundness Theorem, $S' \models \perp$. Meaning that S' has no model, which contradicts the assumption about S. □

4.8 Applications of completeness/compactness

Question.

Can we axiomatise the theory of finite groups? In other words, does there exist a first-order theory T in a suitable language such that every finite group is a model of T and every model of T is a finite group?

Consider, for each $n \in \mathbb{N}$, the following sentence t_n defined in any language L:

$$(\exists x_1)(\exists x_2) \cdots (\exists x_n)(x = x_1 \vee x = x_2 \vee \cdots \vee x = x_n)$$

Note that t_n is satisfied in an $L-$structure A if and only if A has at most n elements. So we would like to take T to be the theory of groups (in the language of groups) and add the 'sentence' $t_1 \vee t_2 \vee \cdots$. Obviously, this is not possible since sentences are finite strings of symbols

Corollary 4.11. *The theory of finite groups is not axiomatisable as a first-order theory.*

Proof. Assume that T is a first-order theory whose models are the finite groups. Let $S = T \cup \{\neg t_1, \neg t_2, ...\}$ where the t_n are the sentences defined above. Note that $\neg t_n$ says that there are more than n elements in any model. Thus, every finite subset of S has a model: *e.g.*, the cyclic group of order N for sufficiently large N. By the Compactness Theorem, S has a model, which is absurd. □

Corollary 4.12. *Let S be a theory in a first-order language. If S has arbitrarily large finite models, then S has an infinite model.*

Proof. Set $S' = S \cup \{\neg t_1, \neg t_2, ...\}$ (t_n as before). Observe that, by assumption, every finite subset of S' has a model, and thus S' has a model by the Compactness Theorem. A model of S' is an infinite model of S.

□

Corollary 4.13 (The Upward Lowenheim-Skolem Theorem)**.** *If a first-order theory S has an infinite model, then it has an uncountable model.*

Proof. □

Note. For any set X we can take $I = \gamma(X)$(from Hartogs' Lemma). This shows that S has models that do not inject into X.

Remark. We can easily write down uncountable groups or vector spaces, but already for fields, the Upward Lowenheim–Skolem Theorem is not obvious.

Corollary 4.14 (The Downward Lowenheim-Skolem Theorem)**.** *Let S be a first-order theory in a countable language (meaning that $\Omega \cup \Pi$ is countable). If S has a model, then it has a countable model.*

Proof. □

4.9 Peano Arithmetic

We finish this chapter with another worked example. Our aim is to axiomatise the set of natural numbers. The key defining property of $\mathbb{N}$ is induction which we try to emulate with an axiom-scheme.

The language consists of $\Omega = \{0, s, +, \times\}$ with arity $0, 1, 2, 2$, respectively, and $\Pi = \emptyset$.

Peano Arithmetic (PA) (also known as *formal number theory*) is the theory in the language above with sentences as follows.

$$(\forall x)\neg(sx = 0)$$

$$(\forall x)(\forall y)(sx = sy \Rightarrow x = y)$$

$$(\forall x)(x + 0 = 0)$$

$$(\forall x)(\forall y)(x + sy = s(x + y))$$

$$(\forall x)(x \times 0 = 0)$$

$$(\forall x)(\forall y)(x \times sy = (x \times y) + x)$$

$$(\forall y_1) \cdots (\forall y_n)\big((p[0/x] \wedge (\forall x)(p \Rightarrow p[sx/x])) \Rightarrow (\forall x)p\big)$$
for any formula p with $FV(p) = \{x, y_1, ..., y_n\}$

Remark. The last axiom is the axiom-scheme for induction. The variables $y_1, ..., y_n$ are parameters. To see why they are needed, consider the following formula $p : (x+y)+z = x + (y + z)$ with free variables x, y, z. We can prove $(\forall x)(\forall y)(\forall z)p$ by induction on z

with x, y treated as parameters. Formally, we verify that

$$p[0/z] \wedge (\forall z)(p \Rightarrow p[sz/z])$$

holds in any model of PA, and hence it is provable by completeness. We then use the induction-scheme to deduce that $(\forall z)p$ holds, and hence so does $(\forall x)(\forall y)(\forall z)p$ by Generalization.

Chapter 5

Set Theory

This is just another piece of mathematical theory like group theory, topology, etc. We will axiomatise set theory as a first-order theory. So we could think of this chapter as just another worked example of first-order logic. Since any model of set theory should contain all of mathematics, it will obviously be a very complicated example of a first-order theory.

Zermelo–Fraenkel Set Theory (ZF). This is the version of set theory we will be studying.

5.1 Construction of ZF

The language has no operation symbols ($\Omega = \emptyset$) and a single predicate $\in$ of arity 2: $L = L(\emptyset, \{\in\})$. There will be 9 axioms (i.e., sentences of the theory): 2 to get started, 4 for building new sets and 3 further ones.

- A model of ZF will be denoted by V . This is a non-empty set together with an interpretation $\in_V \subset V \times V$ of the predicate $\in$ in which the 9 axioms of ZF are satisfied.
- Elements of V will be called 'sets'. When (a, b) is in $\in_V$, we say that'a belongs to b' or that 'a is a member of b'. We refer to V as the 'set-theoretic universe' or the 'universe of sets'.

So 'set' and 'belongs to' now have technical meanings inside the universe, but they retain their usual meaning in the world of mathematics of which V is a part. We will be interested in the question of what the universe of all sets looks like. We begin by listing the axioms of ZF.

5.1.1 Axiom of Extensionality (Ext)

'If two sets have the same members, then they are equal.'

$$(\forall x)(\forall y)\big((\forall z)(z \in x \iff z \in y) \Rightarrow x = y\big)$$

5.1.2 Axiom of Separation (Sep)

'Can form a subset of a set.' This is in fact am axiom-scheme:

$$(\forall t_1)\cdots(\forall t_n)(\forall x)(\exists y)(\forall z)(z \in y \iff (z \in x \wedge p))$$

for any formula p with $FV(p) = \{z, t_1, ..., t_n\}$, where $t_1, ..., t_n$ should be thought of as parameters.

Breaking down the above definition: given any parameters $t_1, ..., t_n$ and any starting set x, there is always a set y such that for every set z, z is a member of y if and only if z is a member of x and z satisfy the formula p.

The set y whose existence is asserted in the sentence above is unique by (Ext)(Why) and is denoted by $\{z \in x : p\}$:

Note that z represents the potential object that belongs to y, $t_1, ..., t_n$ are arbitrary set that passed into the condition p. Consider y_1, y_2 both satisfies the above (Sep) scheme

(Formally, we are adding an $(n+1)-$ary operation symbol to the language of ZF.)

Note. Parameters are needed. For example, if we take $p(z, t) \equiv t \in z$, then in a model of ZF we can form, given sets t and x, the subset $\{z \in x : t \in z\}$ of x.

5.1.3 Empty-Set Axiom (Emp)

'There is a set with no members.'

$$(\exists x)(\forall y)\neg(y \in x)$$

The set x whose existence this axiom asserts is unique by (Ext)(Why). We call this set the empty set denoted by $\emptyset$. (Formally, we add the constant $\emptyset$ to the language of ZF with the sentence $(\forall y)\neg(y \in \emptyset)$.)

Strictly speaking, this axiom is not needed as it follows from (Sep): in a structure $V \neq \emptyset$, we can pick at least one set $a \in V$, using $p(z) \equiv \neg(z = z)$ and applying (Sep) $(\exists y)(\forall z)(z \in y \iff (z \in a \wedge z \neq z))$. Since $z \neq z$ is always false, $z \in a \wedge z \neq z.$ is false for all z, thus $\forall z \neg(z \in y)$.

However, if in first-order logic we allow the empty set as a structure, then (Emp) is needed . Since consider $V = \emptyset$, then for any sentence starting with universal quantifier would be vacuously true, hence the example like Axiom of Extensionality (Ext) and also Axiom of Separation (Sep) would hold, however by definition of existential quantifier $(\exists x)p$, it is defined as $\neg(\forall x)\neg p$, which since $(\forall x)\neg p$ holds true, $(\exists x)p \equiv \neg(\forall x)\neg p$ would hold false. So (Emp) is strictly necessary to rule out $V = \emptyset$ and force the set-theoretic universe to contain at least one set.

5.1.4 Pair-Set Axiom (Pair)

'For any sets x, y, we can form $\{x, y\}$.'

$$(\forall x)(\forall y)(\exists z)(\forall t)\big(t \in z \iff (t = x \vee t = y)\big)$$

The unique (by (Ext)(Why)) set z whose existence is asserted here is denoted by $\{x, y\}$. We shall write $\{x\}$ for $\{x, x\}$. (Formally, we add a binary operation $\{,\}$ and a unary operation $\{\}$ to the language of ZF.)(What)

It follows from (Ext)(What??) that $\{x, y\} = \{y, x\}$ for all x, y. Thus, (Pair) gives us unordered pairs. Ordered pairs can be constructed using the following device(Why??)(due, independently, to K.Kuratowski and N. Weiner): for sets x, y define $(x, y) = \{\{x\}, \{x, y\}\}$. This satisfies:

$$(\forall x)(\forall y)(\forall z)(\forall t)\big((x, y) = (z, t) \iff (x = z \wedge y = t)\big)$$

as one would expect. We now introduce a number of abbreviations.

- 'x is an ordered pair' means

$$(\exists y)(\exists z)(x = (y, z))$$

- 'f is a function' means

$$(\forall x)(x \in f \Rightarrow \text{`}x \text{ is an ordered pair'})\wedge(\forall x)(\forall y)(\forall z)\Big(\big((x, y) \in f \wedge (x, z) \in f\big) \Rightarrow y = z\Big)$$

- 'f is a function from x to y (written $f : x \to y$) means'

$$(\text{`}f \text{ is a function'})\wedge(\forall s)\big((\exists t)((s, t) \in f) \iff s \in x\big)\wedge(\forall t)\big((\exists s)((s, t) \in f) \Rightarrow t \in y\big)$$

5.1.5 Union Axiom(Un)

5.1.6 Power-Set Axiom (Pow)

5.1.7 Axiom of Infinity (Inf)

5.1.8 Axiom of Replacement (Rep)

5.1.9 Axiom of Foundation(Fnd)

5.1.10 ZF

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